PEARSON EDEXCEL INTERNATIONAL A LEVEL BIOLOGY · WBI15

Unit 5: Respiration, Internal Environment, Coordination and Gene Technology

Revision notes, spec point by spec point (Topics 7 and 8)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

THE 20-MARK ARTICLE QUESTION

The last question on every WBI15 paper (Q8, or Q9 in Jan 2026) is based on a scientific article released before the exam. You have a clean copy in the exam. It is worth 20 of the 90 marks.
Each part names the paragraph it uses, and the parts follow the order of the article. Most marks are for ordinary Unit 1 to 5 biology applied to the article's context: the article gives you the setting, not the answers.

- **Prepare early.** Number the paragraphs, look up every unfamiliar term, and mark every spec term (receptor, transcription factor, ATP, antigen, mutation, plasmid). Examiners say the students who studied the article in detail did well and the rest often left parts blank.
- **Rehearse the standard chains** for each spec term you find: gene expression (stimulus → transcription factor → mRNA → translation → named protein), synapse and impulse, recombinant DNA, microarray and bioinformatics, inflammation, mutation → changed protein shape. Expect a definition of a term from the first paragraphs as the opener.
- **Never copy the article.** Text lifted from the article, or facts already in the stem, score nothing. Add the biological step the article leaves out, and name the gene, drug, tissue or organism it is about.
- **Read the command word.** "Explain" is the most common: give cause → effect in 3 to 4 linked steps. "Suggest" wants a definite mechanism ("fewer vesicles fuse"), not "it affects the brain". "Describe" needs no

reasons.

- **Use exact words:** impulses (not signals), membrane, ions, gene or allele, water potential (not concentration), increases or decreases (not "affects").
- **Practise under time.** Do past article questions in about 25 minutes, then mark them against the mark scheme, taking marks off for copied text and generic answers.

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Topic 7 – Respiration, Muscles and the Internal Environment

SPEC 7.1 Aerobic respiration: the overall reaction

WHAT THIS SPEC POINT ASKS

(i) Know that aerobic respiration splits the respiratory substrate, releasing CO_2 as a waste product, and joins the hydrogen from it to oxygen, releasing a large amount of energy. (ii) Know that respiration happens in many small steps, each catalysed by its own intracellular enzyme. You don't need enzyme names.

7.1 The overall equation

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (about 2870 kJ released per mole of glucose)

Read it the way the spec does:

- **Splitting the substrate.** The carbon skeleton of glucose is broken down step by step. Its carbon atoms leave as **CO_2**, a waste product, in the decarboxylation reactions of the link reaction and Krebs cycle.
- **Reuniting hydrogen with oxygen.** Hydrogen atoms are removed from the substrate (dehydrogenation) and carried by the coenzymes **NAD** and **FAD** to the electron transport chain. There they finally combine with **oxygen** from the air to make **water**. This is where most of the energy is released.
- The energy isn't "produced". It is **released** from the substrate and **transferred** to make ATP from ADP + Pi. Much of it (about 60%) is released as **heat**, which helps keep mammals warm.

7.1 Oxidation and reduction

	Oxidation	Reduction
Hydrogen	lost	gained
Electrons	lost	gained
Oxygen	gained	lost

- Glucose is **oxidised** (it loses hydrogen). Oxygen is **reduced** (it gains hydrogen and electrons) to water.
- NAD + 2H → reduced NAD. Reduced NAD is the carrier "loaded" with hydrogen; it becomes NAD again when it hands the hydrogen on.

7.1 Why respiration happens in steps

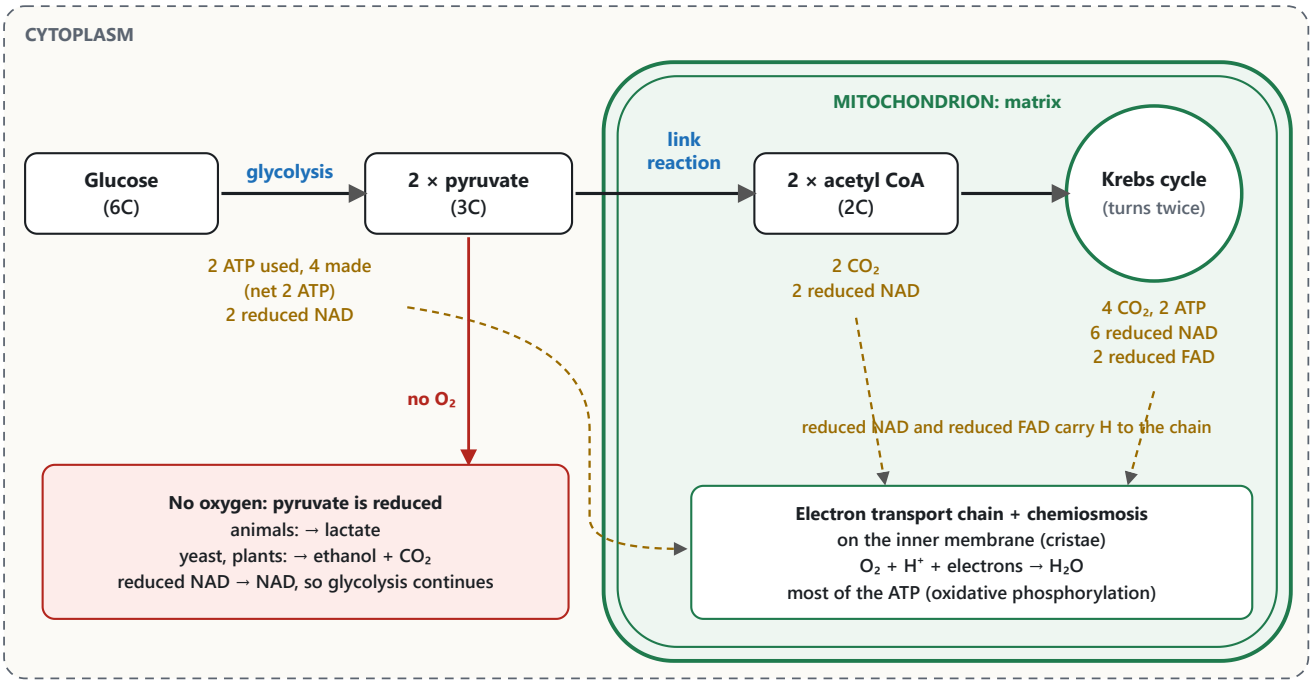
- Respiration is a **series of reactions**: the product of one step is the substrate of the next. Each step is catalysed by a **specific intracellular enzyme**, because each enzyme's active site fits only one substrate.
- Releasing the energy in **small amounts**:
  - lets much of it be captured in ATP at several points, instead of all being lost as heat at once (as when glucose burns),
  - avoids a sudden temperature rise that would denature enzymes,
  - allows control: the rate of a key step can be slowed when ATP is plentiful,
  - provides **intermediates** that other pathways use, and entry points for other substrates (fatty acids, amino acids, glycerol).

7.1 ATP: the energy currency

- **ATP** (adenosine triphosphate) = adenine + ribose + three phosphate groups.

- **Hydrolysis** of ATP to ADP + Pi, catalysed by **ATPase**, releases energy in small, usable amounts exactly where it's needed: active transport, muscle contraction, protein synthesis, the Na^+/K^+ pump in neurones.
- **ATP synthase** makes ATP from ADP + Pi (a condensation reaction) using energy released in respiration.
- ATP is soluble, moves easily around the cell and is remade very quickly. It is an immediate energy source, not a long-term store.

7.1 The four stages at a glance



Aerobic respiration of one glucose molecule. Glycolysis happens in the cytoplasm; the link reaction and Krebs cycle in the mitochondrial matrix; the electron transport chain on the inner membrane. Without oxygen, only glycolysis continues.

Stage	Where	Needs O_2?	Main products per glucose
Glycolysis	cytoplasm	no	2 pyruvate, net 2 ATP, 2 reduced NAD
Link reaction	mitochondrial matrix	yes (indirectly)	2 acetyl CoA, 2 CO_2, 2 reduced NAD
Krebs cycle	mitochondrial matrix	yes (indirectly)	4 CO_2, 2 ATP, 6 reduced NAD, 2 reduced FAD
Oxidative phosphorylation	inner mitochondrial membrane	yes	water, most of the ATP

WORKED EXAMPLE: EFFICIENCY OF AEROBIC RESPIRATION

The question gives the figures: 1 mol glucose releases 2867.48 kJ; up to 38 ATP are made per glucose; hydrolysis of 1 mol ATP releases 30.51 kJ.
Energy captured in ATP = 38 × 30.51 = **1159.38 kJ**.
Efficiency = 1159.38 ÷ 2867.48 × 100 = **40.43%** (WBI15 Jun 2021).
The other energy is released as **heat**.
Same method for a lipid: 122 × 30.51 = 3722.22 kJ; 3722.22 ÷ 12 244 × 100 = **30.4%** to 3 s.f. (WBI15 Jan 2024). "30.40%" was marked wrong: give exactly the s.f. asked for.

MARK-SCHEME LANGUAGE

- Equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$. One mark for the correct formulae, one for balancing. ATP/ADP in the equation are ignored. A correct word equation is accepted.
- ATP use: "**hydrolysis** of ATP (to ADP and Pi)" that "**releases energy**", then name the process it powers (active transport, muscle contraction, protein synthesis).
- Energy not trapped in ATP: "lost as **heat**", "raises body temperature".
- Oxygen is "the **final electron acceptor**"; water forms by the **reduction** of oxygen.

Often asked as: write and balance the equation (for glucose or a given lipid); describe how ATP supplies energy; efficiency calculations from supplied data; which molecule first contains labelled oxygen.

COMMON MISTAKE

"Energy is **produced**." Energy is released or transferred; ATP is produced.
Writing glucose as $C_6H_{12}O_2$, balancing by putting a 6 in front of glucose, or drawing the arrow backwards.
A GCSE-level ATP answer ("ATP gives energy for life processes"). Name the process and link it to **hydrolysis** of ATP.
Thinking the oxygen you breathe in ends up in CO_2. It ends up in **water**, so labelled O_2 first appears in water (WBI15 Oct 2022).

BEYOND THE SPEC, BUT EXAMINED

- Balancing a lipid equation: $C_{57}H_{104}O_6 + 80O_2 \rightarrow 57CO_2 + 52H_2O$ (WBI15 Jun 2021). Balance C first (gives the CO_2), then H (gives the H_2O), then O last.
- "Stepped and enzyme-controlled" is tested through **inhibitors** that block one step (cyanide, amytal, hydramethylnon), not through enzyme names (WBI15 Oct 2021, Jan 2025). See 7.4.

SPEC 7.2

Glycolysis

WHAT THIS SPEC POINT ASKS

Understand what glycolysis does in both aerobic and anaerobic respiration: phosphorylation of a hexose, ATP made by substrate-level phosphorylation, reduced NAD, pyruvate and (in animals without oxygen) lactate. You don't need the intermediate stages or compounds.

7.2 Where and when

- Glycolysis happens in the **cytoplasm**. It doesn't need oxygen, so it is the first stage of **both** aerobic and anaerobic respiration.
- Glyco-lysis means "sugar splitting": one 6C glucose becomes two 3C pyruvate.

7.2 The steps (per glucose)

1. **Phosphorylation.** Two ATP are hydrolysed and their phosphate groups are added to glucose, making a phosphorylated hexose (hexose bisphosphate). This makes the sugar more **reactive** (lowers the activation energy for splitting it), and the charged phosphates stop it leaving the cell.
2. **Splitting.** The phosphorylated hexose splits into **two triose phosphate** molecules (3C).
3. **Oxidation.** Hydrogen is removed from each triose phosphate (dehydrogenation). **NAD** picks it up, making **2 reduced NAD**.

4. **ATP formation.** Phosphate groups are transferred from the intermediates directly to ADP, making **4 ATP** by **substrate-level phosphorylation**. The end product is **2 pyruvate** (3C).

Net yield per glucose: **2 ATP** (4 made − 2 used), **2 reduced NAD**, **2 pyruvate**.

- **Substrate-level phosphorylation**: ATP made by transferring a phosphate group **directly** from a substrate molecule to ADP. It happens in glycolysis and the Krebs cycle and needs no electron transport chain. (Compare oxidative phosphorylation, 7.4.)

7.2 What happens to pyruvate next

	Oxygen available	No oxygen (anaerobic)
Animals	actively transported into the mitochondrial **matrix** → link reaction	**reduced to lactate** by reduced NAD (one step, no CO_2)
Yeast and plants	as in animals	decarboxylated to ethanal (**CO_2 released**), then ethanal reduced to **ethanol** by reduced NAD

7.2 Why anaerobic respiration makes lactate (or ethanol)

- Without oxygen, the electron transport chain stops, so reduced NAD can't be reoxidised there.
- A cell has only a little NAD. Once it is all reduced, step 3 of glycolysis can't happen and glycolysis would stop.
- Turning pyruvate into lactate uses the hydrogen from reduced NAD, so **NAD is regenerated**. Glycolysis can keep going and keep making **2 ATP per glucose**.
- That is far less ATP than aerobic respiration, because pyruvate isn't oxidised any further: most of the energy stays locked in lactate (or ethanol).

MARK-SCHEME LANGUAGE

- Pyruvate in anaerobic conditions: "**reduced / converted to lactate**"; "reduced NAD is **oxidised**", so NAD is regenerated and "**glycolysis can continue**"; "lactate transported to the liver". ACCEPT "H atoms added to pyruvate".
- Products of glycolysis: "pyruvate", "reduced NAD", "ATP" (net 2).
- Why cyanide doesn't stop anaerobic ATP production: "anaerobic respiration involves just **glycolysis**", "ATP made in glycolysis (by **substrate-level phosphorylation**)", "in the **cytoplasm**, so mitochondria / the ETC aren't needed".
- Less ATP: anaerobic respiration must be linked to **less ATP per glucose**. "Anaerobic respiration" on its own is not credited.

Often asked as: MCQs on where glycolysis happens and on its first step (phosphorylation of hexose); identify ADP, ATP, NAD and reduced NAD on a pathway diagram; why cyanide doesn't affect anaerobic ATP production; why a cheetah can sprint only briefly.

COMMON MISTAKE

"Anaerobic respiration produces no ATP." It makes a net **2 ATP** per glucose.
"Lactate is made to produce ATP." Its job is to **regenerate NAD**; the ATP comes from glycolysis.
Saying CO_2 is released when lactate forms. Only the ethanol pathway releases CO_2.
Writing NADP or NADPH (those belong to photosynthesis). Write "reduced NAD" to match the spec.
"The phosphate comes from pyruvate." It comes from the phosphorylated intermediates.

BEYOND THE SPEC, BUT EXAMINED

- ATP yields are supplied in questions, e.g. anaerobic 2 v aerobic 36 per glucose (WBI15 Jun 2023). Glucose released from **glycogen** is already phosphorylated without using ATP, so it gives **3 ATP** and 2 lactate (WBI15 Oct 2021).
- Cheetah sprints: fast-twitch fibres, anaerobic respiration, lactate build-up and oxygen debt; also heat builds up faster than homeostasis can remove it (WBI15 Oct 2022).
- Log-scale graphs (lactate against body mass in crocodiles). Say aerobic respiration "cannot keep up" with ATP demand in vigorous exercise, and link every trend to aerobic or anaerobic respiration (WBI15 Oct 2021, 6-mark).

SPEC 7.3 Link reaction and Krebs cycle

WHAT THIS SPEC POINT ASKS

Understand how the link reaction and Krebs cycle complete the oxidation of glucose: CO_2 formed by decarboxylation, ATP by substrate-level phosphorylation, and reduced NAD and reduced FAD by dehydrogenation. Both happen in the mitochondria (matrix), while glycolysis happens in the cytoplasm. You don't need the names of other compounds.

7.3 Mitochondrion structure

Part	Feature	Role
Outer membrane	smooth	pyruvate is actively transported across it
Inner membrane	folded into **cristae** (large surface area)	holds the electron carriers and **ATP synthase**: oxidative phosphorylation
Intermembrane space	narrow	H^+ builds up here
Matrix	fluid containing enzymes, mitochondrial DNA and ribosomes	**link reaction** and **Krebs cycle**

7.3 Three kinds of reaction to recognise

- **Decarboxylation**: a carboxyl group is removed as **CO_2**.
- **Dehydrogenation**: hydrogen is removed (an oxidation) and picked up by **NAD** or **FAD**, which are reduced.
- **Substrate-level phosphorylation**: a phosphate group is transferred directly to ADP, making ATP.

7.3 Link reaction (once per pyruvate, so twice per glucose)

1. Pyruvate (3C) is actively transported into the **matrix**.
2. It is **decarboxylated** (CO_2 removed) and **dehydrogenated** (hydrogen to NAD → reduced NAD). What's left is a 2C **acetyl group**.
3. The acetyl group combines with **coenzyme A** to form **acetyl CoA**, which carries it to the Krebs cycle.

- No ATP is made in the link reaction.

7.3 Krebs cycle (once per acetyl group, so twice per glucose)

1. The 2C acetyl group joins a **4C** compound to make a **6C** compound. Coenzyme A is released to collect another acetyl group.
2. The 6C compound is **decarboxylated** and **dehydrogenated**: a **5C** compound, CO_2 and reduced NAD are made.
3. The 5C compound is decarboxylated and dehydrogenated: a **4C** compound, CO_2 and reduced NAD.
4. The 4C compound is converted back into the original 4C acceptor. On the way, **1 ATP** is made by substrate-level phosphorylation, and **1 reduced FAD** and **1 more reduced NAD** are formed.

- It's a **cycle** because the 4C acceptor is regenerated, ready for the next acetyl group.
- Per turn: 2 CO_2, 1 ATP, 3 reduced NAD, 1 reduced FAD. Per glucose, double it.

7.3 Follow the carbon

6C glucose → 2 × 3C pyruvate → 2 × 2C acetyl + **2 CO_2** (link reaction) → **4 CO_2** (two turns of the Krebs cycle). That's 6 CO_2: every carbon atom of glucose has left as CO_2.

7.3 Tally per glucose

Stage	Where	CO_2	ATP (substrate-level)	Reduced NAD	Reduced FAD
Glycolysis	cytoplasm	0	2 (net)	2	0
Link reaction (× 2)	matrix	2	0	2	0
Krebs cycle (× 2)	matrix	4	2	6	2
Total		**6**	**4**	**10**	**2**

Most of the energy from glucose is now held in the **reduced coenzymes**. They take it to the electron transport chain (7.4).

EXAM TIP

Check whether a question means "per turn" or "per glucose". Per turn of the Krebs cycle: 2 CO_2, 1 ATP, 3 reduced NAD, 1 reduced FAD. Per glucose: twice that.

MARK-SCHEME LANGUAGE

- Fate of pyruvate in aerobic respiration (any 3): "moves into the **matrix**"; "**dehydrogenated** / oxidised by NAD"; "**decarboxylated** / CO_2 removed" to "a **2C acetyl** group"; "combines with **coenzyme A** to form **acetyl CoA**"; "enters the **Krebs cycle**".
- Krebs cycle products (only the first three you list are marked): "carbon dioxide", "ATP", "reduced NAD", "reduced FAD". Numbers like "6 ATP" are ignored. Reduced NADP is not accepted.
- Drawing a mitochondrion: "**double membrane** with **cristae**", "**matrix** labelled as the site of the Krebs cycle".
- Data on isolated mitochondria: pyruvate is "used in the link reaction / aerobic respiration"; "**ADP** is needed to make ATP" (respiration slows if ADP runs out); an inhibitor of pyruvate transport "stops pyruvate entering the matrix". Refer to each sample by name (WBI15 Jun 2022).

Often asked as: MCQs on locations and on how substrates enter the Krebs cycle; name three products of the Krebs cycle; describe the fate of pyruvate; label the site of the Krebs cycle; magnification or width of a mitochondrion from an electron micrograph.

COMMON MISTAKE

Putting the Krebs cycle on the cristae, or just "in the mitochondria". The link reaction and Krebs cycle are in the **matrix**; the electron transport chain is on the **inner membrane**.
Mixing up decarboxylation (CO_2 removed) and dehydrogenation (hydrogen removed).
Saying the link reaction makes ATP.
Reciting the Krebs cycle in a data question without using the data.

BEYOND THE SPEC, BUT EXAMINED

- Names the spec says you don't need are still credited: **acetyl CoA**, coenzyme A, **oxaloacetate** (the 4C acceptor), **citrate** (the 6C compound), "TCA cycle" (WBI15 Jan 2025).
- Other substrates: fatty acids are broken into 2C units that enter as acetyl CoA; amino acids enter as pyruvate, acetyl CoA or Krebs intermediates; glycerol enters glycolysis as triose phosphate.
- Magnification of a mitochondrion in standard form: convert mm to nm before dividing; watch the sign of the power of 10 (WBI15 Jan 2022, Jun 2022).

SPEC 7.4

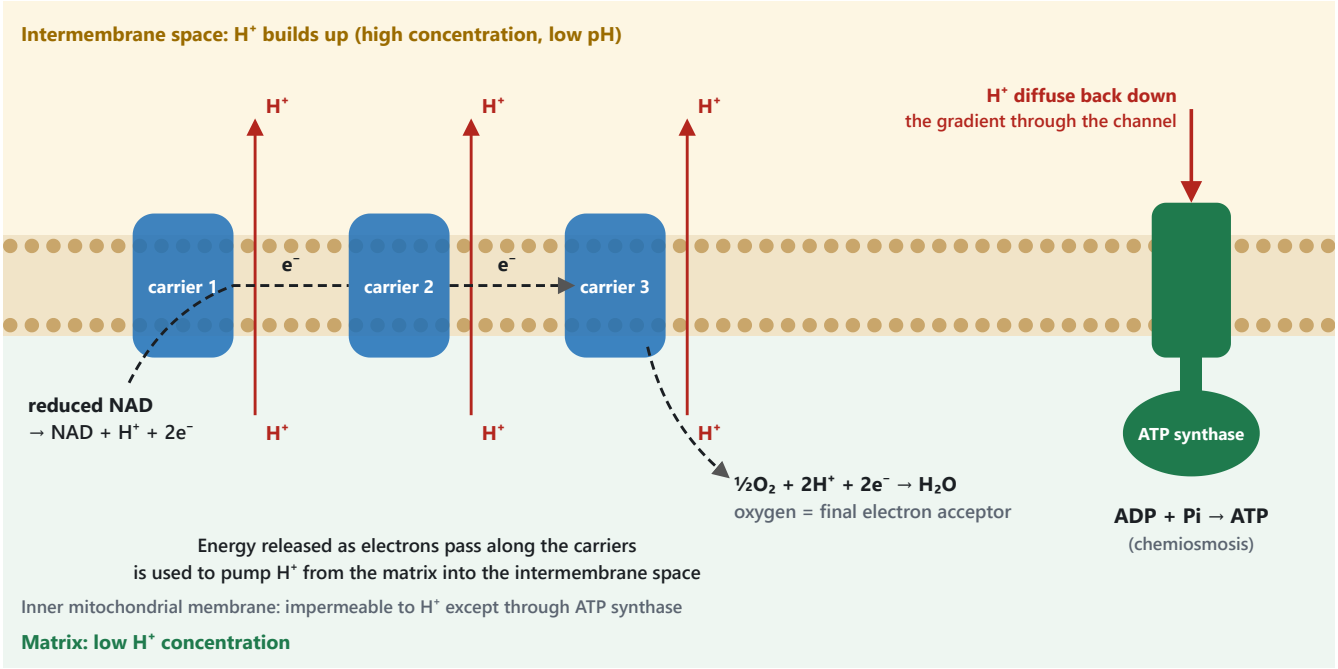
Oxidative phosphorylation: the electron transport chain and chemiosmosis

WHAT THIS SPEC POINT ASKS

Understand how ATP is made by oxidative phosphorylation, linked to the electron transport chain in the mitochondria, including the roles of chemiosmosis and ATP synthase.

7.4 Step by step

1. Reduced NAD and reduced FAD deliver hydrogen atoms to the **electron transport chain (ETC)**: a series of electron carrier proteins in the **inner mitochondrial membrane** (cristae). The coenzymes are **reoxidised**, so NAD and FAD can return to the link reaction and Krebs cycle.
2. Each hydrogen atom splits into a **proton (H^+)** and an **electron (e^-)**.
3. The electrons pass from carrier to carrier in a series of redox reactions (each carrier is reduced, then oxidised). Each carrier is at a **lower energy level** than the one before, so **energy is released** at each step.
4. That energy is used to **pump (actively transport) H^+** from the **matrix** into the **intermembrane space**.
5. H^+ builds up in the intermembrane space: a **proton gradient** (an electrochemical gradient) forms across the inner membrane, which is impermeable to H^+.
6. H^+ **diffuses** back into the matrix down the gradient, but only through the channel in **ATP synthase**. This flow drives ATP synthase to join **ADP + Pi → ATP**. Making ATP using this flow of protons is **chemiosmosis**.
7. At the end of the chain, **oxygen is the final electron acceptor**. It combines with electrons and H^+ to form **water**: $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$.



Electrons from reduced NAD pass along carriers in the inner membrane. The energy released pumps H^+ into the intermembrane space; H^+ diffuses back through ATP synthase, which makes ATP. Oxygen accepts the electrons at the end, forming water.

7.4 Why oxygen matters

- No oxygen → no final electron acceptor → electrons can't leave the last carrier → the chain stops → no H^+ is pumped → no chemiosmosis → very little ATP.
- Reduced NAD and reduced FAD can't be reoxidised, so the cell runs out of NAD and FAD. The link reaction and Krebs cycle stop too. Only glycolysis continues (anaerobic respiration).

7.4 How much ATP per glucose?

Source	Older textbook count	Modern estimate
Substrate-level phosphorylation (glycolysis + Krebs, net)	4	4
10 reduced NAD	10 × 3 = 30	10 × about 2.5 = 25
2 reduced FAD	2 × 2 = 4	2 × about 1.5 = 3
Total	**38**	**about 30–32**

- Real yields are lower than 38: some H^+ leaks back across the membrane without passing through ATP synthase, and energy is used to move pyruvate, ADP and Pi into the mitochondrion.
- Edexcel questions **give you the numbers**: 38 ATP and 30.51 kJ per ATP (WBI15 Jun 2021); 2 anaerobic v 36 aerobic (WBI15 Jun 2023). Use the values in the question.

MARK-SCHEME LANGUAGE

- Chemiosmosis (any 5): "hydrogen (atoms) carried by **reduced NAD / reduced FAD** to the ETC"; "split into protons and electrons"; "**electrons** pass along the chain, **releasing energy**"; "energy used to **pump protons** into the **intermembrane space**"; "protons **diffuse** back (down the electrochemical gradient) into the matrix through **ATP synthase**"; "forming ATP from ADP and Pi".
- Movement words: protons are "**pumped / actively transported**" out and "**diffuse**" back. "Stalked particle" is accepted for ATP synthase; "ATPase" is not. "ADP + Pi → ATP" with no mention of ATP synthase is ignored.
- Oxygen: "**terminal / final electron acceptor**", "accepts electrons and H^+ to form **water**", "no oxygen means electrons cannot pass along the chain".
- Cyanide (any 4): "electrons not passed on, so reduced coenzymes are not oxidised"; "cells respire **anaerobically** / glycolysis only"; "**less ATP** for muscle contraction"; "pyruvate converted to **lactate**"; "**pH falls**"; "enzymes denature".

Often asked as: describe the role of chemiosmosis in ATP synthesis (2–5 marks); describe oxidative phosphorylation; the role of reduced NAD or of oxygen; effects of inhibitors (cyanide, carbon monoxide, amytal, hydramethylnon); 6-mark interpretation of oxygen-consumption data for ETC defects.

COMMON MISTAKE

Mixing up the active and passive steps: protons are **pumped** into the intermembrane space and **diffuse** back through ATP synthase.
Confusing the **inner membrane** with the **intermembrane space**.
Starting a sentence with "It…" so the examiner can't tell whether you mean NAD or reduced NAD. Name the molecule.
"Affects ATP production." Say **increases** or **decreases**.
Stopping at "anaerobic respiration" in an inhibitor question. Carry on: **less ATP**, so less muscle contraction (or whatever the context needs).

BEYOND THE SPEC, BUT EXAMINED

- Named ETC complexes (I to IV), coenzyme Q and cytochrome c appear in data questions. You only need to follow where electrons enter and where a blocker stops them (WBI15 Jan 2021, 6-mark).
- **Carbon monoxide** binds to the haem in haemoglobin more strongly than O_2, so less O_2 reaches the cells to act as the final electron acceptor, and less ATP is made (WBI15 Jun 2023).
- An ETC inhibitor such as **amytal** reduces oxygen use and Krebs cycle activity, because NAD is no longer regenerated (WBI15 Jan 2025).
- Mitochondrial DNA codes for some ETC proteins, so mutations in it reduce ATP production (WBI15 Jan 2023 article).

SPEC 7.5

What happens to lactate after anaerobic respiration

WHAT THIS SPEC POINT ASKS

Understand what happens to the lactate made in animals after a period of anaerobic respiration.

7.5 During hard exercise

- Muscles need ATP faster than oxygen can be delivered, so extra ATP comes from anaerobic respiration (glycolysis → lactate) and from creatine phosphate.
- Lactate builds up in the muscle and passes into the blood. Lactic acid releases H^+, so **blood pH falls**. Low pH inhibits enzymes and contributes to muscle **fatigue**.

7.5 After exercise

1. Lactate **diffuses into the blood** and is carried to the **liver** (heart muscle also takes some up and uses it).
2. Once oxygen is available, lactate is **oxidised back to pyruvate** (its hydrogen goes to NAD).
3. The pyruvate is then either:
   - **respired aerobically**: link reaction, Krebs cycle and ETC, giving CO_2, water and ATP; or
   - **converted to glucose** in the liver, which goes into the blood or is stored as **glycogen**.

- The extra oxygen needed for this (and to remake ATP and creatine phosphate and reload myoglobin with O_2) is the **oxygen debt**. That's why breathing rate and heart rate stay high for a while after exercise stops.

WORKED EXAMPLE: EXPLAINING A LACTATE GRAPH

A cyclist's blood lactate rises over the first 15 minutes of a ride, then stays roughly constant from 15 to 60 minutes.

The question asks about 15 to 60 minutes only. Lactate is still being made by anaerobic respiration, but it is being **removed at the same rate** (converted back to pyruvate, or taken to the liver), so its concentration stays about the same (WBI15 Oct 2024). Don't explain the first 15 minutes too.

MARK-SCHEME LANGUAGE

- Fate of lactate (any 2–3): "lactate **diffuses into the blood**"; "**transported to the liver**"; "**converted / oxidised back to pyruvate**" (ACCEPT dehydrogenation); "pyruvate then used in the **link reaction** / aerobic respiration"; "converted to **glucose or glycogen** in the liver".
- Lactate must become **pyruvate** first. "Lactate used in the link reaction / Krebs cycle" is ignored.
- "Aerobic respiration after the exercise repays the **oxygen debt**."
- Lactate concentration rises, so blood **pH falls**.

Often asked as: describe what happens to lactate at the end of a race; explain lactate graphs (cyclists, diving seals); compare the relative changes in lactate and heart rate; explain why breathing stays fast after exercise.

COMMON MISTAKE

Leaving out the **liver**.

Going down an "oxygen debt" route ("breathing removes lactate") instead of describing the **conversion** of lactate.

Explaining the whole graph when the question names a time window.

"An unfit person has more lactate." Compare **rates**: makes lactate sooner, or removes it more slowly.

BEYOND THE SPEC, BUT EXAMINED

- Comparing two lines on different y-axes: use **percentage change**. Lactate rose by 308.5% while heart rate rose by only 22.5% (WBI15 Jan 2022).
- Seals diving: muscle lactate falls at the end of a dive as it moves into the blood and to the liver (WBI15 Oct 2025, 6-mark).

SPEC 7.6 Respiratory quotient (RQ)

WHAT THIS SPEC POINT ASKS

Understand what is meant by the respiratory quotient (RQ).

7.6 Definition

RQ = volume of CO_2 produced ÷ volume of O_2 used (over the same time). Moles can replace volumes, since equal numbers of moles of gas take up equal volumes at the same temperature and pressure.

7.6 Typical values (questions usually supply them)

Respiratory substrate	RQ
Carbohydrate (e.g. glucose)	1.0
Protein	about 0.8–0.9
Lipid	about 0.7
Mixture of carbohydrate and lipid	between 0.7 and 1.0
Anaerobic respiration that releases CO_2 (yeast, plants)	above 1

7.6 Why lipids give a lower RQ

- Lipids contain proportionally **more hydrogen** and much less oxygen than carbohydrates.
- More O_2 is needed to turn all that hydrogen into water, so more O_2 is used for every CO_2 made: RQ is below 1.
- Glucose: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$, so RQ = 6 ÷ 6 = **1.0**.

7.6 Reading RQ values

- **Above 1**: some anaerobic respiration that releases CO_2 (yeast, plants), or carbohydrate being converted into fat. Lactate fermentation in animals releases no CO_2 itself.
- **Falling over time**: the substrate is switching from carbohydrate to fat (e.g. a long cycle ride as glycogen stores run down).
- Measuring RQ needs **aerobic** respiration: the CO_2 comes from the link reaction and Krebs cycle, which only keep going when oxygen is there to act as the final electron acceptor (WBI15 Jan 2026).

WORKED EXAMPLE: RQ FROM AN EQUATION AND FROM DATA

From an equation: $2C_{18}H_{34}O_2 + 51O_2 \rightarrow 36CO_2 + 34H_2O$.
RQ = 36 ÷ 51 = **0.71**, so a lipid is being respired (WBI15 Jan 2024).
From measurements: a cockroach uses 1.7 cm^3 O_2 h^{-1} and produces 1.3 cm^3 CO_2 h^{-1}.
RQ = 1.3 ÷ 1.7 = 0.76 → **0.8** to 1 d.p. (WBI15 Oct 2021). Likely protein or a mixture of substrates.
Working backwards: O_2 uptake is 2.20 dm^3 min^{-1} and the RQ is 0.88.
CO_2 produced = RQ × O_2 used = 0.88 × 2.20 = **1.94 dm^3 min^{-1}** (WBI15 Jan 2023).

MARK-SCHEME LANGUAGE

- Definition (2 marks): "**carbon dioxide produced ÷ oxygen used / consumed**", as volumes or moles. "Amount" and "number of molecules" are ignored.
- Trend with a mixture: "as the proportion of lipid increases, the RQ decreases", supported by a calculation from the table.
- Cyclist: "RQ **decreased**" + "switching to **lipid / fat** rather than carbohydrate". Anaerobic respiration is ignored here.
- Why oxygen is needed to calculate RQ: "CO_2 is produced in the **link reaction / Krebs cycle**" and "oxygen is the **final electron acceptor** / needed for oxidative phosphorylation".

Often asked as: calculate RQ from an equation or gas volumes; work out CO_2 from the RQ and O_2 uptake; identify the substrate; interpret a changing RQ (a cyclist, germinating barley over 30 days).

COMMON MISTAKE

Inverting it ($O_2 \div CO_2$).
Describing the numbers but never naming the **substrate**.
Rounding too early, or giving too many decimal places.
"An RQ above 1 always means anaerobic respiration." It does only where anaerobic respiration releases CO_2 (yeast, plants); it can also mean fat is being made from carbohydrate.

BEYOND THE SPEC, BUT EXAMINED

- Germinating barley over 30 days: carbohydrate is respired at first, lipid from about day 5 to day 10 (RQ falls), then the proportion of carbohydrate rises again. Describe each stage separately (WBI15 Jan 2026).
- A falling RQ during exercise means a **gradual change of substrate**; few candidates said this (WBI15 Jan 2023).

SPEC 7.7

Core Practical 15: a redox indicator and respiration in yeast

WHAT THIS SPEC POINT ASKS

Use an artificial hydrogen carrier (a redox indicator) to investigate respiration in yeast.

7.7 Principle

- **Dehydrogenase** enzymes in glycolysis and the Krebs cycle remove hydrogen from respiratory substrates. Normally NAD (or FAD) accepts it.
- A **redox indicator** is an **artificial hydrogen acceptor**: it takes the hydrogen instead, and **changes colour when it is reduced**.
  - methylene blue: blue → colourless
  - DCPIP: blue → colourless
  - TTC (triphenyltetrazolium chloride): colourless → red
- The faster the yeast respires, the faster the colour changes. **Rate = 1 ÷ time** for the colour change.

CORE PRACTICAL 15: EFFECT OF TEMPERATURE ON RESPIRATION IN YEAST

1. Make a suspension of yeast in glucose solution (activated in warm glucose beforehand). Use the same volume and concentration in every tube.

2. Put tubes of yeast suspension, and **separate** tubes of indicator (e.g. 1 cm^3 of methylene blue), in **thermostatically controlled water baths** at a range of temperatures, e.g. 20, 25, 30, 35 and 40 °C. Leave them 10 minutes to **equilibrate**.
3. Add the indicator to the yeast, mix once by inverting, and start the timer.
4. **Don't shake again**: oxygen from the air would re-oxidise the indicator. (A thin layer of oil on top keeps air out.)
5. Stop the timer when the blue colour has gone, judged against a **colour standard** (yeast suspension with water in place of the indicator).
6. Repeat three times at each temperature. Calculate the mean time and the rate (1 ÷ mean time).

Controls: boiled yeast + indicator (no colour change: shows living yeast and active enzymes are needed); yeast with no glucose (much slower).

7.7 Variables

- **Independent:** temperature (or glucose concentration, or type of sugar).
- **Dependent:** time taken for the indicator to change colour, converted to a rate (s^{-1}).
- **Control:** volume and concentration of yeast suspension; volume and concentration of glucose; volume and concentration of indicator; pH (use a buffer); how the tube is mixed; same person judging the end point.

WORKED EXAMPLE: RATE FROM TIME

Mean time for methylene blue to lose its colour: **250 s** at 25 °C and **125 s** at 35 °C.
Rate at 25 °C = 1 ÷ 250 = **4.0×10^{-3} s^{-1}**. Rate at 35 °C = 1 ÷ 125 = **8.0×10^{-3} s^{-1}**.
The rate doubles for a 10 °C rise: molecules have more kinetic energy, so more enzyme–substrate complexes form. Above the optimum, enzymes denature (active sites change shape) and the rate falls.

7.7 Errors and improvements

- The end point is **subjective** (yeast is cream-coloured) → use a colour standard and the same observer, or a **colorimeter** reading absorbance at set times.
- Indicator re-oxidised by air → don't shake; oil layer.
- Yeast settles → stir the stock suspension before taking each sample.
- Safety: methylene blue stains (gloves); take care with hot water baths.

MARK-SCHEME LANGUAGE

- Applied version, the MTT assay (MTT turns yellow → purple when reduced): "use **living / viable** cells"; "**reduced NAD** is produced in respiration and reduces the dye"; "same concentration of each drug; same number of cells and conditions"; "record the **time** for the colour to change (or use a colorimeter)"; "the **longer** it takes to turn purple, the **more toxic** the drug" (WBI15 Jan 2022 article).
- Planning language (from Unit 6): "range of five temperatures", "**thermostatic** water bath", "equilibrate yeast and indicator **separately**", "named indicator", "add **glucose** (respiratory substrate)", "repeats and means".

Often asked as: not set as a method question in WBI15 from 2020 to 2026; its logic appears in unfamiliar contexts such as the MTT toxicity assay. Unit 6 asks for the full plan.

COMMON MISTAKE

Forgetting the glucose, or "thermostatically controlled" for the water bath. Shaking the tube while timing. Reversing the conclusion: a drug that slows respiration slows the colour change, so a **longer** time means a **more** toxic drug. Few candidates mentioned using living cells.

SPEC 7.8

Core Practical 16: respirometer, rate of respiration and RQ

WHAT THIS SPEC POINT ASKS

Use a simple respirometer to find the rate of respiration and the RQ of a suitable material, such as germinating seeds or small invertebrates.

7.8 Principle

- Organisms in a sealed tube take in O_2 and give out CO_2.
- **Soda lime** (or potassium hydroxide) **absorbs the CO_2**, so the only change in gas volume is the **O_2 taken in**. The gas volume and pressure fall, and the coloured liquid in the capillary tube moves **towards** the organisms.
- Volume of O_2 = distance moved × cross-sectional area of the capillary = **$\pi r^2 l$** (r = radius of the bore, l = distance moved).
- A **control tube** contains **glass beads** of the same volume as the organisms (and the same soda lime). Any movement in it is caused by changes in temperature or pressure, so it can be used to correct the readings.

CORE PRACTICAL 16: RATE OF RESPIRATION AND RQ

1. Weigh the organisms (e.g. 2.0 g of germinating seeds, or woodlice or maggots). Put soda lime in the tube, a **wire gauze** above it, then the organisms on the gauze.
2. Set up the control tube with glass beads of equal volume.
3. Stand both tubes in a **thermostatically controlled water bath** (e.g. 25 °C) with the tap open for 10 minutes to **equilibrate**.
4. Close the tap. Mark the liquid's position and measure the **distance it moves in a set time** (e.g. 5 minutes). Reset it with the syringe.
5. Repeat at least three times and calculate the mean. Rate = $\pi r^2 l$ ÷ time (÷ mass for a rate per gram).
6. **For the RQ:** repeat with the **same organisms**, but with the soda lime replaced by the same volume of glass beads, so CO_2 isn't absorbed.

7.8 Getting the RQ from the two runs

- With soda lime, movement **x** = O_2 taken in.
- Without soda lime, movement **y** = O_2 taken in − CO_2 given out (the net change).
- So CO_2 given out = x − y, and **RQ = (x − y) ÷ x**.
- If the liquid doesn't move without soda lime, CO_2 out = O_2 in, so RQ = 1 (carbohydrate).
- If it moves **away** from the organisms without soda lime, more CO_2 is given out than O_2 taken in: add that distance instead of subtracting it (RQ above 1).

WORKED EXAMPLE: MAGGOTS

Capillary radius 0.5 mm; 1.5 g of maggots; 25 °C; 5 minutes per reading.
With soda lime, the liquid moves **25 mm** towards the maggots. Without soda lime, it moves **5 mm** towards them.
O_2 used = $\pi r^2 l$ = 3.14 × 0.5^2 × 25 = **19.6 mm^3** in 5 min → 19.6 ÷ 5 = **3.9 mm^3 min^{-1}** → 3.9 ÷ 1.5 = **2.6 mm^3 g^{-1} min^{-1}**.
CO_2 produced = 25 − 5 = 20 mm of capillary. RQ = 20 ÷ 25 = **0.80**, so protein (or a mixture of substrates) is being respired.
The same capillary is used for both runs, so πr^2 cancels in the RQ and you can use the distances directly.

7.8 Control variables, safety and errors

- **Control:** mass and type of organism; temperature; volume of soda lime; capillary bore; equilibration time; age of the seeds or stage of the larvae.
- **Safety and ethics:** soda lime and KOH are **corrosive** (gloves, eye protection); the gauze stops organisms touching them; handle invertebrates gently, keep runs short and return them to their habitat.
- **Errors:** leaks at the bung (seal with petroleum jelly); temperature changes (water bath and control tube); soda lime becoming saturated (use fresh); small movements are hard to read (finer capillary, longer time, or a gas syringe or pressure sensor); seeds may switch to anaerobic respiration as O_2 runs low (keep runs short).

MARK-SCHEME LANGUAGE

- "CO_2 absorbed by **soda lime / KOH**, so movement shows **O_2 uptake**". Sodium hydrogencarbonate is **not** a CO_2 absorber.
- "**Control tube** with glass beads of the same volume" corrects for temperature and pressure changes.
- Glass beads replace soda lime in the second run "**so carbon dioxide is not absorbed**" (and the volume stays the same).
- RQ = "volume of CO_2 produced ÷ volume of O_2 used".
- Rate answers need the unit conversions done (cm^3 to dm^3, minutes to hours) and the s.f. or standard form asked for.

Often asked as: not set as a method question in WBI15 from 2020 to 2026; instead, RQ and rate calculations from given values (WBI15 Oct 2021, Jan 2023, Jan 2026). The full plan is a favourite in Unit 6.

COMMON MISTAKE

Saying a respirometer measures CO_2 output. It measures **O_2 uptake**; the CO_2 comes from the difference between the two runs.
Using the **diameter** instead of the radius in πr^2.
A control tube with soda lime only. It needs **inert glass beads** of the same volume as the organisms.

SPEC 7.9 Muscles, tendons, ligaments and the skeleton

WHAT THIS SPEC POINT ASKS

Know how muscles, tendons, the skeleton and ligaments interact to produce movement, including antagonistic muscle pairs, extensors and flexors.

7.9 The parts

Structure	Joins	Properties	Job
Bone	–	rigid and strong (collagen + calcium phosphate)	lever; support; attachment for muscles
Tendon	**muscle to bone**	strong, **inelastic** (mostly collagen)	transmits the pull of the muscle to the bone without losing force
Ligament	**bone to bone**	strong, **slightly elastic** (collagen + more elastic fibres)	holds the bones of a joint together, keeps it **stable**, allows some movement
Cartilage	covers the ends of bones	smooth, tough, flexible	reduces friction, absorbs shock
Synovial fluid	fills the joint cavity	–	lubricates the joint

7.9 Antagonistic pairs

- Muscles can only **pull** (contract). They can't push. So each movable joint needs two muscles that pull in opposite directions: an **antagonistic pair**.
- A **flexor** bends (flexes) a joint. An **extensor** straightens (extends) it.
- When one contracts, its antagonist **relaxes** and is **stretched** back to its original length by the contracting muscle.
- Elbow: **biceps** = flexor; **triceps** = extensor. Knee: hamstrings = flexors; quadriceps = extensors.
- To hold a joint **steady** at a fixed angle, **both** muscles contract with equal force, and ligaments hold the bones in place.

WORKED EXAMPLE: DESCRIBE HOW THE ARM IS STRAIGHTENED (2 MARKS)

The **triceps (extensor) contracts** and pulls on the lower arm bone (ulna) through its tendon.
The **biceps (flexor) relaxes** and is stretched. The arm straightens at the elbow.

MARK-SCHEME LANGUAGE

- Ligament: "joins **bone to bone**", "maintains joint **stability**". Tendon: "joins **muscle to bone**", "**transmits force**".
- "Flexor and extensor muscles are **antagonistic**"; "one **contracts** while the other **relaxes**". The words "joint" and "straightening" from the stem are ignored.
- Name **both** muscles of the pair: naming only one scored 0 (WBI15 Jun 2024).
- Composition: both are mostly **collagen** (strength); ligaments contain more **elastic** fibres so the joint can stretch a little; tendons stay stiff so force isn't lost.
- Ligament damage needs a stated cause (a twisting injury, overextension, a tackle). "Injury" alone gains nothing.

Often asked as: MCQs on tendons and ligaments; label the flexor, extensor and ligament on an elbow diagram; describe how the arm is straightened; how the knee is held steady; tendon and ligament composition and stiffness data.

COMMON MISTAKE

Swapping tendons and ligaments, or flexors and extensors (labelling the triceps as a flexor).
Writing about biceps and triceps when the question is about the knee.
"The muscle pushes the bone back." Muscles only pull: the antagonist makes the return movement.
Stopping at "holds bones together" without adding "keeps the joint **stable**".

BEYOND THE SPEC, BUT EXAMINED

- Collagen and elastin percentages and stress–strain slopes of tendons and ligaments, e.g. a slope ratio of 11.25 : 1 (WBI15 Oct 2025).
- Tendons get less stiff with age, so less force reaches the bone, contributing to falls in older people (WBI15 Jun 2021 article).
- Skeletal muscle is **neurogenic** (it contracts only when stimulated by motor neurones), not myogenic (WBI15 Jan 2022).

SPEC 7.10

Skeletal muscle fibres; fast and slow twitch

WHAT THIS SPEC POINT ASKS

(i) Know the structure of a mammalian skeletal muscle fibre. (ii) Understand the structural and physiological differences between fast-twitch and slow-twitch muscle fibres.

7.10 From muscle to filament

muscle → bundles of **muscle fibres** (the cells) → each fibre packed with **myofibrils** → each myofibril is a chain of **sarcomeres** → made of **actin** (thin) and **myosin** (thick) filaments.

7.10 The muscle fibre

Part	What it is	Why it matters
Multinucleate fibre	one long cell formed when many cells fuse	many nuclei to make proteins along its length; no membranes between sections, so the whole fibre contracts together
Sarcolemma	the cell surface membrane	carries the action potential
T-tubules (transverse tubules)	infoldings of the sarcolemma running deep into the fibre	carry the action potential into the fibre, close to the sarcoplasmic reticulum
Sarcoplasm	the cytoplasm	contains glycogen and myoglobin
Sarcoplasmic reticulum (SR)	specialised endoplasmic reticulum wrapped around each myofibril	stores **Ca^{2+}**, releases it when stimulated, pumps it back afterwards
Many mitochondria	packed between the myofibrils	ATP for contraction and for pumping Ca^{2+}
Myofibrils	bundles of actin and myosin filaments	the contractile units

7.10 The sarcomere and its bands

- A **sarcomere** is the repeating unit between two **Z lines** (Z discs).

- **A band** (dark): the length of the **myosin** filaments, including where they overlap with actin.
- **I band** (light): **actin only**. Each I band spans a Z line, so half of it lies in each neighbouring sarcomere.
- **H zone**: **myosin only**, in the middle of the A band.
- **M line**: the centre of the sarcomere, where the myosin filaments are held together.
- The alternating dark and light bands make skeletal muscle look **striated** (striped). See the figure in 7.11.

7.10 Fast-twitch v slow-twitch fibres

	Slow twitch (type I)	Fast twitch (type II)
Contraction	slower, less powerful, **sustained**	faster, more powerful, short bursts
Fatigue	resistant	fatigue quickly
Main ATP source	**aerobic** respiration	**anaerobic** respiration and creatine phosphate
Mitochondria	**many**	fewer
Myoglobin	**lots** (dark red)	little (pale)
Capillaries	**dense** network	fewer
Glycogen and creatine phosphate	less	**more**
Sarcoplasmic reticulum	less	**more** (fast Ca^{2+} release)
Fibre diameter	smaller (short diffusion distance for O_2)	larger
Examples	posture muscles in the back; marathon runners' legs; chicken leg	sprinters' legs; eyelid muscles; chicken breast

- **Myoglobin** is a red, oxygen-binding protein in muscle. It has a **higher affinity for O_2** than haemoglobin, so it takes O_2 from the blood and **stores** it, releasing it only when the muscle's O_2 level is very low.

EXAM TIP

"Explain why a marathon runner needs slow-twitch fibres" needs **comparatives** ("more mitochondria", "more capillaries") **and** a link to the athlete: many mitochondria → more aerobic respiration → more ATP over a long time without lactate building up → less fatigue.

WORKED EXAMPLE: SARCOMERE LENGTH FROM AN ELECTRON MICROGRAPH

Three sarcomeres, measured from the first Z line to the fourth, are **66 mm** long on a micrograph at × 10 000.
One sarcomere on the image = 66 ÷ 3 = 22 mm = 22 000 µm.
Actual length = image ÷ magnification = 22 000 ÷ 10 000 = **2.2 µm**.
Always measure from Z line to Z line, and convert the units before dividing.

MARK-SCHEME LANGUAGE

- Fibre structure: "single **multinucleate** cell", "repeating bands / sarcomeres", plus sarcolemma, SR, T-tubules, many mitochondria.
- Fast twitch compared with slow twitch (comparatives essential; converse accepted): "**fewer** mitochondria", "**less** myoglobin", "**fewer** capillaries", "**larger** diameter", "**more** sarcoplasmic reticulum", "**more** glycogen", "**more** creatine phosphate". "Less blood supply" alone is ignored because it isn't a structure.
- Slow-twitch benefits: "many mitochondria give more **aerobic** respiration / ATP"; "myoglobin **stores** oxygen"; "large capillary network supplies O_2 and glucose"; "slow, sustained contraction, so **less fatigue**".
- Sprinter: "fast speed over a short time"; "fast-twitch fibres **contract more quickly**"; "rely on **anaerobic** respiration or **creatine phosphate**, not on oxygen supply".
- Sarcomere: "the repeating unit **between Z lines**". Type 2 / IIa / IIb / IIx are accepted for fast twitch.

Often asked as: describe a muscle fibre; compare fast- and slow-twitch fibres; why sprinters or marathon runners have more of one type; the role of myoglobin; sarcomere length, ratios and magnification from an electron micrograph; training and fibre type.

COMMON MISTAKE

Slow-twitch fibres have capillaries "to store O_2". Capillaries **deliver** O_2; **myoglobin stores** it.
Listing properties without comparatives, or without linking a feature to what the athlete needs.
Giving function when the question asks for **structure** ("fatigue quickly" is not a structure).
"More mitochondria" the wrong way round (slow twitch has more).
Putting units on a magnification, or rounding it wrongly (15 500 is not 16 000).

BEYOND THE SPEC, BUT EXAMINED

- **Intermediate** (type IIa) fibres: endurance training converts some fast fibres into intermediate fibres; slow fibres don't change; more training gives a bigger switch (WBI15 Jan 2021).
- **Creatine phosphate** in fast-twitch fibres regenerates ATP quickly by giving its phosphate to ADP (WBI15 Jun 2021 article).
- **Sarcopenia** (age-related loss of muscle), satellite cells, loss of motor units and percentage-change calculations (WBI15 Jun 2021, Jan 2024 articles).
- Ratio of relaxed to contracted sarcomere length from a micrograph, e.g. 1.8 : 1 (WBI15 Jun 2024).

SPEC 7.11 Sliding filament theory

WHAT THIS SPEC POINT ASKS

Understand how skeletal muscle contracts according to the sliding filament theory, including the roles of actin, myosin, troponin, tropomyosin, calcium ions (Ca^{2+}), ATP and ATPase.

7.11 The proteins

- **Myosin** (thick filament): long tails with **heads** that stick out. Each head can bind to actin and has **ATPase** activity (it hydrolyses ATP).
- **Actin** (thin filament): has **binding sites** for myosin heads.

- **Tropomyosin**: a long, thin protein that winds along the actin filament and **covers the myosin-binding sites** when the muscle is relaxed.
- **Troponin**: a protein attached to tropomyosin that can **bind Ca^{2+}**.

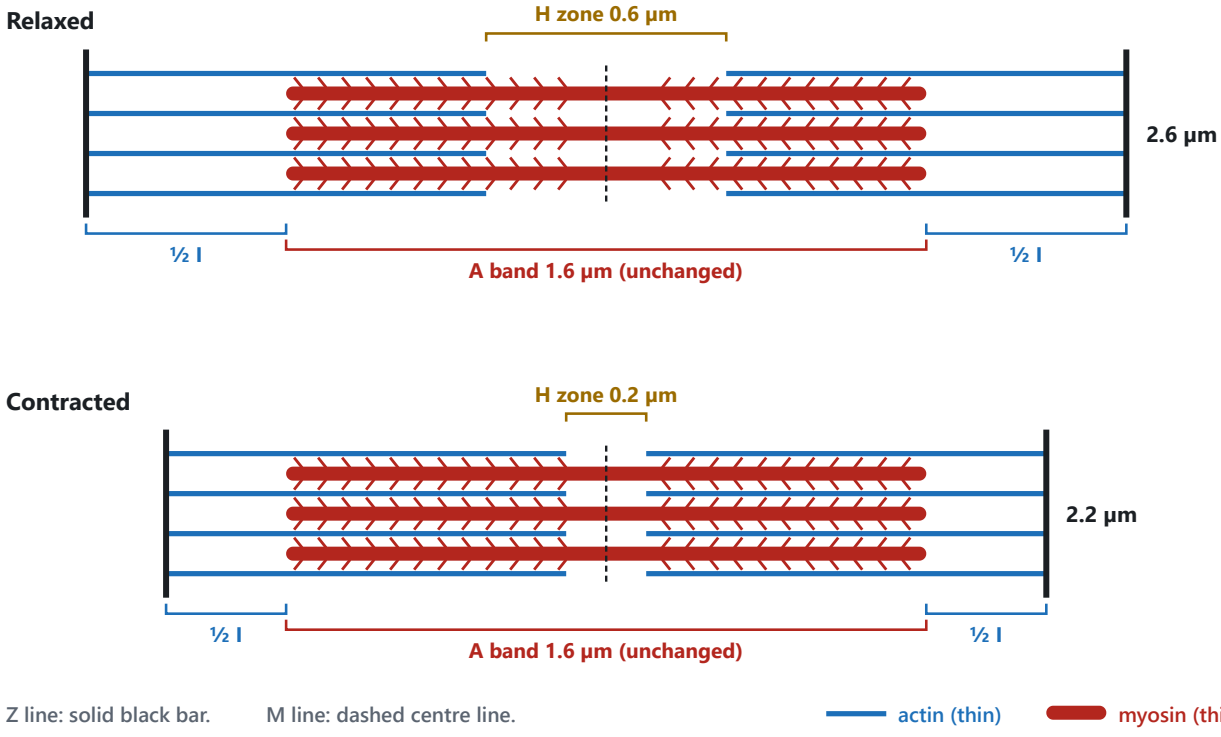
7.11 Contraction, step by step

1. An impulse arrives at the **neuromuscular junction**. Acetylcholine causes the **sarcolemma** to depolarise, and the action potential spreads along it and down the **T-tubules**.
2. This makes the **sarcoplasmic reticulum release Ca^{2+}** into the sarcoplasm.
3. Ca^{2+} **binds to troponin**, which **changes shape** and **pulls tropomyosin** away from the binding sites. The **myosin-binding sites on actin are exposed**.
4. A **myosin head** (already carrying ADP + Pi) **binds** to a site on actin, forming a **cross-bridge**.
5. The head **tilts** (the **power stroke**), releasing ADP + Pi and **pulling the actin filament towards the M line** (the centre of the sarcomere).
6. A new **ATP binds** to the myosin head. This makes the head **detach** from actin, breaking the cross-bridge.
7. **ATPase** on the myosin head **hydrolyses the ATP** to ADP + Pi. The energy released **returns the head to its upright ("cocked") position**.
8. The head binds to a new site further along the actin, and the cycle repeats as long as Ca^{2+} and ATP are present. Thousands of heads cycle out of step, so the pull is smooth.

7.11 Relaxation

- When impulses stop, Ca^{2+} is **actively transported back into the SR** (this uses ATP).
- Troponin loses its Ca^{2+} and tropomyosin moves back to **cover the binding sites**. No cross-bridges can form, so the muscle relaxes and is pulled back to its full length by its antagonist.

7.11 What changes length



One sarcomere, relaxed (2.6 µm) and contracted (2.2 µm). The actin and myosin filaments keep their length and slide past each other: the sarcomere, the I band and the H zone shorten; the A band stays the same.

	On contraction
Sarcomere (Z line to Z line)	**shorter**
I band	**shorter**
H zone	**shorter** (may disappear)
A band	**no change** (it is the length of the myosin)
Actin and myosin filaments	**no change**: they slide past each other

WORKED EXAMPLE: BAND LENGTHS

In the figure, the myosin filaments are 1.6 µm long and each actin filament reaches 1.0 µm from its Z line.
Relaxed sarcomere, 2.6 µm: I band (two halves) = 2.6 − 1.6 = **1.0 µm**; H zone = 2.6 − (2 × 1.0) = **0.6 µm**.
Contracted sarcomere, 2.2 µm: I band = 2.2 − 1.6 = **0.6 µm**; H zone = 2.2 − 2.0 = **0.2 µm**.
A band = **1.6 µm** in both. Ratio of relaxed to contracted length = 2.6 ÷ 2.2 = **1.2 : 1** (1 d.p.).

7.11 Two jobs for ATP

- **Detaching** the myosin head from actin (when ATP binds) and **re-cocking** it (when ATPase hydrolyses ATP).
- **Pumping Ca^{2+}** back into the SR so the muscle can relax.
- **Rigor mortis**: after death no ATP is made, so myosin heads can't detach and the muscles lock rigid.
- **Creatine phosphate** in muscle remakes ATP quickly: creatine phosphate + ADP → creatine + ATP.

MARK-SCHEME LANGUAGE

- Calcium route: "Ca^{2+} **released from the sarcoplasmic reticulum**"; "**binds to troponin**, which **changes shape**"; "**tropomyosin moves**"; "**exposes** (myosin) **binding sites on actin**"; "myosin head binds, forming a **cross-bridge**".
- ATP route, in logical order: "ADP and Pi released, myosin head **pulls actin** towards the centre of the sarcomere"; "**ATP binds** to the myosin head and **breaks the cross-bridge**"; "ATP **hydrolysed** (by ATPase on myosin), providing energy to **return the head** to its original position"; "head can bind again".
- Say "**myosin head**", not just "myosin". "The power stroke" is **not** accepted for the ATP-driven recovery step.
- Less Ca^{2+}: less binds to troponin, tropomyosin stays in place, **fewer cross-bridges**, the muscle stays relaxed or contracts weakly.
- Force against velocity: faster shortening means fewer cross-bridges attached at any moment, so **less force** ("inversely proportional" or "negative correlation" accepted).

Often asked as: describe the sliding filament theory (4 marks); the role of Ca^{2+}, troponin or ATP on its own; which part of the sarcomere shortens; why less Ca^{2+} disrupts contraction; isolated fibres in ATP or glucose solutions.

COMMON MISTAKE

Mixing up the power stroke (head tilts, ADP + Pi released) with the recovery stroke (ATP hydrolysed, head returns upright). This was the commonest error in the examiner reports.
Saying the actin or myosin filaments **shorten**. They slide; the sarcomere shortens.
Saying Ca^{2+} binds to tropomyosin, or that tropomyosin "frees the myosin heads". Ca^{2+} binds **troponin**, and tropomyosin uncovers the binding sites **on actin**.
Writing out the whole theory when the question asks only about ATP, or only about troponin.
"Myoglobin head." It's the **myosin** head.

BEYOND THE SPEC, BUT EXAMINED

- Isolated muscle fibres: in an **ATP** solution they shorten straight away (heads can detach and re-cock); in **glucose** they shorten less, because ATP must first be made by respiration. The solutions had the same water potential as the cytoplasm so the fibres didn't change length by osmosis (WBI15 Jan 2026).
- Creatine phosphate as a fast ATP regenerator (WBI15 Jun 2021 article).
- T-tubules and the Ca^{2+} release channels of the SR (ryanodine receptors) in an article context (WBI15 Jun 2021).

SPEC 7.12

The myogenic heart, its conduction system and ECGs

WHAT THIS SPEC POINT ASKS

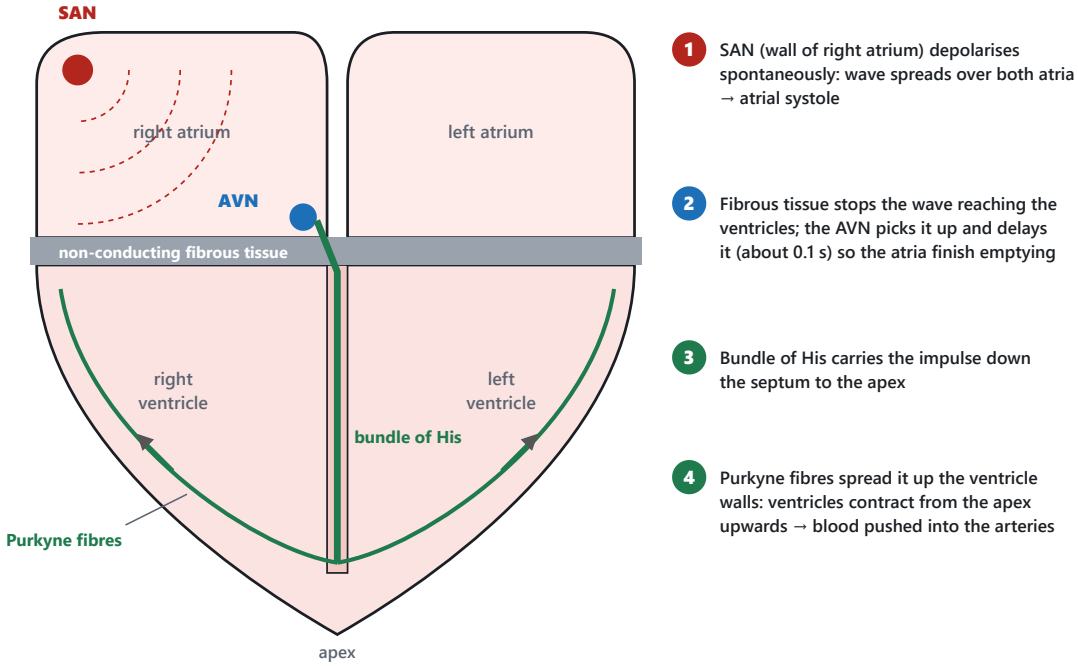
(i) Know that cardiac muscle is myogenic. (ii) Understand how the heart's own electrical activity coordinates the heartbeat: the sinoatrial node (SAN), atrioventricular node (AVN), bundle of His and Purkyne fibres. (iii) Understand how electrocardiograms (ECGs) help diagnose abnormal heart rhythms.

7.12 Myogenic

- **Myogenic**: cardiac muscle starts each contraction **itself**, without nerve impulses. Nerves and hormones only **change the rate** (7.13).
- Skeletal muscle is **neurogenic**: it contracts only when a motor neurone stimulates it.

7.12 How the heartbeat is coordinated

1. The **sinoatrial node (SAN)**, a patch of specialised cardiac muscle in the wall of the **right atrium**, depolarises spontaneously and regularly. It is the **pacemaker**.
2. A **wave of excitation** (depolarisation) spreads across both **atria**, which contract together (**atrial systole**), pushing blood into the ventricles.
3. A layer of **non-conducting fibrous tissue** between the atria and ventricles stops the wave spreading straight into the ventricles.
4. The **atrioventricular node (AVN)** picks up the wave and **delays it** by about 0.1 s. This gives the atria time to **finish contracting and emptying** before the ventricles contract.
5. The AVN passes the excitation down the **bundle of His**, in the **septum**, to the **apex** at the bottom of the ventricles.
6. **Purkyne fibres** carry it up through the ventricle walls. The ventricles contract **from the apex upwards** (**ventricular systole**), squeezing blood up and out into the aorta and pulmonary artery.
7. The muscle repolarises and relaxes (diastole) until the SAN fires again.



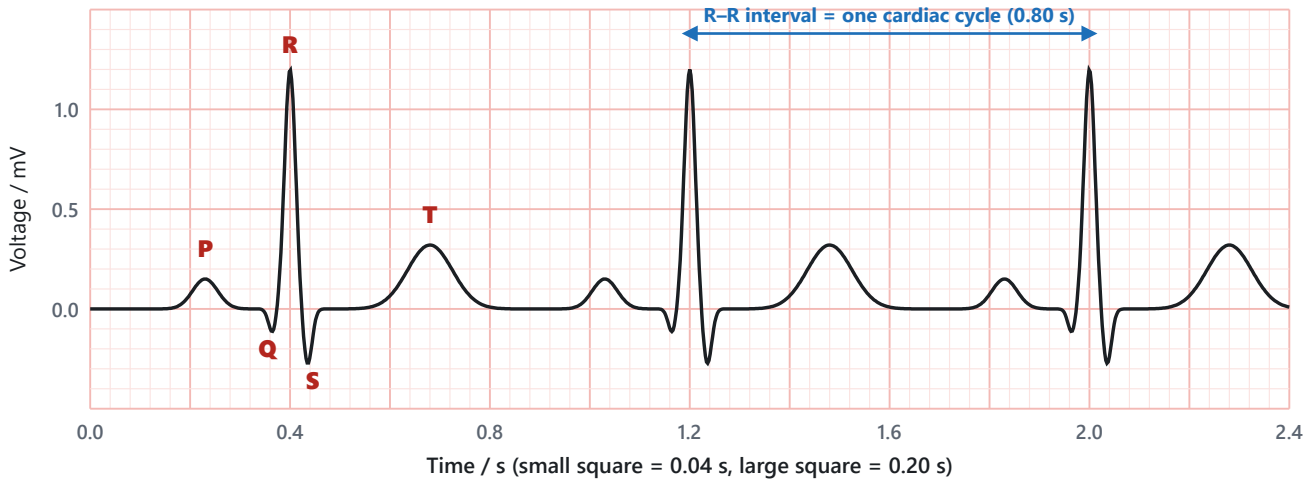
The conduction pathway (schematic; the right side of the heart is drawn on the left). The fibrous layer forces the wave through the AVN, which delays it; the bundle of His and Purkyne fibres make the ventricles contract from the apex upwards.

7.12 The ECG

An **electrocardiogram** records the electrical activity of the heart using electrodes on the skin.

Feature	What's happening
P wave	depolarisation of the **atria** → atrial systole
PR interval	time for the impulse to cross the atria and the AVN delay
QRS complex	depolarisation of the **ventricles** → ventricular systole (the biggest wave: ventricles have the most muscle)
T wave	**repolarisation** of the ventricles → ventricular diastole begins
R–R interval	one complete cardiac cycle

(The atria repolarise during the QRS complex, so this doesn't show as a separate wave.)



A normal ECG. Each large square is 0.20 s. The R–R interval is 0.80 s, so the heart rate is 60 ÷ 0.80 = 75 beats per minute.

7.12 Heart rate from an ECG

- **Heart rate (beats min^{-1}) = 60 ÷ time for one cycle in seconds**, measured from one R peak to the next (or between any two matching points).
- Or count beats: heart rate = number of beats × 60 ÷ time in seconds.

WORKED EXAMPLE: HEART RATE FROM AN ECG

In the figure, R to R = four large squares = 4 × 0.20 = 0.80 s. Heart rate = 60 ÷ 0.80 = **75 beats min^{-1}**. Counting instead: 19 beats in 7.0 s → 19 × 60 ÷ 7 = 162.8 = **163 beats min^{-1}** (WBI15 Jun 2021). Heart rate is a whole number, never a fraction.
From paper speed: at 25 mm s^{-1}, an R–R distance of 20 mm is 20 ÷ 25 = 0.80 s → 75 beats min^{-1}.

7.12 Abnormal rhythms

Rhythm	What the ECG shows
Tachycardia	fast resting rate (above about 100 beats min^{-1}): R peaks close together
Bradycardia	slow resting rate (below about 60 beats min^{-1}): R peaks far apart. Normal in many trained athletes
Atrial fibrillation	no clear P waves, an irregular wavy baseline, irregular spacing of QRS complexes
Ventricular fibrillation	a chaotic trace with no recognisable P, QRS or T: the ventricles quiver and don't pump (needs a defibrillator)
Ectopic beat	an extra, early beat followed by a longer gap
Heart block	a long PR interval, or P waves not followed by a QRS complex (the AVN delays or blocks the impulse)

- Doctors compare a patient's trace with a normal trace: the shape, height and duration of each wave, the intervals between them, the rate and the regularity. ECGs can also show heart muscle damaged by a heart attack (for example a changed S–T segment or T wave).

MARK-SCHEME LANGUAGE

- Myogenic: contraction "**initiated within the muscle** itself", "without nervous stimulation".
- Comparing ECGs: a comparison in the **same sentence**, using the labels P, QRS, T, R: "P wave smaller in A", "R peak sharper", "QRS duration shorter", "T wave larger", "heart rate lower in the normal trace", "shorter T–P gap in the patient". Converse accepted. Unlabelled descriptions gain nothing.
- Heart rate: show the count and the time (19 beats in 7 s → 163 bpm); a whole number.
- Impulse route: SAN → AVN → bundle of His → Purkyne fibres; the "**delay** at the AVN **allows the atria to empty**" (ventricles to fill).
- Designing an ECG study: record ECGs **before and after** the stimulus (or in two matched groups) and compare heart rate or the interval between the same peaks.

Often asked as: define myogenic; label the SAN and AVN; calculate heart rate from an ECG; compare two ECG traces; name the stage of the cycle at T; design an ECG investigation (posture, energy drinks).

COMMON MISTAKE

"The ventricles contract from the top (base) downwards." They contract **from the apex upwards**, so blood is pushed up towards the arteries.
Saying the cardiovascular centre or a nerve **starts** each heartbeat. The SAN starts it; nerves only change the rate.
Describing trace A, then trace B, without comparing them point by point.
Measuring between different points on each beat, or giving a heart rate as a decimal.
Not knowing that the **T wave** marks ventricular repolarisation (the start of ventricular **diastole**).

BEYOND THE SPEC, BUT EXAMINED

- Named arrhythmias haven't been required: abnormal rhythm is tested by **comparing traces** (WBI15 Jun 2021, Jan 2022, Oct 2023, Jan 2026).
- **POTS** (postural orthostatic tachycardia syndrome): heart rate rises by 30 beats min^{-1} or more, or above 120, within 10 minutes of standing up (WBI15 Jun 2022 article).
- Other measures of cardiovascular health: blood pressure, CRP in the blood, angiograms (WBI15 Jun 2025 article).
- A slower heart rate on an ECG shown as "fewer waves of depolarisation" per minute (WBI15 Oct 2025).

SPEC 7.13

Cardiac output; control of heart rate and ventilation

WHAT THIS SPEC POINT ASKS

(i) Calculate cardiac output. (ii) Understand how changes in ventilation and cardiac output deliver O_2 to the tissues and remove CO_2 quickly, including how heart rate and ventilation rate are controlled by the cardiovascular control centre and the ventilation centre in the medulla oblongata.

7.13 Cardiac output

Cardiac output = stroke volume × heart rate

- **Stroke volume**: volume of blood pumped out of the left ventricle in one beat (cm^3).
- **Heart rate**: beats per minute.
- **Cardiac output**: volume of blood pumped per minute ($cm^3\ min^{-1}$ or $dm^3\ min^{-1}$). About 5 $dm^3\ min^{-1}$ at rest; 20–30 $dm^3\ min^{-1}$ in hard exercise.

WORKED EXAMPLE: CARDIAC OUTPUT

One cardiac cycle takes 0.80 s, so heart rate = 60 ÷ 0.80 = 75 beats min^{-1}. Stroke volume = 72 cm^3.
Cardiac output = 72 × 75 = 5400 $cm^3\ min^{-1}$ = **5.4 $dm^3\ min^{-1}$** (÷ 1000).
Rearranged: cardiac output 4500 $cm^3\ min^{-1}$ and heart rate 51 beats min^{-1} give stroke volume = 4500 ÷ 51 = **88.24 cm^3** (WBI15 Jun 2021).
Percentage increase from rest (4000 $cm^3\ min^{-1}$) to exercise (21 250 $cm^3\ min^{-1}$) = (21 250 − 4000) ÷ 4000 × 100 = **431%**.

- **Trained athletes** have a larger, stronger left ventricle, so a bigger stroke volume. They get the same resting cardiac output with a **lower resting heart rate**, and reach a higher maximum cardiac output.

7.13 Why cardiac output and ventilation rise together in exercise

- Working muscles respire faster, so they use more **O_2** and make more **CO_2** (and heat).

- Higher **cardiac output** delivers O_2 and glucose faster and carries CO_2 away faster.
- Higher **ventilation** (faster, deeper breathing) keeps alveolar air high in O_2 and low in CO_2, so the diffusion gradients across the alveolar walls stay steep.
- Blood is redistributed: arterioles supplying skeletal muscle (and the skin, for heat loss) dilate, while those supplying the gut and kidneys constrict.

7.13 Control of heart rate

- The **cardiovascular control centre** is in the **medulla oblongata**.
- It receives impulses from:
  - **chemoreceptors** in the aortic bodies, carotid bodies and medulla, which detect **rising CO_2 / falling pH** (CO_2 dissolves in plasma, forming carbonic acid, which releases H^+);
  - **baroreceptors** in the aortic arch and carotid sinuses, which detect **blood pressure**;
  - stretch receptors in muscles and joints, which detect movement at the start of exercise.
- It responds by changing the impulses sent to the **SAN**:
  - more impulses along the **sympathetic (accelerator) nerve** → noradrenaline released at the SAN → SAN **depolarises more frequently** → heart rate **rises** (and the ventricles contract more strongly, raising stroke volume);
  - more impulses along the **parasympathetic vagus nerve** → acetylcholine released at the SAN → SAN fires less often → heart rate **falls**.

Change detected	Receptors	More impulses along	Heart rate
CO_2 rises, pH falls (exercise)	chemoreceptors	sympathetic nerve to SAN	rises
CO_2 falls, pH rises (after exercise)	chemoreceptors	vagus (parasympathetic) nerve to SAN	falls
Blood pressure rises	baroreceptors	vagus nerve to SAN	falls
Blood pressure falls	baroreceptors	sympathetic nerve to SAN	rises

7.13 Control of ventilation

- The **ventilation centre** is in the **medulla oblongata**. It sends impulses rhythmically to the **external intercostal muscles** and **diaphragm**, which contract to breathe in.
- When CO_2 rises (pH falls), **chemoreceptors** in the medulla and in the carotid and aortic bodies send more impulses to the ventilation centre. It sends **more frequent impulses** to the intercostal muscles and diaphragm, so breathing becomes **faster and deeper**.
- **Stretch receptors** in the walls of the bronchioles send impulses when the lungs inflate. These **inhibit** inspiration, so breathing out follows and the lungs don't over-inflate.
- At rest, breathing out is mostly passive: the muscles relax and the lungs recoil elastically.
- When CO_2 falls back after exercise, fewer impulses reach the breathing muscles, and the rate and depth fall.

MARK-SCHEME LANGUAGE

- Heart rate rising (any 4): "**increase in blood CO_2 / decrease in pH**"; "detected by **chemoreceptors**"; "impulses to the **cardiovascular control centre** in the **medulla oblongata**"; "**more impulses** via the **sympathetic** nerve to the **SAN**"; "SAN depolarises **more frequently**"; "heart muscle contracts more often". Baroreceptors and adrenaline are accepted alternatives.
- Slowing: pH rises after exercise; impulses along the **parasympathetic / vagus** nerve slow the SAN.
- Ventilation: "**chemoreceptors**"; "impulses to the **ventilation centre** in the medulla"; "**more impulses to the intercostal muscles and diaphragm**"; "increased rate (and depth) of breathing".
- Cardiac output: "heart rate × stroke volume"; it "measures the volume pumped per minute, so allows for different heart sizes". Athletes: "higher stroke volume, so lower resting heart rate for the same output".
- Write "impulses", never "signals" or "messages". Write "more" impulses.

Often asked as: cardiac output, stroke volume or heart rate calculations (with units and s.f.); how heart rate rises in exercise or falls afterwards; how breathing rate is controlled; 6-mark questions on the heart and exercise (athletes v non-athletes); blood redistribution to skin and muscles.

COMMON MISTAKE

Describing what happens **during** exercise when the question asks about **recovery** (or surfacing after a dive).
Saying the vagus nerve "sends impulses to the medulla". It carries impulses **from the medulla to the SAN**.
Saying impulses reach the diaphragm along "sympathetic nerves". The breathing muscles are stimulated by motor neurones (intercostal and phrenic nerves), not the autonomic system.
Finishing a ventilation answer by talking about heart rate.
Thinking breathing rate rises *because* heart rate rises (or the reverse). Both respond to the same rise in CO_2.
Forgetting stroke volume: cardiac output rises in exercise because **both** heart rate and stroke volume rise.

BEYOND THE SPEC, BUT EXAMINED

- **Cardiac index** = cardiac output ÷ body surface area, in $dm^3\ min^{-1}\ m^{-2}$ (WBI15 Jun 2022).
- **Alveolar ventilation** = (tidal volume − dead space) × breathing rate: (500 − 150) × 12 = 4200 $cm^3\ min^{-1}$ = 4.2 $dm^3\ min^{-1}$; × 60 = **250 $dm^3\ h^{-1}$** to 2 s.f. Units missing or wrong capped it at 2 marks (WBI15 Jan 2022).
- Lung stretch receptors inhibiting inspiration (WBI15 Jun 2022 article).
- Diving seals: the heart rate slows a lot during a dive (WBI15 Oct 2025).

SPEC 7.14

Adrenaline and the fight-or-flight response

WHAT THIS SPEC POINT ASKS

Understand the role of adrenaline in the fight-or-flight response.

7.14 What triggers it

- A threat (or the anticipation of exercise) is detected. The hypothalamus activates the **sympathetic nervous system**.

- Sympathetic neurones stimulate the **adrenal medulla** (the inner part of each adrenal gland, on top of the kidneys) to secrete **adrenaline** into the **blood**.
- Adrenaline reaches every tissue, but it only affects **target cells** that have **adrenaline receptors** on their cell surface membranes.

7.14 How it acts on a target cell

- Adrenaline can't cross the membrane, so it **binds to receptors on the cell surface membrane**.
- This activates an enzyme (adenylyl cyclase) that makes a **second messenger**, **cAMP**, inside the cell. cAMP activates a cascade of enzymes, so one adrenaline molecule has a large, fast effect. Peptide hormones work the same way (7.22).
- Adrenaline is an **amine** hormone (made from the amino acid tyrosine), not a peptide or a steroid.

7.14 The effects: ready to fight or run

Effect	How it helps
Heart rate and **stroke volume** rise (adrenaline acts on the SAN and the ventricle muscle)	higher cardiac output: O_2 and glucose reach the muscles faster
Vasodilation of arterioles to skeletal muscle; **vasoconstriction** of arterioles to the gut and skin	blood redirected to the muscles
Bronchioles dilate; breathing becomes faster and deeper	more O_2 in, more CO_2 out
Liver **glycogen → glucose**	more blood glucose for respiration
Pupils dilate (radial muscles of the iris contract, circular muscles relax)	more light enters the eye
Digestion slows (gut muscle relaxes)	blood and energy diverted to the muscles
Greater mental alertness	faster reactions

MARK-SCHEME LANGUAGE

- Role (any 3): "released **into the blood** from the **adrenal glands**"; "**binds to receptors** on target cells / organs"; then named effects: "increased **heart rate**", "**vasodilation** in muscle", "increased **ventilation** rate", "**pupil dilation**", "relaxation of gut muscle / decreased digestion", "**glycogen to glucose** in the liver".
- Pupil: "sympathetic / motor neurones stimulated", "**circular** muscles **relax**", "**radial** muscles **contract**".
- Combination data (heart rate of *Daphnia*): heart rate rose with adrenaline and fell with nicotine; highest rate with the most adrenaline and least nicotine. Use the **combination** of chemicals, as the question asked (WBI15 Oct 2023).

Often asked as: MCQs naming the hormone and its effects on breathing and heart rate; describe the role of adrenaline in fight or flight; adrenaline and the pupil; 6-mark question on ageing and the heart's response to adrenaline.

COMMON MISTAKE

Leaving out the two points few candidates gave: adrenaline is **released into the blood from the adrenal glands**, and it **binds to receptors**.
Calling adrenaline a **peptide** hormone, or saying it enters the cell. It's an amine that acts at the cell surface.
Telling the whole iris reflex story when the question is about adrenaline.
Confusing adrenaline (a hormone from the adrenal medulla) with noradrenaline (the neurotransmitter released by sympathetic neurones at the SAN).

BEYOND THE SPEC, BUT EXAMINED

- Adrenaline acting directly on the **SAN** is credited as an alternative route for raising heart rate (WBI15 Oct 2022, Oct 2023).
- Ageing reduces the heart's response to adrenaline: use all three graphs (heart rate, cardiac index, blood pressure), compare older and younger people, and comment on the error bars (WBI15 Jun 2022, 6-mark).
- Adrenaline also counted as a neurotransmitter in an MCQ pair (WBI15 Jan 2026).

SPEC 7.15 Core Practical 17: exercise and spirometer traces

WHAT THIS SPEC POINT ASKS

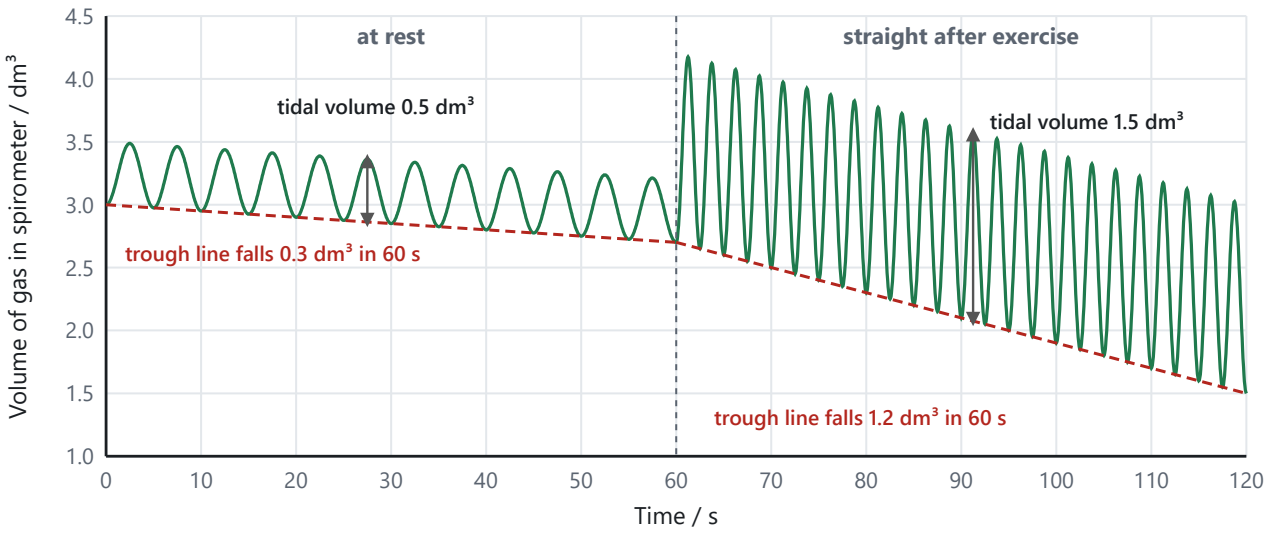
Investigate the effects of exercise on tidal volume, breathing rate, minute ventilation and oxygen consumption, using data from spirometer traces.

7.15 How a spirometer works

- A chamber of air (or medical-grade oxygen) floats over water. The subject breathes in and out of it through a mouthpiece, wearing a **nose clip**. The chamber rises and falls, and a pen or sensor records a trace of volume against time.
- **Soda lime** absorbs the CO_2 breathed out, so it isn't breathed back in. As the subject uses O_2, the **total volume** of gas in the chamber **falls**, so the trace slopes downwards. The **slope** gives the rate of O_2 consumption.

7.15 Reading the trace

Measurement	Meaning	How to read it
Tidal volume	volume of air breathed in (or out) in one normal breath	height from a **peak to the next trough**
Breathing rate	breaths per minute	count the peaks in a known time, then convert to per minute
Minute ventilation	volume of air breathed in per minute	tidal volume × breathing rate
O_2 consumption	volume of O_2 used per minute	fall in a line through the troughs (or peaks) ÷ time
Vital capacity	largest volume breathed out after the deepest possible breath in	biggest peak-to-trough height



A spirometer trace. After exercise the breaths are deeper (bigger tidal volume) and more frequent, and the trace falls more steeply because more O_2 is being used.

WORKED EXAMPLE: READING THE TRACE ABOVE

At rest: 12 peaks in 60 s → breathing rate = **12 breaths min^{-1}**. Peak to trough = **0.5 dm^3** (tidal volume). Minute ventilation = 0.5 × 12 = **6.0 dm^3 min^{-1}**.
The trough line falls from 3.0 to 2.7 dm^3 in 60 s → O_2 consumption = **0.3 dm^3 min^{-1}** (5.0 × 10^{-3} dm^3 s^{-1}).
After exercise: 24 peaks in 60 s → **24 breaths min^{-1}**; tidal volume = **1.5 dm^3**; minute ventilation = 1.5 × 24 = **36 dm^3 min^{-1}**, six times the resting value.
The trough line falls from 2.7 to 1.5 dm^3 in 60 s → O_2 consumption = **1.2 dm^3 min^{-1}**, four times the resting rate.

CORE PRACTICAL 17: REST AND EXERCISE

1. Check the subject is healthy (no asthma, heart or breathing problems) and has given **informed consent**.
2. Fill the spirometer. Check the **soda lime** is fresh. Fit a **sterilised or new mouthpiece** and a **nose clip**.
3. Let the subject breathe normally to get used to the apparatus, then record for **1–2 minutes at rest**.
4. The subject does a **standard exercise**, e.g. step-ups at a fixed rate (set by a metronome) for 2 minutes.
5. Reconnect **immediately** and record for 1–2 minutes. Keep recording to find how long breathing takes to return to the resting rate (recovery time).
6. Repeat with other subjects (resting between trials) and calculate means.

7.15 Why the values rise in exercise

- Muscles respire faster to make more ATP, so they use **more O_2** and make **more CO_2**.
- CO_2 lowers blood pH. **Chemoreceptors** detect this, and the **ventilation centre** sends more impulses to the diaphragm and intercostal muscles: breathing becomes **deeper** (tidal volume up) and **faster**, so minute ventilation rises.
- O_2 consumption stays above the resting rate for a while after exercise (the **oxygen debt**, 7.5).

7.15 Variables, safety and errors

- **Control:** use the same subject, or match age, sex, mass and fitness; same type, intensity and length of exercise; same room temperature.
- **Safety:** sterilise the mouthpiece; use fresh soda lime (otherwise CO_2 builds up and the subject becomes dizzy); stop if the subject feels unwell; soda lime is corrosive.
- **Errors:** leaks at the mouthpiece or nose; breathing unnaturally through a mouthpiece (practise first); exhausted soda lime makes O_2 use look too low; a sloping trace is hard to read by eye (use a digital spirometer, or average several breaths).

MARK-SCHEME LANGUAGE

- Minute ventilation = "tidal volume × breathing rate". Alveolar ventilation uses a formula given in the question.
- Give the s.f. asked: 0.54 breaths s^{-1} × 60 = 32.4 breaths min^{-1}; × 0.04 dm^3 = **1.3 dm^3 min^{-1}**. "1.30" scored only 1 mark (WBI15 Jun 2023).
- O_2 consumption from the **slope** of the trace, with units (e.g. dm^3 s^{-1}).
- Ventilation rises because "chemoreceptors detect the fall in pH / rise in CO_2", "impulses to the intercostal muscles and diaphragm", "muscles need more ATP, so more O_2 for aerobic respiration".

Often asked as: not shown as a trace in any WBI15 paper from 2020 to 2026, so a likely future question; ventilation calculations from given numbers (WBI15 Jan 2022, Jun 2023). Unit 6 asks for trace reading and the plan.

COMMON MISTAKE

Measuring tidal volume from zero, or from the baseline, to a peak. It's **peak to the next trough**.
Forgetting why the trace falls: O_2 is used up and the CO_2 is absorbed.
Not converting cm^3 to dm^3 (÷ 1000), or not rounding to the s.f. asked.

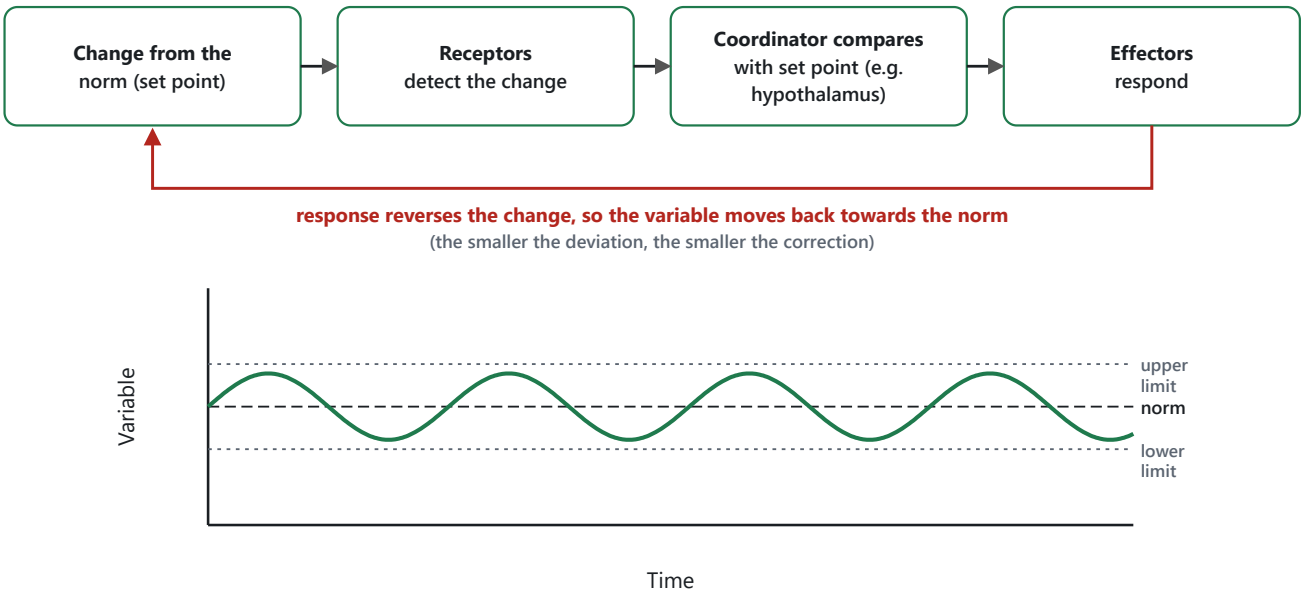
SPEC 7.16 Negative and positive feedback

WHAT THIS SPEC POINT ASKS

(i) Understand what is meant by negative feedback and positive feedback. (ii) Understand how negative feedback keeps systems within narrow limits.

7.16 Negative feedback

- A change away from the **norm (set point)** is detected by **receptors**. A **coordinator** (e.g. the hypothalamus) compares it with the set point and sends impulses or hormones to **effectors**. Their response **reverses the change**, bringing the variable back towards the norm.
- Once the variable is back at the norm, the receptors stop detecting a change, so the corrective response is switched off.



Negative feedback. The response reverses the change, so the variable fluctuates a little around the norm but stays between narrow limits.

7.16 How negative feedback keeps conditions within narrow limits

- Receptors monitor the variable **all the time**, so even a **small** change is detected and corrected before it grows.
- The bigger the change, the bigger the correction. As the variable nears the norm, the correction shrinks, so there's little overshoot.
- There are usually **separate mechanisms** for correcting a rise and a fall (sweating v shivering; insulin v glucagon). Having both gives more control and a faster response in either direction.
- The result is a **dynamic equilibrium**: the variable fluctuates slightly around the set point but stays within narrow limits.

7.16 Positive feedback

- A change triggers a response that **increases the change further**, moving the variable even further from where it started. It **amplifies** the change instead of reversing it.
- It is useful only when a fast, large change is needed and something else then ends it:
  - **Depolarisation of a neurone**: a few Na^+ channels open → Na^+ diffuses in → the membrane depolarises → more voltage-gated Na^+ channels open → more depolarisation, until an action potential is reached.
  - **Childbirth**: contractions of the uterus → more oxytocin released → stronger contractions, until the baby is born.
  - **Blood clotting**: activated platelets release substances that activate more platelets.
- It is harmful when homeostasis breaks down. In hypothermia, a lower body temperature slows metabolism, so less heat is released and the temperature falls further.

WORKED EXAMPLE: CALCULATING A RANGE

Blood calcium is kept within 6% of 91 mg dm^{-3}. 6% of 91 = 0.06 × 91 = 5.46.
Range = 91 − 5.46 to 91 + 5.46 = **85.54 to 96.46 mg dm^{-3}** (WBI15 Oct 2024). Calculate **both** limits.

MARK-SCHEME LANGUAGE

- Difference: "positive feedback **increases** the change in the variable; negative feedback **decreases / reverses** the change, returning it to normal".
- Negative feedback (2): "change **detected by a receptor**"; "**effector** stimulated that **counteracts** the change".
- Narrow limits: "constant monitoring", "small deviations detected and corrected", "separate mechanisms to correct a rise and a fall", "response switched off once back at the norm".
- Depolarisation as positive feedback (any 4): "neurotransmitter binds, a few **Na^+ channels open**"; "Na^+ enters and **depolarises** the membrane slightly"; "**more Na^+ channels open**"; "the membrane depolarises further"; "continues until an action potential". Write "sodium ions"; potassium ions are ignored.

Often asked as: describe the difference between negative and positive feedback; explain how negative feedback keeps a system within narrow limits; why depolarisation is positive feedback; 6-mark negative feedback of blood glucose; calculate the limits of a range.

COMMON MISTAKE

Defining negative feedback, then giving only an **example** for positive feedback with no definition.
Thinking negative feedback only lowers a variable. It **opposes** a change in **either** direction; positive feedback reinforces it.
Giving only the detection, or only the effector, in a 2-mark answer.
Writing "negative feedback" when the question asks for the name of the process (homeostasis, osmoregulation).

BEYOND THE SPEC, BUT EXAMINED

- Blood glucose control (insulin, glucagon, glycogenesis, glycogenolysis) is the example in a 6-mark negative feedback question. The top level needs the **switch-off**: less insulin (or glucagon) once glucose is back to normal. Don't confuse glucagon with glycogen (WBI15 Jan 2023).
- Positive feedback is examined at the **action potential**, not childbirth or clotting (WBI15 Jan 2026).

SPEC 7.17

Homeostasis and thermoregulation

WHAT THIS SPEC POINT ASKS

Understand what is meant by homeostasis and why it matters for keeping the body in dynamic equilibrium during exercise, including the role of the hypothalamus in thermoregulation.

7.17 Homeostasis

- **Homeostasis**: maintaining a **constant internal environment**, within narrow limits, despite changes inside and outside the body. It is a **dynamic equilibrium**: values fluctuate slightly around a set point and are constantly corrected by negative feedback.
- Why it matters:
  - **Temperature**: enzymes work best near 37 °C. Too hot and they **denature**; too cold and reactions slow down.
  - **pH**: changes alter the charges on amino acid R groups, changing the shape of enzymes and other proteins.

- **Water potential of the blood**: too low and cells **shrink** as water leaves by osmosis; too high and cells swell and may burst.
- **Blood glucose**: enough for respiration, but not so much that it lowers the water potential of the blood.

7.17 Dynamic equilibrium during exercise

Exercise pushes several variables away from the norm at once. Homeostatic mechanisms bring them back:

Exercise causes	Detected by	Correction
more heat from respiration: core temperature rises	thermoreceptors → hypothalamus	sweating; vasodilation in the skin
more CO_2: blood pH falls	chemoreceptors → medulla	faster, deeper breathing; higher heart rate
glucose used faster: blood glucose falls	pancreas	glucagon (and adrenaline): liver glycogen broken down to glucose
water lost in sweat: blood water potential falls	osmoreceptors → hypothalamus	more ADH, less urine (7.21)

7.17 Thermoregulation: the role of the hypothalamus

- **Thermoreceptors** in the **hypothalamus** detect the temperature of the blood flowing through it (core temperature). Thermoreceptors in the **skin** detect changes in the outside temperature early.
- The **thermoregulatory centre** in the hypothalamus (with a heat loss centre and a heat gain centre) compares the temperature with the set point (about 37 °C) and sends **impulses along motor and sympathetic neurones** to the effectors.

Too hot: heat loss centre	Too cold: heat gain centre
Vasodilation: arterioles supplying the skin capillaries dilate, so more blood flows near the surface and more heat is lost by radiation	**Vasoconstriction**: skin arterioles constrict, so blood is diverted through shunt vessels away from the surface and less heat is lost
Sweating: water evaporates from the skin, using heat energy from the body	Less sweat produced
Hair erector muscles relax: hairs lie flat, trapping less insulating air	Hair erector muscles contract: hairs stand up and trap insulating air (little effect in humans)
Metabolic rate lowered	**Shivering**: rapid, involuntary muscle contractions; respiration releases heat
	More **adrenaline** and **thyroxine** raise the metabolic rate

MARK-SCHEME LANGUAGE

- Homeostasis: "maintaining the body in a state of **dynamic equilibrium**" or "maintaining a **constant internal environment**". A described example is ignored.
- Thermoregulation (any 3–4): "**negative feedback**"; "**thermoreceptors** detect an increase above 37 °C"; "**impulses** to the **thermoregulatory centre / hypothalamus**" ("heat loss centre" accepted); "impulses via **motor / sympathetic neurones** to a **named effector**"; "a response that increases heat loss or reduces heat gain" (vasodilation, sweat glands, hair erector muscles, lower metabolic rate). "Signal" is rejected; "control centre" on its own is ignored.
- Heat loss in exercise (graph): rises, then reaches a maximum and levels off. "Comment" means describe, not explain.

Often asked as: state what homeostasis means; describe the role of the hypothalamus in keeping temperature steady; temperature control during exercise (humans, kangaroo rats); describe heat-loss graphs; how a seizure raises body temperature.

COMMON MISTAKE

"Capillaries dilate" or "blood vessels move closer to the skin." The **arterioles** dilate or constrict; no vessels move.
Describing the response (more sweating) without saying **how the effector is stimulated**: impulses along neurones from the hypothalamus.
"Signals" or "messages" instead of impulses.
Giving an example of homeostasis instead of the definition, or "keeping conditions constant" with no idea of narrow limits.

BEYOND THE SPEC, BUT EXAMINED

- The **heat loss centre** of the hypothalamus is credited by name (WBI15 Jan 2024, Oct 2024).
- Blood redistributed by vasodilation to the skin (heat loss) and the muscles (oxygen) during exercise, so less reaches the kidneys (WBI15 Jan 2026).
- Insulin falls during exercise: read the negative values on a "change in insulin" axis carefully, and watch the time units (WBI15 Jan 2025).

SPEC 7.18

Structure of the mammalian kidney

WHAT THIS SPEC POINT ASKS

Know the gross (whole-organ) and microscopic structure of the mammalian kidney.

7.18 Gross structure (a kidney cut lengthways)

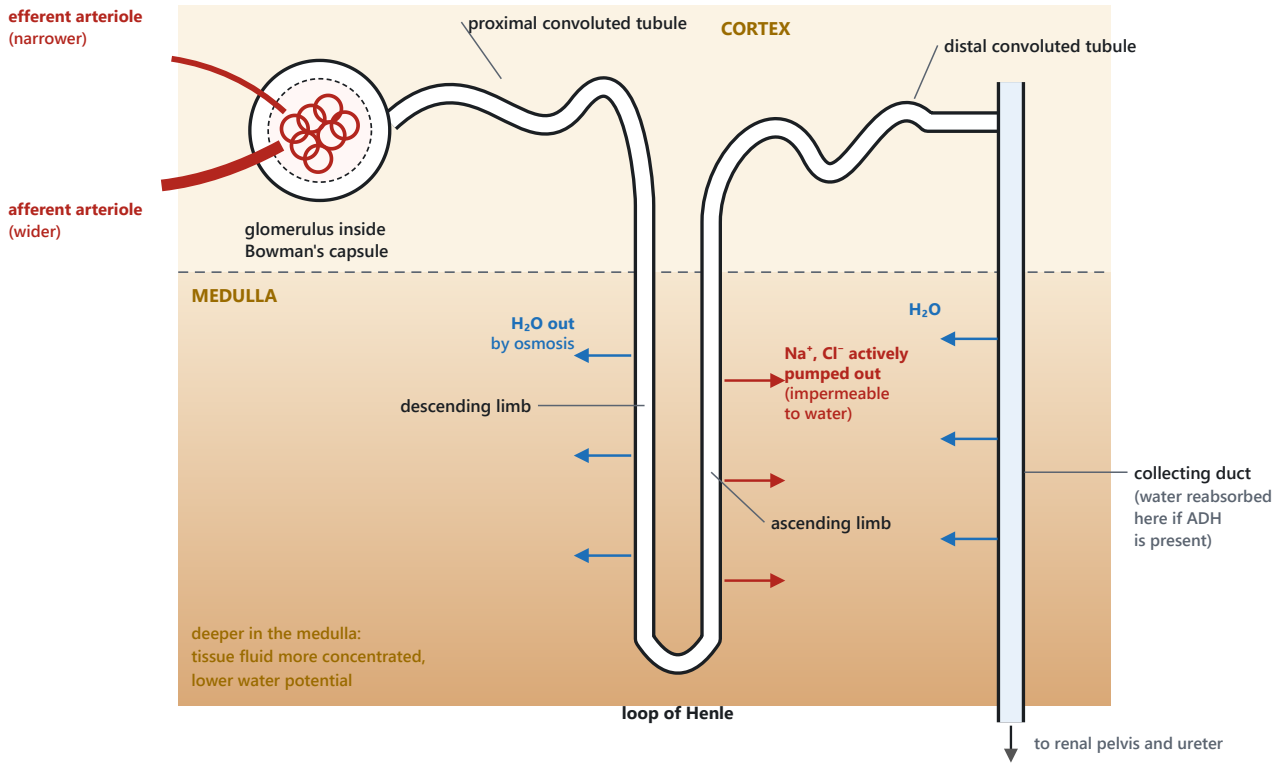
Region	What's there
Fibrous capsule	tough outer layer that protects the kidney
Cortex	outer region: **glomeruli, Bowman's capsules, proximal and distal convoluted tubules**
Medulla	inner region, arranged in cone-shaped pyramids: **loops of Henle** and **collecting ducts** (plus blood vessels)
Renal pelvis	funnel-shaped space where the collecting ducts empty urine; leads into the **ureter**
Renal artery	brings blood (containing urea) from the aorta
Renal vein	takes filtered blood to the vena cava
Ureter	carries urine to the bladder

7.18 The nephron

Each kidney contains about a million **nephrons**. Follow the filtrate:

1. **Bowman's capsule** (cortex): a cup around the **glomerulus**, a knot of capillaries. Ultrafiltration happens here (7.19).
2. **Proximal convoluted tubule (PCT)** (cortex): selective reabsorption of glucose, amino acids, most ions and much of the water (7.20).
3. **Loop of Henle** (a descending limb and an ascending limb dipping into the medulla): makes the medulla tissue fluid very concentrated, so water can be reabsorbed later (7.20).
4. **Distal convoluted tubule (DCT)** (cortex): fine-tunes the balance of ions and pH; its water permeability is controlled by ADH.
5. **Collecting duct** (cortex down through the medulla): water is reabsorbed by osmosis when ADH is present. It carries urine to the renal pelvis.

Blood supply: renal artery → **afferent arteriole** (wider) → **glomerulus** → **efferent arteriole** (narrower) → capillaries around the tubules and alongside the loop of Henle → renal vein.



A nephron (schematic, not to scale). Bowman's capsule, the PCT and the DCT are in the cortex; the loop of Henle and collecting duct run down into the medulla. Ions are pumped out of the ascending limb; water leaves the descending limb and the collecting duct by osmosis.

7.18 Microscopic features to recognise

Structure	Features	Link to function
Glomerular capillary wall	**endothelium with pores** (fenestrations)	lets plasma through, holds back blood cells
Basement membrane	a mesh of collagen and glycoproteins	the **filter**: stops plasma proteins
Inner wall of Bowman's capsule	**podocytes**: cells with foot-like extensions and **filtration slits** between them	filtrate passes between them into the capsule
PCT cells	**microvilli** (brush border) on the side facing the lumen; many **mitochondria** near the base; infolded basal membrane; tight junctions between cells	large surface area for reabsorption; ATP for active transport

- **Juxtamedullary** nephrons have their glomeruli near the medulla and **long loops** of Henle that reach deep into it. Most nephrons (cortical nephrons) have short loops.

MARK-SCHEME LANGUAGE

- Path of a water molecule: **Bowman's capsule → proximal convoluted tubule → loop of Henle → distal convoluted tubule → collecting duct**.
- Drawing Bowman's capsule: a **cup** around the glomerulus with the PCT leaving it; labels such as efferent arteriole, podocytes, basement membrane, squamous epithelium. Label lines must **end on** the structure. Don't label the structure the question already names.
- PCT cell (3 marks): "**microvilli / brush border** on the **lumen** side give a large surface area"; "**mitochondria** in the **basal** cytoplasm supply ATP"; describing the orientation (apical v basal) is a separate mark.
- "The **basement membrane** prevents plasma proteins from being filtered."

Often asked as: locate the cortex, medulla, pelvis, ureter, renal artery and vein; where Bowman's capsule is (the cortex); draw a labelled Bowman's capsule; which part responds to ADH (collecting duct); PCT cell micrographs; magnification and area calculations.

COMMON MISTAKE

Leaving the location of Bowman's capsule blank, or placing it in the medulla. It's in the **cortex**.
Naming microvilli and mitochondria without saying **where** in the cell they are.
Putting units on a magnification.
Using the diameter instead of the radius for the area of a vessel (πr^2).

BEYOND THE SPEC, BUT EXAMINED

- Podocytes, fenestrations, the fibrous capsule and brush-border orientation are credited (WBI15 Jun 2022, Jun 2024).
- Calculations on kidney structures: magnification and volume of a Bowman's capsule (sphere: $\frac{4}{3}\pi r^3$) (WBI15 Oct 2024); area of a renal vein from its diameter (WBI15 Jun 2021).

SPEC 7.19

Urea production and ultrafiltration

WHAT THIS SPEC POINT ASKS

Understand how urea is made in the liver from excess amino acids (no ornithine cycle detail needed) and how it is removed from the blood by ultrafiltration.

7.19 Making urea in the liver

- Amino acids can't be stored. Excess amino acids are carried in the blood to the **liver**.
- **Deamination**: the **amine group ($-NH_2$)** is removed, forming **ammonia (NH_3)**. Ammonia is very soluble and very **toxic**.
- The rest of the amino acid (a keto acid) is **respired** (it can enter the Krebs cycle) or converted to **glucose, glycogen or fat** for storage.
- Ammonia is combined with **CO_2** to make **urea**, which is far less toxic, in the **ornithine cycle** (you don't need the details):
  $2NH_3 + CO_2 \rightarrow CO(NH_2)_2 + H_2O$
- Urea dissolves in the plasma and is carried to the kidneys.

7.19 Ultrafiltration in the glomerulus

- The **afferent arteriole** has a **wider** diameter than the **efferent arteriole**. Blood enters the glomerulus faster than it can leave, so a **high hydrostatic pressure** builds up in the glomerular capillaries.
- The pressure forces small molecules out of the blood into the Bowman's capsule, through three layers:
  1. pores (fenestrations) in the capillary **endothelium**,
  2. the **basement membrane** (the actual filter),
  3. the **filtration slits** between the **podocytes**.
- It's filtration **by size, under pressure**. It isn't selective about whether a substance is useful: that's sorted out later by selective reabsorption.

Filtered (in the glomerular filtrate)	Not filtered (stays in the blood)
water, glucose, amino acids, urea, ions (Na^+, Cl^-, K^+), vitamins, creatinine	red and white blood cells, platelets, **large plasma proteins** (albumin, fibrinogen, prothrombin)

- About **125 cm^3** of filtrate is formed per minute (about 180 dm^3 a day), and almost all of it is reabsorbed.

7.19 Comparing plasma, filtrate and urine

Substance	Plasma / g dm^{-3}	Glomerular filtrate / g dm^{-3}	Urine / g dm^{-3}
Protein	80	0	0
Glucose	1.0	1.0	0
Urea	0.3	0.3	20
Na^+	3.2	3.2	about 3.5 (varies)

(Typical values.)

WORKED EXAMPLE: WHAT THE TABLE SHOWS

Protein: in plasma but not in the filtrate, so it is too large to pass through the basement membrane.
Glucose: the same in plasma and filtrate (freely filtered) but absent from urine, so it is **all reabsorbed** (in the PCT).
Urea: urine ÷ filtrate = 20 ÷ 0.3 = **67 times** more concentrated, because most of the **water** has been reabsorbed (even though some urea diffuses back).
Water: 180 dm^3 of filtrate a day but only 1.5 dm^3 of urine: (180 − 1.5) ÷ 180 × 100 = **99.2%** of the water is reabsorbed.
Filtration rate: 6000 cm^3 of filtrate per hour × 24 = 144 000 cm^3 = **144 dm^3** per day (WBI15 Jan 2025).

MARK-SCHEME LANGUAGE

- Excess amino acids (any 3): "transported to the **liver**"; "**amine group removed** by **deamination**"; "converted to **urea**"; "in the **ornithine cycle**". ("Ammonia" and "amino group" are ignored.) The remainder may become keto acids, glucose or pyruvate.
- Ultrafiltration (2–3): "**high (hydrostatic) pressure in the glomerulus**"; "**afferent arteriole** has a **wider** diameter than the **efferent** arteriole"; "small molecules forced through **pores / fenestrations** and the **basement membrane**". "Arteriole" is accepted, "artery" is not. "Pressure" alone isn't enough: say where it is high.
- Large molecules (haemoglobin, prothrombin, plasma proteins) are not filtered; the **basement membrane** is the filter.

Often asked as: describe what happens to excess amino acids; explain how urea (or creatinine) is removed from the blood; the effect of a wider afferent arteriole; which substances are filtered; filtration-rate calculations with unit conversions.

COMMON MISTAKE

Describing the **digestion** of protein to amino acids when the question asks what happens to the amino acids.
Vague "pressure" answers that never say the pressure is high **in the glomerulus**, or why (the arteriole diameters).
Saying urea is "actively secreted" into Bowman's capsule. It is filtered under pressure.
Unit errors: g → mg is × 1000 (not ÷ 1000); hours → seconds is × 3600.

BEYOND THE SPEC, BUT EXAMINED

- The ornithine cycle is credited as a mark point even though its detail isn't required (WBI15 Jan 2025).
- **Creatinine**: a small molecule that is filtered and not reabsorbed, so the rate it appears in urine measures the filtration rate (WBI15 Oct 2021).
- Mass of glucose filtered per second, with g → mg and h → s conversions (WBI15 Oct 2024).

SPEC 7.20

Selective reabsorption and the loop of Henle

WHAT THIS SPEC POINT ASKS

Understand how solutes are selectively reabsorbed in the proximal tubule, and how the loop of Henle acts as a countercurrent multiplier to increase the reabsorption of water.

7.20 Selective reabsorption in the proximal convoluted tubule

- About **two-thirds** of the water and Na^+ in the filtrate, **all** the glucose and amino acids (normally), and some urea are reabsorbed here.
- How glucose is reabsorbed (**co-transport**, an indirect form of active transport):
  1. **Na^+/K^+ pumps** in the **basal membrane** (the side next to the blood capillary) **actively transport Na^+** out of the PCT cell into the blood, using ATP. This keeps the **Na^+ concentration inside the cell low**.
  2. Na^+ diffuses from the lumen into the cell, down its concentration gradient, through **co-transporter proteins** in the membrane of the **microvilli**. Each carries a **glucose** (or amino acid) molecule with it, against the glucose concentration gradient.

3. Glucose then leaves the cell on the blood side by **facilitated diffusion**, down its concentration gradient, and enters the capillaries.

- **Cl^-** follows the Na^+. Removing all these solutes lowers the water potential of the tissue fluid and blood, so **water** follows by **osmosis**. Some **urea** diffuses back too.
- Adaptations of PCT cells: **microvilli** (large surface area); **many mitochondria** (ATP for active transport); many co-transporter and carrier proteins; infolded basal membrane; very close to the capillaries.

7.20 The loop of Henle: a countercurrent multiplier

- **Ascending limb**: **impermeable to water**. Na^+ and Cl^- are **actively transported out** of it into the tissue fluid of the medulla.
- This **raises the solute concentration** (lowers the water potential) of the tissue fluid in the medulla.
- **Descending limb**: **permeable to water** (it has aquaporins), but much less permeable to ions. Water leaves it by **osmosis** into the concentrated tissue fluid and is carried away in the blood. The filtrate gets more and more concentrated on the way down.
- The filtrate is most concentrated at the hairpin bend. On the way up the ascending limb, ions are pumped out, so it becomes more dilute again.
- **Countercurrent**: the filtrate in the two limbs flows in **opposite directions**, side by side. At any level, the ascending limb can only make a small difference in concentration, but that small difference is **multiplied** along the length of the loop. The result is a **steep gradient**: the tissue fluid gets more and more concentrated (lower and lower water potential) **deeper** into the medulla.
- Why it matters: the **collecting duct** runs down through this concentrated medulla. Water leaves the collecting duct by **osmosis** all the way down (when ADH has made it permeable), so the urine becomes **concentrated** and water is conserved.
- **Longer loops** → a bigger gradient → more water reabsorbed → **more concentrated urine**. Desert mammals such as the kangaroo rat have very long loops of Henle.

WORKED EXAMPLE: GLUCOSE IN THE URINE

As plasma glucose rises, the mass of glucose filtered rises in proportion. Reabsorption also rises, until every co-transporter is working at its maximum rate; in WBI15 Jan 2026 this happened when plasma glucose rose above **12 mmol dm^{-3}**.

Above that, glucose is filtered faster than it can be reabsorbed, so the extra stays in the filtrate and appears in the **urine** (as in untreated diabetes).

A rise in plasma glucose from 5 to 35 mmol dm^{-3} is a $(35 - 5) \div 5 \times 100$ = **600%** increase (WBI15 Jan 2026).

MARK-SCHEME LANGUAGE

- Glucose reabsorption (any 3): "in the **proximal convoluted tubule**"; "**co-transport** with **sodium ions** using carrier / transport proteins"; "glucose **diffuses** into the blood capillaries down its concentration gradient" (facilitated diffusion accepted). Say "sodium **ions**".
- Countercurrent multiplier (any 4): "**filtrate flows in opposite directions** in the two limbs"; "**ions (Na^+, Cl^-) move out of the ascending limb**"; "by **active transport**"; "**increases** the ion concentration / **decreases** the water potential of the surrounding fluid"; "**water leaves the descending limb** by **osmosis** (through aquaporins)". A concentration gradient "formed" along the loop is accepted.
- Desert mammals: "**longer loop of Henle** (more juxtamedullary nephrons)", "more Na^+ pumped out of the ascending limb", "higher solute concentration in the medulla", "more water reabsorbed", "more mitochondria in the ascending limb". Behaviour (nocturnal, burrowing) is credited too.

Often asked as: describe how glucose is reabsorbed; explain how the loop of Henle acts as a countercurrent multiplier; adaptations of kangaroo rats, camels or other desert mammals; plasma–filtrate–urine data; graphs of glucose filtration v reabsorption.

COMMON MISTAKE

Explaining water movement well but being vague about the multiplier. Say which limb is permeable to what, and which way the ions move.
"Sodium" instead of "sodium **ions**". "The tubes flow in opposite directions": it's the **filtrate** that flows.
"There is a concentration gradient." Say the loop **forms or increases** one.
Reversing the link: a high solute concentration means a **low** (more negative) water potential.
Listing differences between two species without a reason for each.

BEYOND THE SPEC, BUT EXAMINED

- **Secretion** of substances into the tubule is credited in a data question on urine composition (WBI15 Jun 2021).
- Desert v wet-habitat mammals: medulla thickness, urine concentration, metabolic water, percentage calculations (WBI15 Oct 2020, Oct 2022, Jan 2024, Oct 2025).

SPEC 7.21

ADH: controlling plasma concentration and blood volume

WHAT THIS SPEC POINT ASKS

Understand how osmoreceptors in the hypothalamus, the pituitary gland and antidiuretic hormone (ADH) control the solute concentration of the plasma and the blood volume by negative feedback.

7.21 When the blood's water potential falls (sweating, not drinking, salty food)

1. **Osmoreceptors** in the **hypothalamus** detect the fall in water potential: water leaves them by osmosis and they **shrink**.
2. They stimulate **neurosecretory cells** in the hypothalamus. ADH is **made** in these cells, travels down their axons, and is **released** from the **posterior pituitary gland** into the blood. More ADH is released.
3. ADH travels in the blood to the **distal convoluted tubule** and **collecting duct**, and binds to **receptors** on the cell surface membranes of their cells.
4. This activates a **second messenger** (cAMP), which makes vesicles containing **aquaporins** (water channel proteins) **fuse with the membrane** facing the lumen.

5. The walls become **more permeable to water**. More water leaves the filtrate by **osmosis** into the concentrated medulla, and back into the blood.
6. A **small volume of concentrated urine** is produced. The blood's water potential rises back to normal, the osmoreceptors are no longer stimulated, and less ADH is released: **negative feedback**.

- Thirst is stimulated at the same time, so you drink more.

7.21 When the blood's water potential rises (drinking a lot of water)

- The osmoreceptors swell, and **less ADH** is released. Aquaporins are taken back into the cytoplasm in vesicles, so the DCT and collecting duct become **less permeable** to water. Less water is reabsorbed, giving a **large volume of dilute urine**, and the water potential falls back to normal.

7.21 Blood volume

- Reabsorbing more water also raises the **blood volume** (and blood pressure). If blood volume falls (after bleeding or heavy sweating), **baroreceptors** in the aortic arch and carotid arteries detect the lower pressure, and more ADH is released. A rise in blood volume reduces ADH release.

MARK-SCHEME LANGUAGE

- Control chain (any 4): "**osmoreceptors** in the **hypothalamus** detect the **water potential** of the blood"; "impulses to the **pituitary gland**, which releases **more / less ADH**"; "ADH acts on the **distal convoluted tubule / collecting duct**"; "**aquaporins** inserted into the membrane / **increased permeability** to water"; "**more / less water reabsorbed**, restoring the blood water potential to normal".
- Too much water: "less ADH", "fewer aquaporins", "DCT / collecting duct less permeable", "more dilute urine".
- High blood volume is detected by **baroreceptors** in the carotid artery or aortic arch.
- Graphs: as ADH rises, urine concentration rises and the rate of urine production falls.
- "Concentration" on its own is ignored: use "**water potential**". "Tubules" on its own is ignored: name the **DCT or collecting duct**.

Often asked as: where ADH is produced and where it acts; name the receptor (osmoreceptor); describe how ADH changes the blood's water potential; the response to drinking a lot of water; keeping blood sodium within narrow limits; urine volume and concentration data after drinking water or salt solution; MDMA and ADH (6-mark).

COMMON MISTAKE

Writing **chemoreceptors** instead of **osmoreceptors**.
"ADH makes the collecting duct less permeable" or "more permeable means more urine". It's the reverse.
Leaving out the pituitary gland, or the ending: the blood's water potential **returns to normal**.
"ADH affects ultrafiltration." It affects water **reabsorption**.
Explaining the aquaporin mechanism in detail for a 2-mark "describe" question.

BEYOND THE SPEC, BUT EXAMINED

- **Hyponatraemia**: too much ADH means too much water is reabsorbed, so the blood Na^+ concentration falls and water enters brain cells by osmosis (WBI15 Jun 2024).
- **MDMA** (ecstasy) increases ADH release, so less urine is made and water is retained (WBI15 Oct 2020, 6-mark).
- Why plasma solute concentration must stay within narrow limits: cells would shrink, or swell and burst, if the plasma's water potential changed (WBI15 Jan 2025).
- Aldosterone is accepted as another hormone controlling blood sodium (WBI15 Jan 2025).

SPEC **7.22**

Switching genes on and off: transcription factors and hormones

WHAT THIS SPEC POINT ASKS

Understand how genes are switched on and off by transcription factors, including the role of peptide hormones acting from outside the cell and steroid hormones acting inside it.

7.22 Transcription factors

- A **transcription factor** is a **protein** that binds to a specific DNA base sequence, usually in the **promoter** region just before a gene, and controls whether that gene is transcribed.
- **Activators** help **RNA polymerase** bind to the promoter. The gene is transcribed into mRNA, which is translated into a protein: the gene is **switched on**.
- **Repressors** block the promoter (or stop activators binding). RNA polymerase can't bind, so the gene isn't transcribed: it is **switched off**.
- Every cell has the same genes. Which transcription factors are active decides which genes are expressed. That's how cells **differentiate**, and how they respond to signals such as hormones.

7.22 Steroid hormones: act inside the cell (e.g. oestrogen, testosterone)

1. Steroids are **lipid-soluble**, so they **diffuse through the phospholipid bilayer** into the cell.
2. In the cytoplasm or nucleus, the hormone binds to a specific **receptor protein**, forming a **hormone–receptor complex**.
3. The complex acts as a **transcription factor**: in the nucleus, it binds to the promoter of its target genes.
4. It **activates** (helps RNA polymerase bind) or **represses** transcription of those genes, so more (or less) of particular proteins is made.

- Example: oestrogen binds to its receptor; the complex switches on genes that make cells of the uterus lining divide.

7.22 Peptide hormones: act from outside the cell (e.g. insulin, glucagon, ADH)

1. Peptide hormones are **water-soluble** and too large and charged to cross the membrane.
2. The hormone binds to a specific **receptor on the cell surface membrane** (with a complementary shape).
3. This activates a **second messenger** (such as **cAMP**) inside the cell, which starts a **cascade** of enzyme activations that amplifies the signal.
4. The cascade can **activate (or inactivate) transcription factors** in the cytoplasm. Activated transcription factors move into the nucleus and switch their target genes on or off.

- Not every effect of a peptide hormone involves genes. ADH inserting aquaporins (7.21) happens without any new transcription.

	Steroid hormones	Peptide hormones
Examples	oestrogen, testosterone, progesterone	insulin, glucagon, ADH (adrenaline, an amine, acts the same way)
Soluble in	lipids	water
Cross the membrane?	**yes**	**no**
Receptor	**inside** the cell (cytoplasm or nucleus)	on the **cell surface membrane**
Effect on genes	the hormone–receptor complex **is** the transcription factor	a second messenger cascade **activates** a transcription factor
Speed	slower to start, long-lasting	faster

MARK-SCHEME LANGUAGE

- Switching a gene **off** (any 3): "transcription factors are **proteins**"; "**bind to the promoter** / gene"; "cause methylation or deacetylation of histones" (ACCEPT "DNA wrapped more tightly"); "**preventing RNA polymerase binding**".
- Steroid hormones (any 3): "**diffuse across the membrane** into the cell / nucleus"; "bind to an **intracellular receptor**"; "the **hormone–receptor complex acts as a transcription factor**"; "activates or represses transcription". "Binds to cell membrane receptors" is rejected. Don't just repeat "regulate transcription" from the stem.
- Peptide hormone, growth factor or cytokine (any 3): "binds to a **receptor on the cell surface membrane**"; "activates a **second messenger** / signalling cascade and a **transcription factor**"; "the transcription factor **binds to the promoter**, so **RNA polymerase** binds"; "the gene is transcribed and the mRNA translated into the protein".
- Name the protein made when the context gives one (actin, myosin, troponin or tropomyosin for muscle; the enzymes that make dopamine).
- Use "fewer / reduced" when the question gives a fall: less testosterone enters cells → **fewer** hormone–receptor complexes → **less** transcription.

Often asked as: describe how genes are switched off by transcription factors; describe how steroid hormones regulate transcription; hormones, growth factors or cytokines leading to differentiation (muscle cells, stem cells, T cells); oestrogen and cell division; testosterone and muscle protein.

COMMON MISTAKE

Describing genes being switched on and off without ever mentioning **transcription factors**.
Saying a peptide hormone (insulin, glucagon) enters the nucleus or **is** a transcription factor. Peptide hormones act at the cell surface (one mark scheme slipped on this: don't copy it).
"Transcription factor → mRNA → dopamine." The mRNA is translated into an **enzyme**, and the enzyme makes dopamine.
Generic answers that ignore the cell type or hormone named in the question.
Copying phrases from the stem or the article.

BEYOND THE SPEC, BUT EXAMINED

- **Epigenetics**: methylation of DNA (at cytosine bases) and deacetylation of histones pack DNA tightly and silence genes. **Histone acetylation** removes positive charge, so DNA is wrapped less tightly and transcription can happen (WBI15 Jan 2023, Jan 2024).
- Genomic imprinting and Prader–Willi syndrome (WBI15 Jan 2024).
- Alternative splicing producing several forms of a receptor from a few genes (WBI15 Jun 2023).

Topic 8 – Coordination, Response and Gene Technology

SPEC 8.1 Neurones: sensory, relay and motor; Schwann cells and myelin

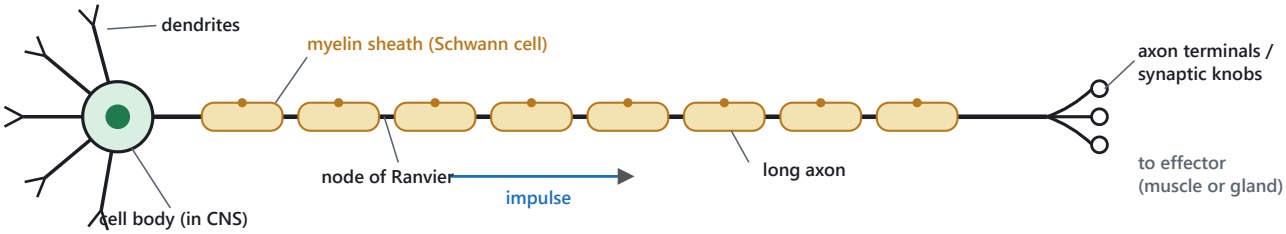
WHAT THIS SPEC POINT ASKS

Know how sensory, relay and motor neurones are built and what each one does, including how Schwann cells form the myelin sheath.

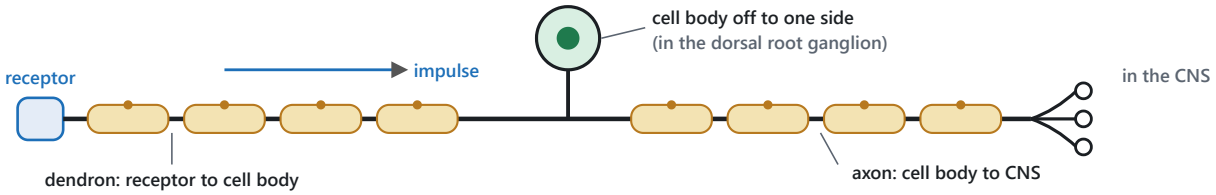
8.1 What every neurone has

- A **neurone** is a cell specialised to carry electrical impulses (**action potentials**). A **nerve** is a bundle of many axons wrapped in connective tissue. One nerve can carry both sensory and motor fibres.
- **Cell body**: contains the nucleus, plus lots of rough ER and mitochondria to make proteins and neurotransmitter.
- **Dendrites**: short, branched extensions that receive impulses from other neurones.
- **Axon**: a long extension that carries impulses away from the cell body.
- **Axon terminals** ending in **synaptic knobs**, which release neurotransmitter onto the next cell.
- Impulses travel one way: dendrites (or dendron) → cell body → axon → synaptic knobs.

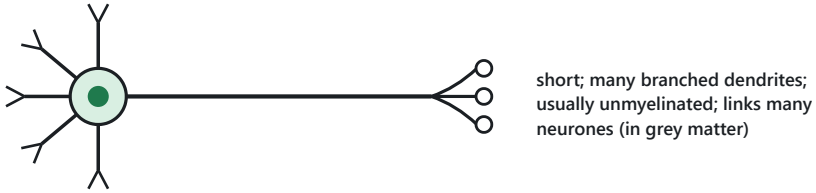
Motor neurone: CNS → effector



Sensory neurone: receptor → CNS



Relay neurone: sensory → motor, inside the CNS



The three types of neurone. Myelin (amber) is made by Schwann cells; the gaps are nodes of Ranvier.

8.1 The three types

	Sensory neurone	Relay neurone	Motor neurone
Carries impulses	from a **receptor** to the **CNS**	from sensory to motor neurones (and to and from the brain)	from the **CNS** to an **effector** (muscle or gland)
Cell body	off to one side of the fibre, in the **dorsal root ganglion** just outside the spinal cord	in the middle, inside the CNS	at one end, **inside the CNS** (ventral horn of the spinal cord, or the brain)
Shape	long **dendron** (receptor to cell body) and a shorter axon (cell body to CNS)	short, with many branched dendrites	many dendrites on the cell body; one long axon
Myelinated?	yes	usually not	yes
Where	mostly in the PNS	entirely within the CNS (grey matter)	cell body in the CNS, axon in the PNS

8.1 Schwann cells and the myelin sheath

- In the peripheral nervous system, **Schwann cells** wrap themselves round an axon many times. Each turn adds two more layers of the Schwann cell's **plasma membrane**, so the sheath is mostly phospholipid: a **fatty electrical insulator** that ions can't cross.
- A Schwann cell is flattened, with little cytoplasm, a lot of membrane and a high lipid content.
- Between neighbouring Schwann cells there are short gaps (about 1–2 μm) where the axon membrane is exposed: the **nodes of Ranvier**. Voltage-gated Na^+ channels are packed into these nodes.
- Myelin makes impulses travel much faster (saltatory conduction, 8.5). Schwann cells also protect and support the axon.

8.1 Structure linked to function

- **Long axon** → carries the impulse a long way (up to about 1 m, spinal cord to foot) with no synapses on the way, so no delays.
- **Many branched dendrites** → receive impulses from many other neurones.
- **Many axon terminals** → pass the impulse on to many cells (e.g. many muscle fibres).
- **Many mitochondria in the synaptic knob** → ATP to make and recycle neurotransmitter and for exocytosis.
- **Myelin** → insulation → faster conduction.

EXAM TIP

Drawing a myelinated motor neurone (3 marks): cell body with dendrites at one end, a long axon with the myelin sheath broken into segments by nodes of Ranvier, and branched terminals with synaptic knobs at the other end. Rule the label lines, make each one touch its part, and only label what you're sure of.

MARK-SCHEME LANGUAGE

- Function comparison: **both** "transmit (electrical) impulses / action potentials" and **both** "release neurotransmitter"; sensory neurones carry impulses "towards the CNS" from "receptors / sense organs"; motor neurones carry them "away from the CNS" to an "effector / muscle / gland". "Spinal cord" is accepted for CNS; "spine" isn't.
- Labels: "dendrites", "cell body" (centron and soma accepted), "axon", "(pre)synaptic knob / axon terminal". Writing "synapse" for the knob is rejected: a synapse is the junction.
- Myelin is made by "Schwann cells"; oligodendrocyte or glial cell was also accepted (WBI15 Oct 2025).
- Drawing: "myelin sheath with nodes of Ranvier", "cell body at one end and synaptic terminal at the other", two correct labels. Maximum 2 marks if you draw a sensory or relay neurone instead.

Often asked as: MCQ on which cell makes myelin; label a neurone diagram; draw a myelinated motor neurone; compare sensory and motor neurone function; pick out sensory, relay and motor neurones on a spinal cord diagram.

COMMON MISTAKE

Comparing **structure** (where the cell body is) when the question asks about **function** (which way the impulse goes and what it connects).
Comparisons in separate paragraphs. Put both halves in one sentence: "sensory neurones carry impulses to the CNS, *whereas* motor neurones carry them away from it."
"Messages" or "signals". Say **impulses** or **action potentials**.
Label lines that stop short of the part, or an extra wrong label: a wrong label cancels a right one.

BEYOND THE SPEC, BUT EXAMINED

- Cell body positions on a spinal cord diagram: sensory in the **dorsal root ganglion** (outside the cord), motor in the **ventral horn** of the grey matter, relay in the grey matter (WBI15 Jan 2026).
- The old names **centron** (cell body) and **dendron** (the long fibre bringing impulses to a sensory cell body) still earn marks.

SPEC 8.2

How the nervous system makes effectors respond

WHAT THIS SPEC POINT ASKS

Understand how the nervous system detects a stimulus and makes an effector (a muscle or a gland) respond.

8.2 The chain

stimulus → receptor → sensory neurone → CNS (relay neurones) → motor neurone → effector → response

- **Stimulus**: a change in the internal or external environment (light, pressure, temperature, a chemical).
- **Receptor**: a cell, or the ending of a sensory neurone, that detects one kind of stimulus. It works as a **transducer**: it turns the energy of the stimulus into an electrical change across its membrane, the **generator potential**.
  - Generator potentials are **graded**: a bigger stimulus gives a bigger generator potential.
  - If the generator potential reaches **threshold**, an action potential starts in the sensory neurone.
  - A stronger stimulus gives a **higher frequency** of action potentials, not bigger ones.

- **CNS**: receives the impulses, combines them with other information, and sends impulses out along the right motor neurones. In a reflex this is just one or two relay neurones; in a voluntary action the brain decides.
- **Effector**: a **muscle** (contracts) or a **gland** (secretes).

8.2 Example: a pressure receptor (Pacinian corpuscle)

1. Pressure squashes the layers of the corpuscle and deforms the membrane of the sensory neurone ending inside it.
2. The stretched membrane opens **stretch-mediated Na^+ channels**.
3. Na^+ diffuses in → depolarisation: a **generator potential**.
4. If it reaches threshold → action potentials travel along the sensory neurone to the CNS.

8.2 Examples of stimulus to response

Stimulus	Receptor	Effector	Response
hand touches a hot plate	pain and heat receptors in the skin	biceps (skeletal muscle)	arm pulled away
bright light	rods and cones in the retina	circular muscle of the iris	pupil constricts
smell of food	chemoreceptors in the nose	salivary glands	saliva secreted
rise in blood CO_2	chemoreceptors in the medulla, aorta and carotid arteries	diaphragm and intercostal muscles	faster, deeper breathing

MARK-SCHEME LANGUAGE

- Pressure receptor: "pressure deforms (cell) membranes (in the Pacinian corpuscle)", "(stretch-mediated) sodium ion channels open", "influx of sodium ions", then depolarisation, threshold, action potential (WBI15 Jun 2021).
- Unfamiliar receptor: "stimulus changes the shape of the membrane" → "ion channel (protein) changes shape / opens" → "influx of sodium (or calcium) ions" → "depolarisation / action potential" (WBI15 Oct 2022).
- Pain: detected by "nociceptors / pain receptors"; impulse along the sensory neurone; "relay neurone transmits the impulse to the brain, where it is perceived as pain". A motor neurone is rejected here.
- Sense organ: "a group of tissues working together" that "detects a stimulus / change in the environment".

Often asked as: never on its own. It comes up as "how does this receptor start an action potential?", receptor data (impulse frequency at different temperatures), and stimulus-to-response chains in the article question.

COMMON MISTAKE

Stopping at "the receptor detects the stimulus". Keep going: channels open, ions diffuse in, depolarisation, threshold, action potential, then the neurones.
Saying a stronger stimulus gives a "bigger" action potential. It gives **more action potentials per second**.

BEYOND THE SPEC, BUT EXAMINED

- **All-or-nothing in receptor data**: at low pressure there was no action potential, while medium and high pressure gave the same peak (+40 mV) (WBI15 Jun 2021).
- Named receptors in new contexts: nociceptors v thermoreceptors firing at different temperatures (WBI15 Jan 2024); TRPV1 (heat) and Piezo (touch) channels (WBI15 Oct 2022 article). Calcium ions can be the depolarising ions.

SPEC 8.3

The spinal reflex arc

WHAT THIS SPEC POINT ASKS

Know the structure and function of a spinal reflex arc, including the grey matter and white matter of the spinal cord.

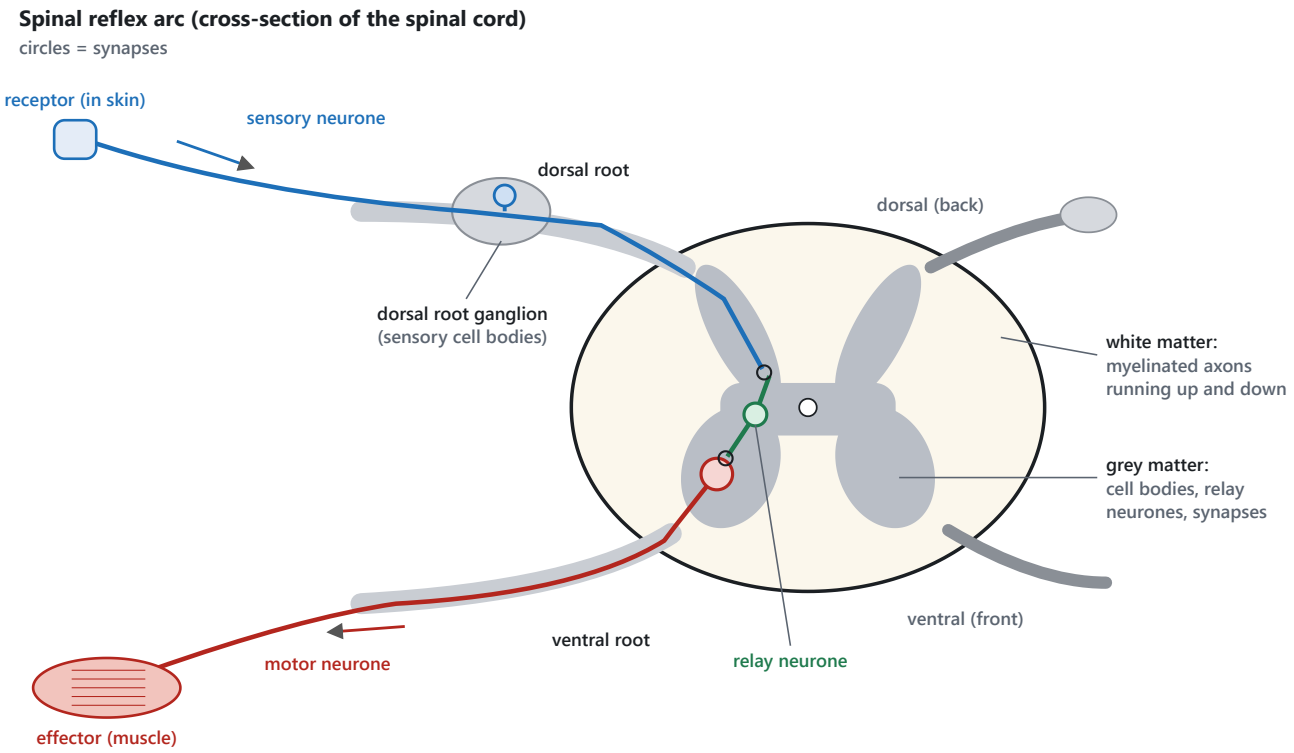
8.3 What a reflex is

- A **rapid, automatic (involuntary)** response to a stimulus, the same every time, with **no conscious thought**. It protects the body: pulling away from heat or a sharp object, blinking, coughing.
- The pathway is a **reflex arc**: receptor → sensory neurone → relay neurone → motor neurone → effector.

8.3 Why it's so fast

- Only **three neurones** and so only **two synapses**. Some reflexes (the knee jerk) have just one synapse: the sensory neurone synapses directly with the motor neurone.
- The impulse doesn't have to be processed by the conscious parts of the brain first, so there's no decision-making delay.
- The sensory and motor neurones are **myelinated**.
- The brain still finds out: the relay neurone also synapses with neurones that run **up the white matter** to the brain. That's why you feel the pain just *after* you've pulled your hand away.

8.3 The spinal cord



Sensory neurone (blue) in through the dorsal root; relay neurone (green) in the grey matter; motor neurone (red) out through the ventral root.

Part	What it contains	Job
Grey matter (central, butterfly- or H-shaped)	cell bodies, dendrites, relay neurones, synapses; mostly unmyelinated	where sensory, relay and motor neurones connect
White matter (outer layer)	**myelinated axons** in bundles (tracts); myelin is fatty, so it looks white	carries impulses up to the brain and down from it
Dorsal root (back)	axons of sensory neurones	carries impulses **into** the cord
Dorsal root ganglion	a swelling on the dorsal root	holds the **sensory neurone cell bodies**
Ventral root (front)	axons of motor neurones	carries impulses **out** to effectors
Central canal	cerebrospinal fluid	not part of the grey matter

Memory aid: **s**ensory **in** at the back (dorsal), **m**otor **out** at the front (ventral).

8.3 A withdrawal reflex, step by step

1. A hot object stimulates pain receptors in the skin of a finger → generator potential → action potentials in the sensory neurone.
2. The impulse travels along the sensory neurone, through the dorsal root (past its cell body in the ganglion) and into the grey matter.
3. The sensory neurone **synapses** with a relay neurone in the **grey matter**.
4. The relay neurone **synapses** with a motor neurone, whose cell body is in the ventral horn.
5. The impulse leaves along the motor neurone axon, through the ventral root, to the effector.
6. At the neuromuscular junction the biceps is stimulated to **contract**: the hand is pulled away.

MARK-SCHEME LANGUAGE

- Every link is a synapse: "receptor cells synapse with the sensory neurone"; "sensory neurone synapses with relay neurone in the grey matter of the spinal cord"; "relay neurone synapses with motor neurone"; "motor neurone synapses with effector" (WBI15 Jan 2023). A correctly labelled diagram is accepted.
- Skin to CNS only: "impulse transmitted via the sensory neurone towards the spinal cord", "passes through the dorsal root ganglion", "synapses with relay neurone", "located in the grey matter" (WBI15 Jan 2024).
- Grey matter is the central butterfly region and white matter surrounds it; the central canal isn't grey matter.

Often asked as: MCQ on where the relay neurone is; describe the structure of a spinal reflex arc; describe the structures an impulse passes through from the skin to the CNS; mark cell body positions on a cord diagram.

EXAM TIP

Only about half of candidates got the grey-matter MCQ right (Jan 2022), and the examiners called the spinal reflex a weak area. Learn the table, and use the word **synapse** at every junction: "the sensory neurone synapses with a relay neurone **in the grey matter**".

COMMON MISTAKE

Describing how the impulse **starts** when the question asks which **structures** it passes through. Going past what was asked (motor neurone and effector when the question stops at the CNS) earns nothing. "A reflex doesn't involve the CNS." The spinal cord **is** part of the CNS; a reflex just doesn't need conscious thought.

BEYOND THE SPEC, BUT EXAMINED

- The **dorsal root ganglion** as the home of the sensory cell body, and a sensory neurone that synapses **directly** with a motor neurone (no relay neurone) (WBI15 Jan 2024).

SPEC 8.4

The nerve impulse (action potential)

WHAT THIS SPEC POINT ASKS

Understand how an action potential is produced and passed along an axon, including how the membrane's permeability to sodium and potassium ions changes.

8.4 The resting potential (about −70 mV)

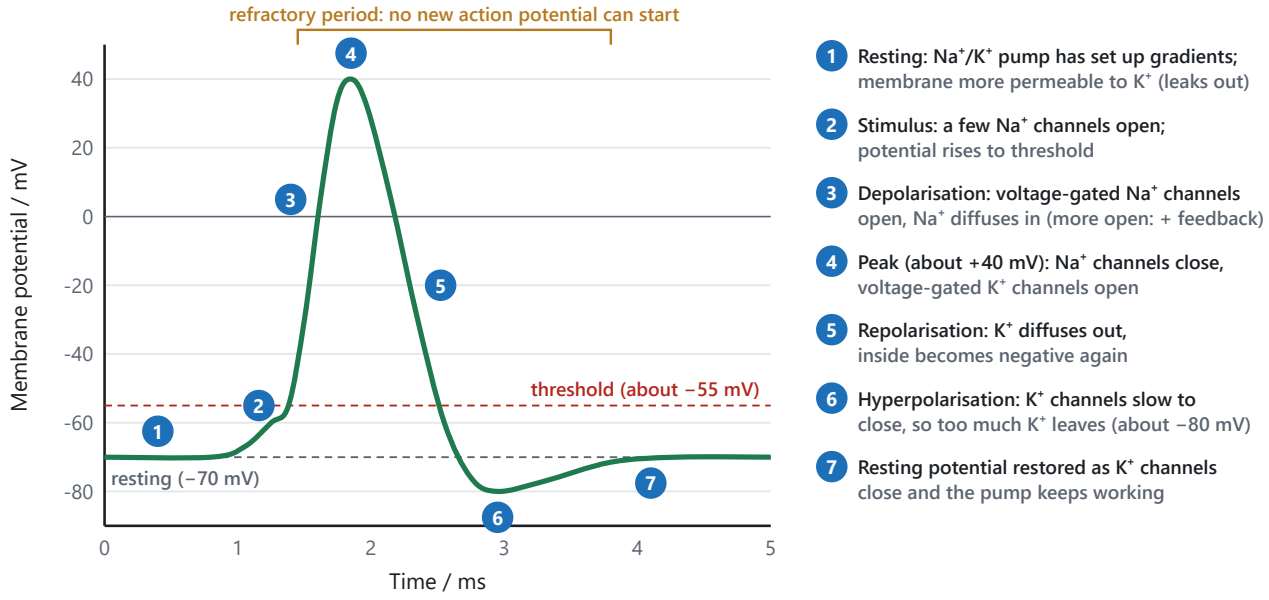
At rest the inside of the axon is **negative** compared with the outside. The membrane is **polarised**. Three things set this up:

1. The **Na^+/K^+ pump** (a carrier protein that uses **ATP**) actively transports **3 Na^+ out** for every **2 K^+ in**. This builds concentration gradients (Na^+ high outside, K^+ high inside) and removes more positive charge than it brings in.
2. **Different permeability**: at rest the voltage-gated Na^+ channels are **closed**, but some **K^+ channels are open**, so the membrane is much **more permeable to K^+**. K^+ **diffuses out** down its concentration gradient, carrying positive charge out.

3. Large negatively charged proteins stay inside.

K^+ stops leaving when the outward concentration gradient is balanced by the attraction of the negative interior (an electrochemical balance).

8.4 The action potential



An action potential recorded at one point on an axon. The numbers match the steps below.

1. **Resting potential** (−70 mV), as above.
2. **Stimulus**: a few Na^+ channels open, Na^+ diffuses in, and the inside becomes less negative. If it reaches the **threshold** (about −55 mV)…
3. **Depolarisation**: **voltage-gated Na^+ channels open**. Na^+ **diffuses in** fast, down its concentration and electrical gradients. Each bit of depolarisation opens more channels (**positive feedback**). The inside becomes positive.
4. **Peak** (about +40 mV): the Na^+ channels **close** (and can't reopen for a short time), and **voltage-gated K^+ channels open**.
5. **Repolarisation**: K^+ **diffuses out**, so the inside becomes negative again.
6. **Hyperpolarisation**: the K^+ channels are slow to close, so too much K^+ leaves and the potential overshoots to about −80 mV.
7. The K^+ channels close and the **resting potential** returns. The Na^+/K^+ pump keeps the concentration gradients topped up over many impulses.

The whole spike lasts about 2 ms.

8.4 How the impulse travels along the axon

- The Na^+ that has rushed in at the active region spreads sideways inside the axon (**local currents**) to the next stretch of membrane.
- That stretch is depolarised to threshold, its voltage-gated Na^+ channels open, and a **new action potential** fires there. So the action potential is **regenerated** at each point: a wave of depolarisation, not a current flowing down a wire.
- **Refractory period**: just behind the action potential, the Na^+ channels are closed and **can't be reopened** for a moment, and the membrane is repolarising or hyperpolarised. This means:
  - impulses travel in **one direction only** (the region behind can't fire again straight away);

- action potentials are **separate (discrete)** and don't merge;
- there's a **maximum frequency** of impulses.

8.4 All-or-nothing

- If threshold is reached, an action potential **always** fires and is **always the same size** (peak about +40 mV). Below threshold, nothing fires.
- The strength of a stimulus is coded by the **frequency** of action potentials (and by how many neurones fire), never by their size.

8.4 What affects how fast an impulse travels

- **Myelination** (8.5): the biggest effect.
- **Axon diameter**: a wider axon has less resistance to the local currents (and less ion leakage per unit length), so it conducts faster. Squid giant axons work this way.
- **Temperature**: warmer → ions diffuse faster and channels open and close faster → faster conduction, up to about 40 °C. Much hotter and the channel proteins **denature**.

WORKED EXAMPLE: CONDUCTION VELOCITY

Two recording electrodes sit 30 mm apart on an axon. The action potential reaches the second electrode 0.50 ms after the first.
Speed = distance ÷ time.
Convert: 30 mm = 0.030 m; 0.50 ms = 0.000 50 s (5.0×10^{-4} s).
Speed = 0.030 ÷ 0.000 50 = **60 m s^{-1}**.
Quick check: mm per ms is the same as m per s, and 30 ÷ 0.50 = 60.

WORKED EXAMPLE: READING THE GRAPH

From the figure: resting potential −70 mV, peak +40 mV.
Change in potential during depolarisation = 40 − (−70) = **110 mV**. Don't drop the minus sign.
Maximum frequency: if the refractory period is 2 ms (0.002 s), at most one action potential can start every 0.002 s, so the maximum frequency = 1 ÷ 0.002 = **500 per second**.
Rate of change: draw a tangent at the steepest part and work out rise ÷ run, e.g. 9 a.u. ÷ 0.5 ms = **18 a.u. ms^{-1}**, to 2 s.f. with units (WBI15 Jan 2025). Show the working: it earns a mark even if the final value is off.

MARK-SCHEME LANGUAGE

- Resting potential: "**active transport** by the sodium-potassium pump" (Na^+ out, K^+ in); "**diffusion of potassium ions out** of the axon"; "so the inside is negatively charged"; "sodium channels closed", membrane "more permeable to K^+". "Axon" is accepted for membrane.
- Why ATP is needed: "ATP is used by the sodium-potassium pump" "to move sodium and potassium ions across the membrane" "to **establish concentration gradients**".
- Producing an action potential (4 marks): "sodium ion channels open", "sodium ions diffuse in", " (positive feedback) more sodium ion channels open", "resulting in depolarisation", "reference to threshold". The word **ion** is needed (WBI15 Jan 2025).
- Reaching threshold: "the membrane potential becomes less negative"; "if it reaches threshold, depolarisation starts". Channels opening on their own isn't enough.
- Repolarisation: "sodium channels close", "potassium ions move **out** via (voltage-gated) potassium channels", "inside becomes negative". K^+ moving in is rejected.
- All-or-nothing: "once threshold is reached", action potentials "either occur fully or not at all".
- Temperature: faster because "ion channels open and close more quickly" or "ions diffuse faster". Slower only through **denaturation** of the channel proteins.
- Toxin that blocks Na^+ channels: "binds to (voltage-gated) sodium channels", "blocks diffusion of sodium ions into the axon", "depolarisation stays below threshold", "no action potential". Say sodium **ions** or sodium **channels**.

Often asked as: MCQs on traces (resting potential, which channels are open, which point is hyperpolarised); why the resting potential needs ATP; how −70 mV is maintained; production of an action potential; effect of a channel-blocking toxin; gradient of a graph; temperature and speed.

COMMON MISTAKE

"The pump gives the ions energy" or "ATP makes energy". Say ATP is used by the pump to set up **concentration gradients**.
K^+ moving **into** the axon during repolarisation. It moves **out**.
Writing "sodium" or "potassium" with no "ions" or "channels".
Long accounts of repolarisation and hyperpolarisation when the question only asked how an action potential is **produced**.
Saying the pump quickly restores the resting potential after each spike. Closing K^+ channels do that; the pump maintains the ion gradients over time.

BEYOND THE SPEC, BUT EXAMINED

- **Tetrodotoxin** (TTX, from pufferfish) blocks voltage-gated Na^+ channels, so no action potentials (WBI15 Jun 2024, Jun 2025).
- Ion concentration tables, e.g. a ratio of K^+ outside : inside (WBI15 Jan 2025).
- A metabolic poison that stops ATP production stops the pump, so the resting potential slowly runs down; washing the poison away restores it.

SPEC 8.5

Myelination and saltatory conduction

WHAT THIS SPEC POINT ASKS

Understand how myelin allows saltatory conduction, and why that makes impulses faster.

8.5 Without myelin

Every patch of membrane along the axon has to depolarise in turn: the local currents depolarise the very next patch, which fires, and so on. This is slow (about 0.5–2 m s^{-1} in a thin axon).

8.5 With myelin

1. The myelin sheath is an **electrical insulator**: ions can't move across the membrane where it's covered.
2. So depolarisation (action potentials) can **only happen at the nodes of Ranvier**, where the voltage-gated Na^+ channels are concentrated.
3. Local currents flow inside the axon from one node to the next (about 1 mm), depolarising the next node to threshold.
4. The action potential seems to **jump from node to node**: **saltatory conduction** (from the Latin for "to jump").
5. Result: up to about 100 m s^{-1} in vertebrate motor neurones.

8.5 It also saves energy

Ions only move at the nodes, so far fewer Na^+ and K^+ ions cross the membrane per impulse. The Na^+/K^+ pump has less to move back, so **less ATP** is used.

	Myelinated	Unmyelinated
Where depolarisation happens	only at the nodes of Ranvier	along the whole membrane
Speed	fast (up to about 100 m s^{-1})	slow (about 0.5–2 m s^{-1} for a thin axon)
Ions moved and ATP used per impulse	less	more
Examples	most vertebrate sensory and motor neurones	many relay and autonomic neurones; invertebrate axons

Below about 1 μm diameter, unmyelinated axons are actually faster; above it, myelinated axons win, and their speed rises much more steeply with diameter.

8.5 When myelin is lost

In **multiple sclerosis** (myelin in the CNS) and **Guillain–Barré syndrome** (myelin in the PNS), the sheath is damaged. Ions leak where the myelin has gone, local currents may not reach threshold at the next node, so impulses are **slower** and **fewer** reach the muscle or the brain. Symptoms: muscle weakness, poor coordination, numbness, blurred vision, difficulty walking.

WORKED EXAMPLE: PERCENTAGE DIFFERENCE IN GRADIENTS

A graph of speed against axon diameter is a straight line for both kinds of axon. The myelinated line rises 120 units over 20 μm: gradient m = 120 ÷ 20 = 6. The gradient for unmyelinated axons is given as 0.5. Percentage difference (relative to the unmyelinated gradient) = (6 − 0.5) ÷ 0.5 × 100 = **1100%** (WBI15 Jan 2022).

Read the stem: here the difference was compared with the unmyelinated value. Many candidates worked out something else.

MARK-SCHEME LANGUAGE

- The chain (3 of 4): "myelin (sheath) acts as an (electrical) insulator"; "ions cannot cross the axon membrane where there is myelin"; "depolarisation / action potentials only occur at the nodes of Ranvier"; "impulse jumps from node to node / saltatory conduction". The converse is accepted: "in an unmyelinated axon depolarisation occurs along the whole length".
- Saltatory conduction: "impulse jumps from one node to the next" or "depolarisation only at the nodes"; property of myelin: "(electrical) insulator" or "impermeable to ions" (WBI15 Oct 2025).
- Demyelination: "loss of myelin" → "slower conduction" → "fewer impulses reach the muscle / CNS" → a named loss of function. "Impulses are lost" and "no impulses" are ignored.
- Data on diameter and myelin: "the larger the diameter, the faster"; "myelination increases speed"; "myelination has a greater effect than diameter" (the point that separated candidates).

Often asked as: "Explain why a myelinated axon conducts faster than an unmyelinated axon of the same diameter" (3 marks, set at least three times); symptoms of a demyelinating disease; speed against diameter graphs; magnification of a nerve section.

COMMON MISTAKE

"The myelinated neurone is insulated." The mark is for **the myelin** acting as an insulator, and for saying what it stops: **ion movement** across the membrane.
"It's faster because the impulse jumps", with no reason. Give the reason: depolarisation only happens at the nodes.
In disease questions, "the neurone is unmyelinated" just repeats the stem. Explain the effect on impulses, then on the named function (walking, vision).

BEYOND THE SPEC, BUT EXAMINED

- Demyelinating diseases as contexts: Guillain–Barré syndrome, ADEM and multiple sclerosis (WBI15 Jun 2022, Oct 2023, Jun 2023).
- Magnification from a scale bar on a nerve section: 9.5 mm = 9500 µm; 9500 ÷ 50 = 190, written **1.9×10^2**, with no units and not "×200" (WBI15 Jan 2024).

SPEC **8.6**

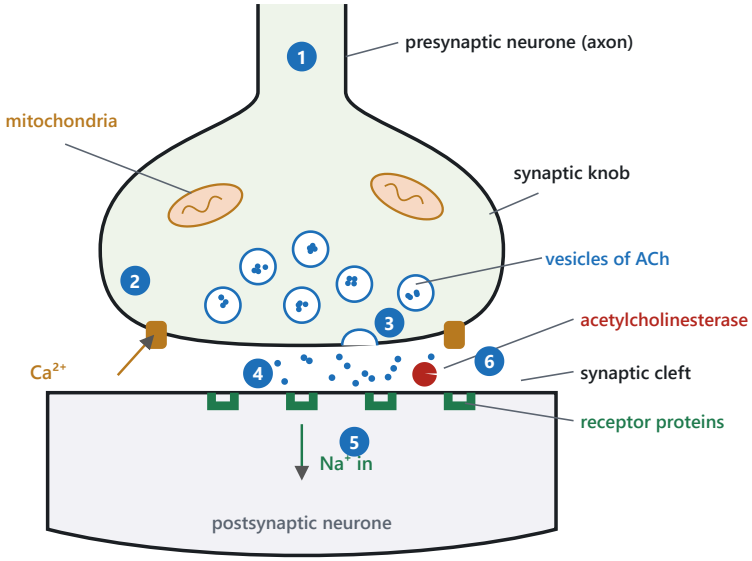
Synapses and the pupil reflex

WHAT THIS SPEC POINT ASKS

(i) Know how synapses are built and how they pass on a nerve impulse, including the role of neurotransmitters such as acetylcholine. (ii) Understand how the pupil widens (dilates) and narrows (constricts).

8.6 Structure of a synapse

- A **synapse** is the junction between two neurones, or between a neurone and an effector cell.
- **Presynaptic knob**: many **mitochondria** (ATP), **vesicles** filled with neurotransmitter, and voltage-gated Ca^{2+} channels in its membrane.
- **Synaptic cleft**: a gap of about 20 nm.
- **Postsynaptic membrane**: **receptor proteins** complementary to the neurotransmitter.
- A synapse that uses **acetylcholine (ACh)** is a **cholinergic** synapse. The junction between a motor neurone and a muscle fibre (the **neuromuscular junction**) works the same way.



1. Action potential arrives at the synaptic knob
2. Voltage-gated Ca^{2+} channels open; Ca^{2+} diffuses into the knob
3. Vesicles move to and fuse with the presynaptic membrane: ACh released by exocytosis
4. ACh diffuses across the cleft and binds to receptors on the postsynaptic membrane
5. Na^+ channels open, Na^+ diffuses in: depolarisation; threshold → action potential
6. Acetylcholinesterase hydrolyses ACh; choline is taken back into the knob and recycled

Transmission across a cholinergic synapse. Only the neurotransmitter crosses the cleft; the vesicles stay in the knob.

8.6 How an impulse crosses a cholinergic synapse

1. An **action potential arrives** at the presynaptic knob and depolarises its membrane.
2. **Voltage-gated Ca^{2+} channels open**, and **Ca^{2+} diffuses into** the knob (there's more Ca^{2+} outside).
3. The Ca^{2+} makes **vesicles** move to and **fuse** with the presynaptic membrane. ACh is released into the cleft by **exocytosis**.
4. ACh **diffuses** across the cleft and **binds** to complementary **receptors on the postsynaptic membrane**.
5. The receptors are part of **Na^+ channels**, which open. **Na^+ diffuses into** the postsynaptic neurone → **depolarisation**. If enough ACh binds for the depolarisation to reach **threshold**, an action potential starts.
6. The enzyme **acetylcholinesterase** in the cleft **hydrolyses** ACh into choline and ethanoate (acetate). These are taken back into the knob and made into ACh again, using ATP from the mitochondria. This stops the postsynaptic cell being stimulated non-stop, and frees the receptors for the next impulse.

8.6 What synapses do

- **One-way transmission**: vesicles are only in the presynaptic knob, and receptors only on the postsynaptic membrane.
- **Summation**: one impulse may not release enough transmitter to reach threshold.
  - **Spatial summation**: several presynaptic neurones release transmitter onto the same postsynaptic neurone at once.
  - **Temporal summation**: one presynaptic neurone fires several impulses in quick succession, so transmitter builds up.
- **Convergence** (many neurones onto one, e.g. rods, 8.8) and **divergence** (one neurone onto many).
- **Inhibitory synapses**: some transmitters open Cl^- or K^+ channels instead. Cl^- in or K^+ out **hyperpolarises** the postsynaptic membrane, so threshold is harder to reach.
- Synapses filter out weak background stimulation, and they are where drugs act (8.7) and where habituation happens (8.9).

8.6 The pupil reflex

The iris has two sets of smooth muscle that work as an **antagonistic pair**:

- **circular muscle** (a ring round the pupil): when it contracts, the pupil **constricts**;
- **radial muscle** (like the spokes of a wheel): when it contracts, the pupil **dilates**.

	Bright light	Dim light
Detected by	rods and cones in the retina (high light intensity)	rods and cones (low light intensity)
Pathway	sensory neurones in the optic nerve → relay neurones in the brain (midbrain) → motor neurones	same route
Motor neurones	**parasympathetic**, to the circular muscle	**sympathetic**, to the radial muscle
Muscles	circular **contracts**, radial **relaxes**	radial **contracts**, circular **relaxes**
Pupil	**constricts**: less light enters, so the retina isn't damaged and not all the rhodopsin is bleached	**dilates**: more light enters, so the rods can still be stimulated

This is a **cranial reflex**: the relay neurones are in the brain, not the spinal cord. It's automatic, and both pupils respond even if light reaches only one eye.

WORKED EXAMPLE: PUPIL AREA

In dim light a pupil's diameter rises from 3.0 mm to 7.0 mm. Area of a circle = πr^2.
Before: r = 1.5 mm, area = $\pi \times 1.5^2 = 7.07$ mm^2. After: r = 3.5 mm, area = $\pi \times 3.5^2 = 38.5$ mm^2.
Percentage increase = $(38.5 - 7.07) \div 7.07 \times 100$ = **444%**.
Use the **radius**, not the diameter. To convert mm^2 to m^2, divide by 10^6.

MARK-SCHEME LANGUAGE

- Synapse (3–4 marks): "impulse causes **calcium ion channels to open** / calcium ions to enter (the presynaptic neurone)"; "**vesicles** (containing neurotransmitter) **fuse** with the presynaptic membrane"; "neurotransmitter released (by exocytosis) into the synaptic cleft and **diffuses** across"; "**binds to receptors on the postsynaptic membrane**"; "sodium ion channels open / sodium ions enter, causing **depolarisation** / an action potential". Any named neurotransmitter is accepted.
- Partly faulty calcium channels: "less calcium influx", "**fewer** vesicles fuse", "less neurotransmitter released", "fewer impulses". The scheme wanted "fewer", not "none".
- Bright light: "**circular muscles contract** and **radial muscles relax**", so the pupil gets smaller and less light enters. Dim light: the reverse. Name the muscles, call them antagonistic and state the light change. Sympathetic and parasympathetic were both credited for the motor neurone. **Ciliary muscles** are rejected.

Often asked as: describe transmission across a synapse (3–4 marks, in most series); the effect of faulty calcium channels or reduced dopamine release; MCQ on which channels open; describe and explain the pupil response; calculate a pupil area or its change.

COMMON MISTAKE

Leaving out the **Ca^{2+} channels**, the **diffusion** across the cleft, or the **binding to receptors on the postsynaptic membrane**: the most-missed marks.
"Binds to the postsynaptic neurone" or "binds to the synapse". Say **postsynaptic membrane**. And vesicles don't cross the cleft: only the neurotransmitter does.
Swapping radial and circular muscles; using "ciliary muscles" (they change the shape of the lens); not saying whether the light got brighter or dimmer.
Absolute statements ("no vesicles fuse") when a channel is only partly faulty.

BEYOND THE SPEC, BUT EXAMINED

- A mutation in a calcium channel gene: changed amino acid sequence → changed channel shape → less Ca^{2+} influx → fewer vesicles fuse → less transmitter (WBI15 Jun 2022).
- Different neurotransmitters need different receptors because each receptor has a **specific 3-D shape** complementary to its transmitter (WBI15 Jun 2022 article).
- Sympathetic neurones release **noradrenaline** at their effectors; parasympathetic neurones release **acetylcholine** (WBI15 Jan 2026 MCQ on a neurotransmitter pair).
- Pupil dilation as a measure of mental effort (WBI15 Oct 2025).

SPEC 8.7

Drugs and synaptic transmission

WHAT THIS SPEC POINT ASKS

Understand how drugs can act by changing nerve impulse transmission: nicotine, lidocaine, cobra venom alpha toxin, L-DOPA for Parkinson's disease, and MDMA (ecstasy).

8.7 Ways a drug can change transmission

Increase transmission	Decrease transmission
mimic the neurotransmitter and activate its receptor (an **agonist**)	bind to the receptor without activating it, blocking the transmitter (an **antagonist**)
supply a **precursor**, so more transmitter is made	stop the transmitter being made
trigger more release of the transmitter	stop its release (e.g. botulinum toxin stops ACh release)
block the enzyme that breaks the transmitter down	block voltage-gated Na^+ channels, so no action potential
block **reuptake** of the transmitter into the presynaptic neurone	

8.7 The five named drugs

Drug	Target	What it does	Effect
Nicotine	nicotinic **ACh receptors** on postsynaptic membranes	similar shape to ACh, so it **binds and opens** the receptor's Na^+ channel → depolarisation and action potentials (agonist). Acetylcholinesterase doesn't break it down, so it acts for longer	stimulant; causes **dopamine release** in the brain's reward pathway (pleasure, addiction); raises heart rate and blood pressure
Lidocaine	**voltage-gated Na^+ channels** in neurone membranes	**blocks** them → Na^+ can't diffuse in → no depolarisation → **no action potentials**	local anaesthetic: impulses from pain receptors don't reach the brain; also used to treat irregular heart rhythms
Cobra venom α-toxin	**ACh receptors** at neuromuscular junctions	**binds** very tightly but **doesn't open** the channel, so ACh can't bind (antagonist) → no depolarisation of the muscle fibre	muscles can't contract: **paralysis**, including the diaphragm and intercostal muscles, so breathing stops
L-DOPA	dopamine-releasing neurones in the brain	a **precursor** of dopamine; it crosses the **blood–brain barrier** (dopamine can't) and is converted to dopamine by an enzyme (dopa decarboxylase) in the brain	more dopamine released → restores control of movement in **Parkinson's disease**
MDMA (ecstasy)	**serotonin reuptake transporters** in presynaptic membranes	**blocks reuptake** of serotonin and makes the transporters run backwards, releasing **extra serotonin** into the cleft	serotonin stays in the cleft and keeps binding postsynaptic receptors → more action potentials → raised mood, euphoria

8.7 L-DOPA in more detail

- In Parkinson's disease, dopamine-releasing neurones in the **substantia nigra** (midbrain) die, so less dopamine is released in the parts of the brain that control movement.
- Giving dopamine doesn't work: it can't cross the blood–brain barrier.
- L-DOPA does cross it. The remaining dopamine neurones take it up and convert it to dopamine → more dopamine released → more binds to receptors → more action potentials → **better control of muscle movement** (less tremor and stiffness).
- It treats the symptoms, not the cause: the neurones keep dying, so it works less well over time.
- It's usually given with **carbidopa** (or benserazide, as in co-beneldopa). These **inhibit the decarboxylase enzyme outside the brain** and can't cross the blood–brain barrier themselves. So less L-DOPA is converted to dopamine in the blood and gut, **more L-DOPA reaches the brain**, a **lower dose** works, and there are fewer side effects.

8.7 MDMA in more detail

- Normally, serotonin is released, binds to postsynaptic receptors, and is then taken back into the presynaptic neurone by **transporter proteins** (reuptake).
- MDMA is absorbed into the blood, crosses the blood–brain barrier, and binds to the transporter. Reuptake stops and extra serotonin is released, so the serotonin concentration in the cleft stays high.
- Side effects: raised body temperature, raised heart rate, and extra **ADH** secretion (so less urine; with a lot of water drunk, the blood can become dangerously dilute). Afterwards serotonin stores are used up, so mood drops.

WORKED EXAMPLE: PREDICTING THE EFFECT OF AN UNFAMILIAR DRUG

Drug X inhibits acetylcholinesterase. Predict its effect at a neuromuscular junction.

1. Acetylcholinesterase can't hydrolyse ACh, so ACh stays in the cleft.
2. ACh keeps binding to the receptors, so the Na^+ channels stay open.
3. The muscle fibre membrane stays depolarised and keeps being stimulated → continuous contraction (spasm). Eventually the membrane can't repolarise, so the muscle stops responding.

Method: find **which step** the drug changes → what happens to the **amount of transmitter bound to receptors** → **Na^+ entry / depolarisation** → **action potentials** → the **effect** on the person.

WORKED EXAMPLE: DOSE ON A LOG SCALE

A dose–response graph has $\log_{10}$ (dose) on the x-axis. The dose giving half the maximum effect (ED_{50}) reads as log dose 3.8 for one drug and 3.5 for another.
Convert back with the antilog: $10^{3.8}$ = 6310 and $10^{3.5}$ = 3162. Difference = **3147** (WBI15 Jan 2026).
You can't subtract the logs (3.8 − 3.5 = 0.3) and call that the difference in dose. A log scale is used so a very wide range of doses fits on one graph.

MARK-SCHEME LANGUAGE

- L-DOPA: "dopamine cannot cross the blood–brain barrier"; "L-DOPA can cross the blood–brain barrier"; "converted to dopamine in the brain"; "restores control of muscle movement / reduces tremor". "Reduces symptoms" is in the stem and earns nothing.
- Carbidopa (or the partner drug in co-beneldopa): "prevents L-DOPA being broken down (in the blood / gut)", so "more L-DOPA reaches the brain" and "a lower dose" can be given (WBI15 Oct 2024, Jun 2022).
- Nicotine: "binds to ACh receptors", "similar structure to ACh", "causes action potentials". "Nicotine blocks receptors" is ignored. With the receptor gene knocked out, nicotine has no effect because there are no receptors to bind to (WBI15 Jan 2024).
- Cobra venom: "binds to the ACh receptor" but "does not stimulate impulses" (WBI15 Jan 2026).
- Lidocaine: "inhibits / blocks sodium ion channels" → fewer impulses → less contraction of heart muscle → less ATP and O_2 needed (WBI15 Jan 2025).
- MDMA by mouth: "absorbed into the blood", "crosses the blood–brain barrier", "**MDMA** (not serotonin) binds to neurones", "stimulates release of serotonin" (WBI15 Jun 2023).

Often asked as: MCQs on how each drug acts; how L-DOPA reduces symptoms; why carbidopa is added; knock-out mice and nicotine; dose–response graphs (log scale, ED_{50}, LD_{50}).

COMMON MISTAKE

Mixing up lidocaine and cobra venom. **Lidocaine** blocks **voltage-gated Na^+ channels** in the axon, so no action potential forms at all. **Cobra α-toxin** blocks **ACh receptors** on the postsynaptic membrane: the impulse reaches the synapse but can't cross it.
Thinking carbidopa stops a dopamine overdose. It stops L-DOPA being converted too early, **outside** the brain.
Calling nicotine a neurotransmitter. It's a drug that mimics one.
Starting in the brain. For a drug taken by mouth, start with absorption from the gut into the blood, then the blood–brain barrier, then the synapse.
If strychnine comes up: it blocks inhibitory **glycine** receptors. It is not an acetylcholinesterase inhibitor, whatever some notes say.

BEYOND THE SPEC, BUT EXAMINED

- The **blood–brain barrier**: brain capillary walls have tightly joined endothelial cells, so most drugs (and dopamine) can't pass (WBI15 Jun 2022).
- **LD_{50}**: the dose that kills 50% of a test population (WBI15 Jun 2025).
- **Caffeine**: absorbed into the blood, crosses the blood–brain barrier, blocks adenosine receptors, so more action potentials (WBI15 Jan 2025 article).

SPEC 8.8

Detecting light: rods in the retina

WHAT THIS SPEC POINT ASKS

Understand how rod cells detect light: the roles of rhodopsin (opsin + retinal), sodium ions, cation channels and hyperpolarisation of the rod in producing action potentials in the optic neurones.

8.8 The retina

- Light is focused by the lens onto the **retina**. Light passes through two layers of neurones before it reaches the photoreceptors at the back: **ganglion cells** (whose axons form the **optic nerve**), then **bipolar cells**, then the **rods and cones**.
- **Rods**: about 120 million per eye, mostly round the edge (periphery); work in dim light; black and white only.
- **Cones**: about 6 million, packed into the **fovea**; need bright light; give colour vision.
- **Blind spot**: where the optic nerve leaves the eye. There are no photoreceptors there.

8.8 Rhodopsin

- **Rhodopsin** = **opsin** (a protein) + **retinal** (a light-absorbing molecule made from vitamin A). It sits in the membranes of stacked discs in the rod's outer segment.
- In the dark, retinal is in the **cis** form. When it absorbs light it changes shape to **trans**-retinal, and rhodopsin breaks into **opsin + trans-retinal**: it is **bleached**.
- In the dark, enzymes turn trans-retinal back into cis-retinal, which rejoins opsin. This needs **ATP** and takes time, which is why your eyes take minutes to adapt to the dark.

8.8 In the dark

1. **Cation (Na^+) channels** in the rod's outer membrane are **open**.
2. Na^+ **diffuses in**, while the Na^+/K^+ pump in the inner segment keeps pumping it out. With Na^+ leaking in all the time, the rod is only slightly negative inside (about −40 mV): it is **depolarised**.
3. The depolarised rod keeps releasing a neurotransmitter (**glutamate**) onto the bipolar cell.
4. Glutamate **inhibits** the bipolar cell, so no impulses pass to the optic neurone.

8.8 In the light

1. **Rhodopsin absorbs light**: cis-retinal → trans-retinal, and rhodopsin splits into **opsin + retinal**.
2. The freed opsin starts a chain of reactions in the rod that **closes the cation (Na^+) channels**.
3. The pump keeps pumping Na^+ out, but Na^+ can no longer leak in, so the inside becomes more negative: the rod is **hyperpolarised** (about −70 mV).
4. The hyperpolarised rod **stops releasing glutamate**.
5. The bipolar cell is no longer inhibited. It **depolarises** and releases its own transmitter onto a ganglion cell.

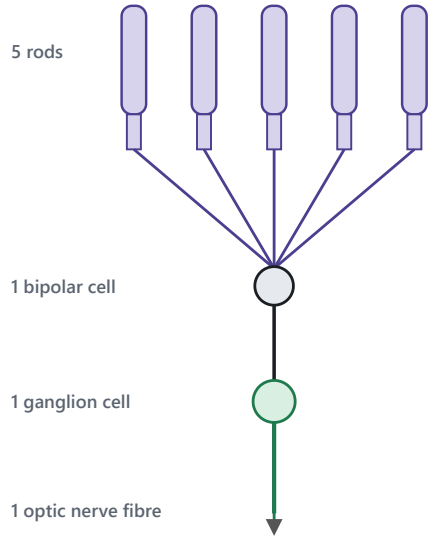
6. The ganglion cell depolarises → **action potentials in the optic neurone** → brain.

	Dark	**Light**
Rhodopsin	intact (cis-retinal)	split into opsin + trans-retinal
Na^+ (cation) channels in the rod	open	**closed**
Rod membrane	depolarised (about −40 mV)	**hyperpolarised** (about −70 mV)
Glutamate release	continuous	stops
Bipolar cell	inhibited	depolarised
Optic neurone	no action potentials	action potentials

The rod is unusual: it is **hyperpolarised** by its stimulus, the opposite of most receptors.

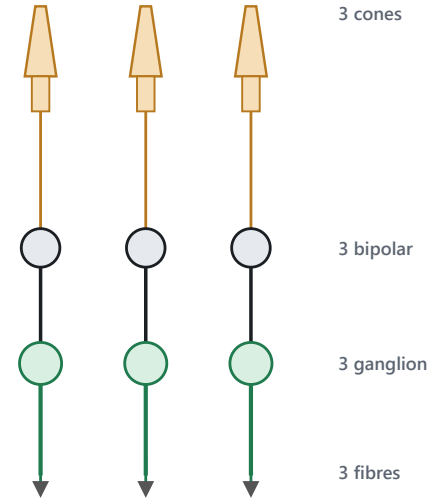
8.8 Rods and cones compared

Rods: convergence



Small generator potentials from many rods add up (spatial summation) and reach threshold in dim light: HIGH SENSITIVITY. The brain can't tell which rod was stimulated: LOW ACUITY. Black and white only.

Cones (fovea): no convergence



Each cone has its own route to the brain, so two points close together are seen separately: HIGH ACUITY. Needs bright light to reach threshold. Three types of cone: colour vision.

Convergence gives rods their sensitivity; separate wiring gives cones their acuity.

	Rods	Cones
Pigment	rhodopsin: broken down by **low** light intensity	iodopsin, three types (red-, green- and blue-sensitive): needs **high** light intensity
Wiring	several rods → one bipolar cell (**retinal convergence**)	one cone → one bipolar cell (at the fovea)
Sensitivity	**high**: small generator potentials from several rods **summate** (spatial summation) to reach threshold in dim light	low
Visual acuity (detail)	**low**: the brain can't tell which rod in the group was stimulated	**high**: each cone has its own route to the brain
Colour	no	yes
Where	periphery of the retina	fovea

WORKED EXAMPLE: NUMBER OF RODS

5000 photoreceptors are counted in a section of retina, and 5/6 of them are rods. Rods = 5/6 × 5000 = 4166.7, so **4167 rods**.
A number of cells must be a **whole number**: 4166.7 lost the mark (WBI15 Jun 2024).

MARK-SCHEME LANGUAGE

- The chain (any 3–4): "rhodopsin absorbs light"; "(cis-retinal converted to trans-retinal) rhodopsin splits into opsin and retinal"; "sodium ion channels (in the membrane) close" (or "permeability to sodium ions decreases"); "more sodium ions are pumped out than diffuse in"; "rod cell becomes hyperpolarised"; "no glutamate (neurotransmitter) released"; bipolar cell depolarises → action potentials in the optic neurone.
- ATP is used for "active transport of sodium ions" (reforming rhodopsin is also accepted) (WBI15 Oct 2025).
- Rhodopsin = opsin + retinal; trans-retinal is the form made when light is absorbed.
- Rods v cones in animals: rods detect low light; cones need bright light and give colour vision; an animal active by day has more cones, one active at night more rods (WBI15 Jun 2024).

Often asked as: describe the role of rhodopsin (3–4 marks, set repeatedly); describe how light is detected by rods; MCQs on the pigment, the channel state and which cell is hyperpolarised; rods and cones data.

COMMON MISTAKE

"Opsin blocks the sodium channels." The examiners reject this: opsin sets off reactions that **close** the channels. (One recent mark scheme let the phrase slip in; don't copy it.)
"Light opens the sodium channels" or "sodium ions are taken up in the light". It's the reverse: light **closes** the channels and the rod **hyperpolarises**.
Mixing up the rod (hyperpolarised in light) and the bipolar cell (depolarised in light).
Going on to describe transmission along the optic nerve when the question only asks how rods detect light.

BEYOND THE SPEC, BUT EXAMINED

- **RPE65**, an enzyme that regenerates cis-retinal: a mutation changes its active site so it no longer fits trans-retinal, rhodopsin can't re-form, and the rods have no working pigment. Gene therapy restored some vision (WBI15 Jan 2021).
- Glutamate as the rod's transmitter, and recordings from bipolar cells that depolarise in light (WBI15 Oct 2021).

SPEC 8.9

Habituation

WHAT THIS SPEC POINT ASKS

Understand what is meant by habituation. (Recommended additional practical: investigate habituation to a stimulus.)

8.9 What it is

- **Habituation** is a **reduced response to a stimulus that is repeated but harmless** (neither threatening nor rewarding). It's a simple form of learning.
- Why it helps: the animal stops wasting energy and time on stimuli that don't matter (wind moving the grass, traffic noise), so it can feed, mate and stay alert to real threats.
- It is **specific to that stimulus**: a new or stronger stimulus still triggers the full response, and the response recovers after a rest. This is how you know the animal isn't simply tired or injured.
- Examples: a sea slug (*Aplysia*) stops withdrawing its gill when its siphon is touched again and again; snails stop pulling in their tentacles; birds ignore a scarecrow; you stop noticing your clothes.

8.9 What happens at the synapse

1. Repeated stimulation makes the **Ca^{2+} channels** in the presynaptic membrane less responsive, so fewer open.
2. **Less Ca^{2+} enters** the presynaptic knob.
3. **Fewer vesicles fuse** with the presynaptic membrane, so **less neurotransmitter** is released.
4. Less transmitter binds to the postsynaptic receptors, so fewer Na^{+} channels open and the depolarisation **doesn't reach threshold**.
5. **No action potential** in the motor neurone, so no response from the effector.

The receptor and the sensory neurone still fire normally: the change is at the synapse.

RECOMMENDED PRACTICAL: HABITUATION IN SNAILS

1. Let each snail settle (acclimatise) for 10 minutes on a damp surface, in the same conditions.
2. Touch the snail gently between the tentacles with a damp cotton bud, using the same force (e.g. the bud on a rod moved a fixed distance) and the same spot each time.
3. Start a stopwatch. Record the time until the tentacles are fully extended again.
4. Touch it again as soon as they're fully out (or at a fixed interval, e.g. every minute). Repeat for 10–15 touches.
5. Repeat with at least five other snails and work out the mean time for each touch number.
6. Plot mean time to re-emerge (y) against touch number (x).
   Expected result: the time to re-emerge falls as the touches are repeated.

8.9 Variables

Independent	number of times the stimulus has been given (touch number)
Dependent	time for the tentacles to extend fully again (or: duration of withdrawal; % of animals responding)
Control	same snail for a whole run; same object, force and spot; same interval between touches; temperature; light intensity; humidity; noise and vibration; snails of similar size from the same place

8.9 Evaluation and ethics

- A touch by hand is hard to standardise, and "fully extended" is a judgement: decide on a clear end point first.
- Other stimuli (shadows, vibration, sound) can make the snail withdraw: work somewhere quiet and still.
- **Show it's habituation, not fatigue**: once the response has faded, give a different stimulus (another spot, a stronger touch). The full response comes straight back.
- Ethics: handle gently, keep the snails moist and cool, use a stimulus that can't hurt them, keep the number of trials low, and return them to where they were found.
- Analysis: a **correlation** test (Spearman's rank) between touch number and time.

WORKED EXAMPLE: PERCENTAGE DIFFERENCE

After habituation, 20 roundworms in group 1 and 12 in group 2 responded to a tap.
Using the formula given (difference ÷ mean of the two values × 100): mean = (20 + 12) ÷ 2 = 16, so (20 − 12) ÷ 16 × 100 = **50%** (WBI15 Jan 2024).
Use whichever formula the question gives. "Percentage change" uses the **starting** value as the denominator.

MARK-SCHEME LANGUAGE

- Definition: "a {repeated / prolonged / non-threatening} stimulus" results in "a decreased response". "Saves energy" isn't credited for the definition (WBI15 Jan 2026).
- Mechanism: "calcium ion channels become less responsive", "less Ca^{2+} enters the presynaptic knob", "fewer vesicles fuse", "less neurotransmitter released", "threshold not reached", "no action potential in the postsynaptic (motor) neurone".
- Recovery data: give the overall trend, the change in steepness ("steeper rise up to 10 min, slower after"), the comparison between groups, and whether any group fully recovered.
- Test for "number of encounters against % response": a **correlation** coefficient (WBI15 Jun 2021).

Often asked as: define habituation; MCQ on its advantage; data on habituation or recovery (crabs, roundworms, rats, fish, children); percentage difference; the right statistical test. The practical is planned in Unit 6 (sea slugs, woodlice, fish).

COMMON MISTAKE

"The animal gets tired" or "the receptor stops working". The receptor still fires; transmission at the synapse falls, and a new stimulus brings the response straight back.
Mixing up the **recovery** curve with the **habituation** curve when reading data.

BEYOND THE SPEC, BUT EXAMINED

- Habituation questions sit next to other kinds of learning, such as rats learning a maze for a food reward (WBI15 Jan 2026).
- Habituation can be harmful: elk that habituate to traffic are more likely to be hit (WBI15 Jan 2026).

SPEC 8.10

The central and peripheral nervous systems

WHAT THIS SPEC POINT ASKS

Know that the mammalian nervous system is made of the central nervous system (CNS) and the peripheral nervous system (PNS).

- **CNS** = **brain + spinal cord**. It contains the relay neurones, combines incoming information and sends out instructions. It's protected by the skull and vertebrae.
- **PNS** = all the nerves **outside** the CNS (cranial and spinal nerves). It contains:
  - **sensory neurones**, carrying impulses from receptors to the CNS;
  - **motor neurones**, carrying impulses from the CNS to effectors.

The motor side of the PNS has two parts:

	Somatic	**Autonomic**
Effectors	skeletal muscle	smooth muscle (gut, iris, blood vessels), cardiac muscle, glands
Control	mostly voluntary (plus spinal reflexes)	involuntary

The autonomic system has two divisions, which often have opposite effects on the same organ. Both are active all the time; the balance between them sets things like heart rate and pupil size.

	Sympathetic ("fight or flight")	**Parasympathetic ("rest and digest")**
Transmitter at the effector	mostly noradrenaline	acetylcholine
Heart (SAN)	speeds up	slows down (via the vagus nerve)
Pupil	dilates (radial muscle contracts)	constricts (circular muscle contracts)
Airways	widen	narrow
Gut	digestion slowed	digestion increased

EXAM TIP

This point is never examined on its own, but the sympathetic/parasympathetic split comes up again and again: heart rate (7.13), the pupil reflex (8.6) and article questions.

MARK-SCHEME LANGUAGE

- The autonomic nervous system and resting breathing rate: "chemoreceptors detect CO_2 / pH" → "impulses to the ventilation centre in the medulla" → "impulses to the intercostal muscles and diaphragm" (WBI15 Jun 2022 article). Note the word "resting".
- A sympathetic nerve acting on an organ: "releases neurotransmitter (e.g. noradrenaline)" which acts on the target cells. Don't skip the neurotransmitter step.

Often asked as: only inside heart-rate, pupil and article questions (2020–2026).

COMMON MISTAKE

Swapping the divisions: **parasympathetic** constricts the pupil and slows the heart; **sympathetic** dilates the pupil and speeds the heart up.
Treating every reflex as autonomic: the withdrawal reflex uses somatic motor neurones to skeletal muscle.

BEYOND THE SPEC, BUT EXAMINED

- The **enteric nervous system** (the gut's own network of neurones) as part of the autonomic nervous system (WBI15 Jun 2022 article).

SPEC 8.11

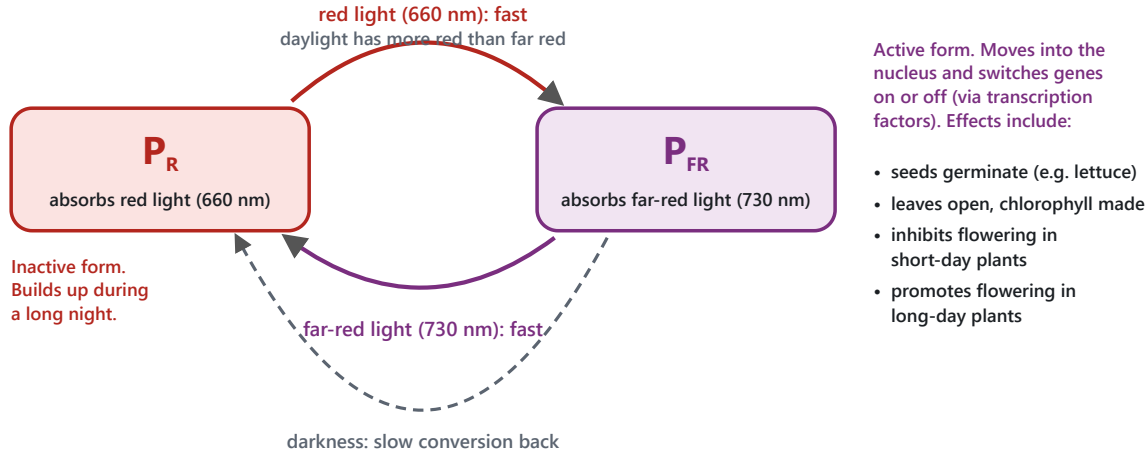
Plant responses: phytochrome, auxin and gibberellins

WHAT THIS SPEC POINT ASKS

Understand how phytochrome, auxin (IAA) and gibberellins bring about responses in plants, including their effects on transcription.

Plants have no nervous system. They respond to stimuli mainly by **changing how they grow**, coordinated by **plant growth regulators** (plant hormones). These are made in growing tissues and move by diffusion, active transport or in the phloem. Many work by switching genes on or off.

8.11 Phytochrome



The two forms of phytochrome. Each form is named after the light it absorbs.

- Phytochrome is a **photoreceptor pigment** (a protein with a light-absorbing part). It is **not** a hormone. It exists in two forms that convert into each other:
  - **Pr** absorbs **red light (660 nm)** and is quickly converted to Pfr;
  - **Pfr** absorbs **far-red light (730 nm)** and is quickly converted back to Pr;
  - in the **dark**, Pfr slowly turns back into Pr.
- **Pfr is the active form.**

8.11 Reasoning with red and far-red light

- **Daylight** contains more red than far-red light, so Pr → Pfr wins: Pfr builds up during the day.

- **At night** there's no light, so Pfr slowly reverts to Pr. The longer the night, the less Pfr is left by morning. The plant "measures" night length by how much Pfr remains.
- **Under other leaves**, the canopy absorbs red light but lets far-red through. Pfr stays low, and the seedling grows tall and thin towards the light (shade avoidance).
- **Underground** there's no light, so no Pfr: the seedling is **etiolated** (long, pale stem; leaves unopened). On reaching light, Pfr forms → leaves open, chlorophyll is made, stem growth slows.

8.11 How Pfr changes transcription

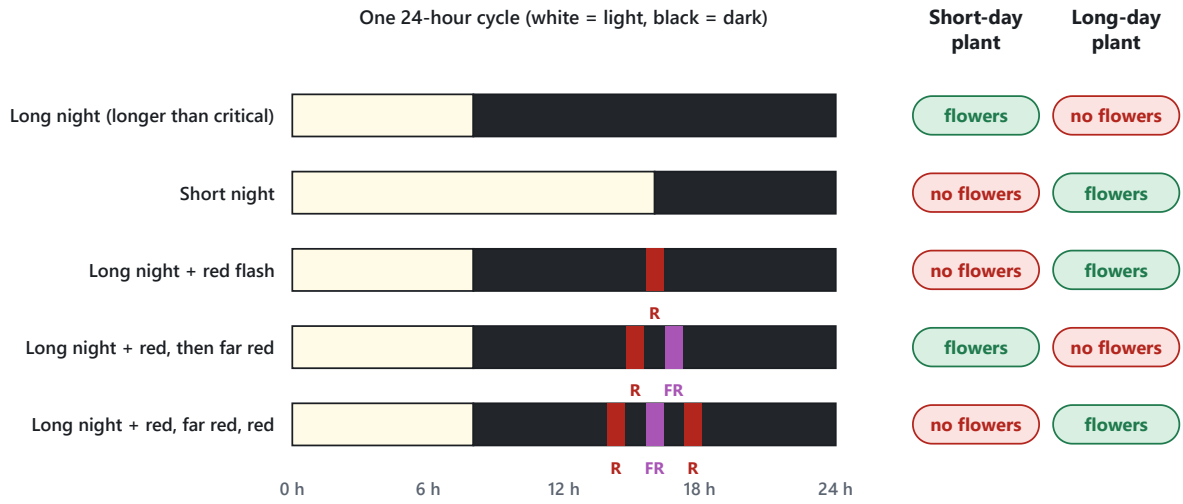
1. Pfr moves from the cytoplasm into the **nucleus**.
2. It binds to **transcription factors** (called PIFs, phytochrome-interacting factors) and changes their activity, so some genes are switched on and others off.
3. In a germinating seed, for example, genes for **gibberellin synthesis** are switched on, and gibberellin then switches on the amylase gene (below).

8.11 Germination

- Small seeds such as **lettuce** have few food reserves and must germinate near the surface. They germinate only when they get **red light**, a sign that they're near the surface and not shaded.
- Red flash → Pfr → **germination**. Red then far-red → Pfr converted back → **no germination**. Red, far-red, red → germination. **The last flash decides.**

8.11 Flowering (photoperiodism)

- Plants flower in the right season by sensing night length. The **critical night length** is the night length that switches flowering on or off.
- **Short-day plants** (better called "long-night" plants, e.g. chrysanthemum): flower when the night is **longer than the critical night length**. A long night lets **Pfr fall** to a low level. In short-day plants **Pfr inhibits flowering**, so low Pfr allows it.
- **Long-day plants** ("short-night" plants, e.g. spinach): flower when the night is **shorter than the critical night length**, so Pfr stays **high**. In long-day plants **Pfr promotes flowering**.
- **Night-break experiments**: a flash of red light in the middle of a long night converts Pr back to Pfr, so Pfr is high again. A short-day plant then **doesn't flower**; a long-day plant **does**. A far-red flash straight after the red converts the Pfr back to Pr and cancels the red flash. Again, **the last flash decides**.



R = red flash, FR = far-red flash. A red last flash means high Pfr: short-day plants don't flower, long-day plants do.

8.11 Auxin (IAA)

- Made in **shoot tips** and young leaves; moves down the shoot by diffusion and active transport.
- **Stimulates cell elongation** in shoots (not cell division). In roots, high concentrations **inhibit** elongation.
- **How it makes a cell elongate (acid growth)**:
  1. IAA binds to a receptor on the cell surface membrane.
  2. This activates **proton pumps** (using ATP): H^+ ions are actively pumped out of the cytoplasm into the **cell wall**.
  3. The wall becomes acidic (about pH 5). Proteins that work at low pH (expansins) break the **hydrogen bonds between cellulose microfibrils**, so the wall loosens and can stretch.
  4. K^+ ions move into the cell, lowering its **water potential**, so water enters by **osmosis**.
  5. The cell swells and the loosened wall stretches: the cell **elongates**.
- Auxin also changes transcription: it triggers the breakdown of repressor proteins, so transcription factors can switch on genes needed for growth.

Phototropism (shoots grow towards light):

1. Light from one side → auxin is moved to the **shaded side** of the shoot tip.
2. More auxin on the shaded side → more **cell elongation** there.
3. The shaded side grows faster, so the shoot **grows towards** the light, and the leaves get more light for photosynthesis.

Gravitropism: in a root lying on its side, auxin collects on the **lower** side, where it **inhibits** elongation. The upper side grows faster, so the root grows **downwards**. In a shoot on its side, auxin also collects on the lower side, but there it stimulates elongation, so the shoot grows upwards.

Apical dominance: auxin from the apical (top) bud **inhibits** the growth of side (lateral) buds. Cut the tip off and the side buds grow, so the plant becomes bushier.

Classic coleoptile (young grass shoot) experiments:

Treatment	Result	What it shows
tip removed	no growth, no bending	the tip makes the growth substance
tip covered with an opaque cap	grows straight up	the tip detects the light
tip cut off, agar (gel) block put between tip and stump	still bends towards light	the substance is chemical and can diffuse through agar
plastic or mica barrier on the shaded side	doesn't bend	the substance moves down the shaded side
agar block that held a tip, placed on one side of a cut stump (in the dark)	bends away from the side with the block	the substance makes cells on that side elongate

8.11 Gibberellins

- Made in young leaves, embryos and seeds.
- **Stem elongation**: they make internodes grow longer (cell elongation and division). Many dwarf varieties can't make gibberellin; spraying gibberellin on them makes them grow tall.
- **Germination of cereal grains** (barley, wheat):
  1. The grain absorbs water, which activates the **embryo**. The embryo makes **gibberellin**.
  2. Gibberellin diffuses to the **aleurone layer** (a protein-rich layer round the endosperm).
  3. There, gibberellin causes **repressor proteins (DELLA proteins) to be broken down**. Without them, a **transcription factor** can bind to the promoter of the **amylase gene**, so the gene is **transcribed**. The

mRNA is translated into **amylase**.

4. Amylase is secreted into the **endosperm**, where it **hydrolyses starch** (glycosidic bonds) into **maltose**, and then glucose.
5. The sugars go to the embryo for **respiration** (ATP for growth and cell division) and to make cellulose for new cell walls.

	Phytochrome	**Auxin (IAA)**	**Gibberellins**
What it is	photoreceptor pigment	plant growth regulator	plant growth regulator
Made / found	throughout the plant	shoot tips, young leaves	embryos, young leaves
Main effects	germination, de-etiolation, flowering	shoot cell elongation, tropisms, apical dominance	stem elongation, seed germination
Gene effect	Pfr in the nucleus changes transcription factor activity	breakdown of repressors → growth genes on	breakdown of DELLA repressors → amylase gene transcribed

MARK-SCHEME LANGUAGE

- Phytochrome (six-mark): "Pr (inactive) absorbs red light (660 nm) and is converted to Pfr"; "Pfr (active) absorbs far-red light (730 nm) and is converted to Pr"; "in the dark Pfr slowly converts to Pr"; "Pfr activates genes / transcription factors needed for germination", e.g. "activates the amylase gene". **"Pfr acts as a transcription factor" is not accepted.** Explain **each** treatment and say "the last flash determines the response".
- Flowering: short-day plants need "a dark period longer than the critical night length"; "Pfr low / Pr high gives flowering in short-day plants"; a red flash in the night "prevents flowering" in short-day plants; long-day plants flower "if the last flash of light is R" (WBI15 Oct 2023).
- Phototropism: auxin "accumulates on the shaded side" and "stimulates cell elongation", so the shoot "grows towards the light". "Bends towards the light" alone is rejected.
- Auxin mechanism (4 of 7): "auxin binds to a receptor"; "H^+ pumped into the cell wall"; "lower pH breaks hydrogen bonds between cellulose microfibrils"; "K^+ uptake lowers water potential"; "water enters by osmosis"; "the cell elongates" (WBI15 Jun 2025).
- Gibberellin: "binds to a receptor / acts via a transcription factor"; "amylase gene activated / transcribed"; "amylase mRNA" → "amylase"; "amylase hydrolyses starch to maltose". Inactivating a repressor is accepted. **Not** "gibberellins break down starch".
- Apical dominance: "auxin from the apical bud inhibits the lateral buds" (WBI15 Jun 2024).

Often asked as: six-mark questions on red/far-red light and germination or flowering (set four times, 2021–2026); coleoptile experiments; how auxin makes cells elongate; how gibberellin controls starch breakdown; growth data with IAA, gibberellin or coloured light.

EXAM TIP

Six-mark plant answers are marked on a three-step ladder: (1) **describe** the results, quoting treatments and values; (2) the **chemical nature** (Pr and Pfr, or auxin as a diffusible substance); (3) the **molecular mechanism** (transcription factors, proton pumps, acid growth). Level 3 needs step 3.

COMMON MISTAKE

Saying gibberellin itself digests starch. It switches on the **amylase gene**; amylase digests the starch. And write **amylase**, not "amylose".
Calling phytochrome a hormone. It's a **photoreceptor**.
Naming Pr and Pfr by what they make: Pr **absorbs** red light, and red light **turns it into** Pfr.
"Auxin causes cell division." In shoots it causes cell **elongation**.
Writing about germination when the question is about **flowering**.

BEYOND THE SPEC, BUT EXAMINED

- Short-day and long-day plants, critical night length, night-break flashes and the 660 nm and 730 nm wavelengths (WBI15 Oct 2023, Jan 2026).
- The acid-growth mechanism in detail (WBI15 Jun 2025).
- Synthetic auxin (NAA) compared with IAA, using algal growth as a test (WBI15 Oct 2022).
- Gibberellins in flowering: flower number rises then falls as gibberellin concentration increases (WBI15 Oct 2023).

SPEC 8.12

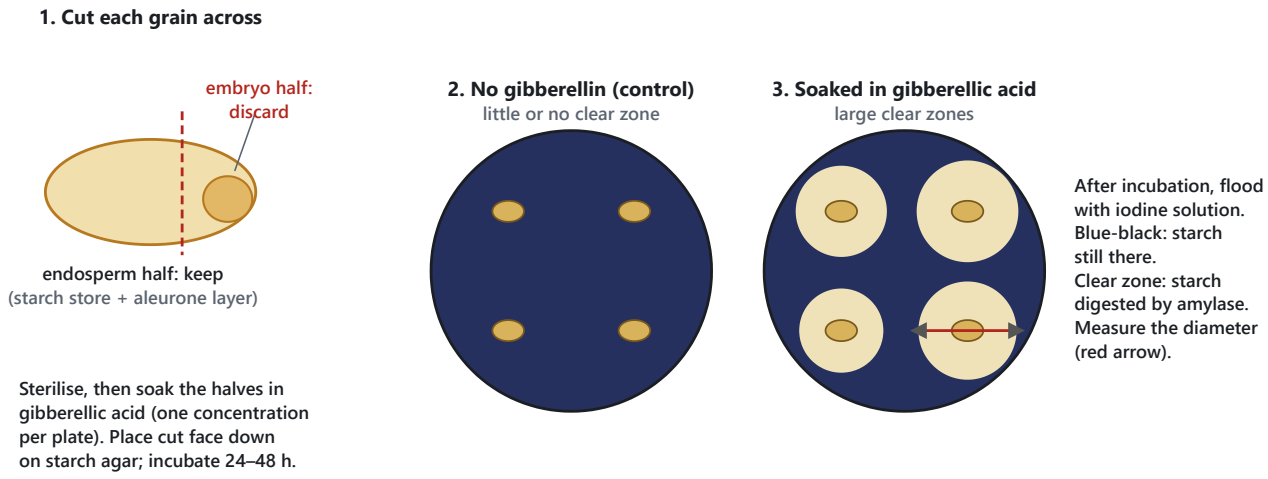
Core Practical 18: amylase in germinating cereal grains

WHAT THIS SPEC POINT ASKS

Investigate how germinating cereal grains produce amylase (and how gibberellin affects this).

8.12 Principle

- A cereal grain (barley or wheat) has an **embryo** at one end and a large starch store, the **endosperm**, wrapped in the **aleurone layer**.
- When the grain takes up water, the embryo releases gibberellin → the aleurone layer makes amylase → amylase digests the starch (8.11).
- Cut off the embryo and the endosperm half has **no gibberellin of its own**. Soaking it in **gibberellic acid** solutions lets you test the effect of gibberellin on amylase production.
- Amylase is detected on **starch agar**. After incubation, the plate is flooded with **iodine**: starch stains **blue-black**, and where amylase has diffused out and digested the starch there's a **clear zone**. A bigger clear zone means more amylase activity.



Half-grains without their embryos make little amylase unless they are given gibberellic acid.

CORE PRACTICAL 18: METHOD IN BRIEF

1. Cut cereal grains in half across the middle with a scalpel, on a tile, cutting away from you. **Discard the embryo halves**; keep the endosperm halves.
2. **Sterilise** the halves in dilute sodium hypochlorite (e.g. 1% for 5–10 minutes), then rinse in sterile water. This kills microorganisms on the surface, which could make amylase themselves.
3. Make a **dilution series** of gibberellic acid (e.g. 0.001, 0.01, 0.1 and 1 g dm^{-3}, diluting ×10 each step), plus sterile water as the 0 control.
4. Soak a batch of halves in each solution for the same time (e.g. 24 hours) at the same temperature.
5. Using aseptic technique, place the same number of halves (e.g. four) **cut face down** on starch agar in labelled Petri dishes. Tape the lids on with two short strips.
6. Incubate at a set temperature (e.g. 25 °C) for **24–48 hours**.
7. Flood each plate with **iodine solution**, lifting the lid only as far as needed, then pour off the excess.
8. Measure the **diameter** of each clear zone (two diameters at right angles, then the mean). Work out the mean for each concentration.
9. Plot mean clear-zone diameter or area (y) against gibberellic acid concentration (x; a log scale suits a ×10 dilution series).

8.12 Variables

Independent	concentration of gibberellic acid (other versions: whole grain v endosperm half; germination time; grain variety)
Dependent	diameter or area of the clear zone after a fixed time
Control	variety, age and batch of grain; size of the half-grains; soaking time and volume; depth and starch concentration of the agar; number of halves per plate; incubation temperature and time; volume and concentration of iodine
Control treatments	halves in water only (0 gibberellic acid): the endosperm alone makes little amylase; whole grains with embryos: positive control; boiled halves: shows the clear zone is caused by an enzyme

8.12 Safety

- **Scalpel**: cut on a tile, away from your fingers.
- **Sodium hypochlorite** is an irritant: eye protection and gloves; rinse spills.
- **Iodine solution** stains and irritates: eye protection.
- **Microorganisms**: aseptic technique; don't open the plates after incubation except briefly to add iodine; dispose of plates by autoclaving.

8.12 Processing the results

WORKED EXAMPLE: CLEAR-ZONE AREA

Four half-grains soaked in 0.1 g dm^{-3} gibberellic acid give clear zones of 18, 22, 20 and 24 mm diameter.
Mean diameter = (18 + 22 + 20 + 24) ÷ 4 = 21 mm, so radius = 10.5 mm.
Area = πr^2 = π × 10.5^2 = **346 mm^2** (3 s.f.).
Halves in water only give a mean diameter of 7 mm: area = π × 3.5^2 = **38.5 mm^2**.
Percentage increase = (346 − 38.5) ÷ 38.5 × 100 = **800%** (using unrounded areas; the ratio of areas is $(10.5 \div 3.5)^2 = 9$).
The common slip: putting the diameter into πr^2, which makes the area four times too big.

8.12 Sources of error and improvements

Problem	Improvement
clear zones aren't circular, or have fuzzy edges	measure two diameters at right angles; or photograph the plate and trace the zone on squared paper to find its area
grains differ in size, age and starch content	use grains from one batch of similar mass; use more halves at each concentration
microorganisms on the grain make amylase	sterilise the halves; use aseptic technique and sterile agar
the iodine colour fades or spreads	measure straight after adding iodine, at the same time for every plate
agar depth differs between plates	pour the same volume into each dish

MARK-SCHEME LANGUAGE

- Why grains need amylase (3 of 4): "amylase hydrolyses glycosidic bonds (in starch)" → "releasing maltose / glucose" → "used in respiration / ATP production" → "for growth / cell division". "Produces energy" is ignored, and a wrong sugar name loses the mark (WBI15 Oct 2025).
- Planning this practical (WBI16 Oct 2024): "aseptic technique (sodium hypochlorite or sterile water)"; "cut seed in half to separate endosperm from embryo"; "soak both halves in gibberellin"; "control not soaked in gibberellin"; "place (cut side down) on starch agar"; "incubate for a stated time (24–48 h) at a stated temperature"; "add iodine (after incubation)"; "compare clear-zone sizes".
- Planning in general: name two controlled variables **and** how you'd control them; test at least five concentrations, with repeats at each.

Often asked as: in WBI15, the biology (why grains need amylase; how gibberellin switches on the amylase gene). The method itself is examined in Unit 6 as a plan for a new variable (abscisic acid, wheat variety, embryo v endosperm).

COMMON MISTAKE

Throwing away one half of the grain when the question asks you to compare the **embryo half and the endosperm half**. Treat both halves the same way and compare them.
Forgetting that the agar must contain **starch**, or calling the clear area a "zone of inhibition" (that's an antibiotic plate).
Saying gibberellin digests the starch.

BEYOND THE SPEC, BUT EXAMINED

- A colorimeter route: grain extract added to starch solution, samples tested with iodine and read against a calibration curve or colour standards (WBI16 Oct 2025).
- **Abscisic acid** works against gibberellin: the higher its concentration, the less amylase is made (WBI16 Jan 2021).

SPEC 8.13

Coordination by nerves and hormones

WHAT THIS SPEC POINT ASKS

Understand how animals coordinate their responses through nervous control and hormonal control.

- **Nervous control**: electrical impulses along neurones, with chemical transmission at synapses.
- **Hormonal control**: **hormones** are made by **endocrine glands** (ductless glands), secreted straight into the **blood**, carried all round the body, and act only on **target cells** that have **complementary receptors**. Peptide hormones bind to receptors on the cell surface membrane; steroid hormones pass through it and bind to receptors inside the cell (7.22).

	Nervous	Hormonal
Signal	electrical impulses (chemical at synapses)	chemical (a hormone)
Route	along neurones	in the blood
Speed	very fast (milliseconds)	slower (seconds to hours)
How long the effect lasts	short: stops when impulses stop and the transmitter is broken down	longer: until the hormone is broken down (e.g. by the liver)
Target	precise: only the cells the neurone connects to	any cell with the right receptor, so it can be widespread
Typical uses	rapid responses: reflexes, muscle contraction	slower or long-term control: blood glucose, water balance, growth, puberty

8.13 The two systems working together

- **Fight or flight**: sympathetic nerves act within a second (heart rate rises), and they also stimulate the **adrenal glands** to release **adrenaline**, which reaches every tissue and keeps the response going (7.14).
- **Heart rate**: the cardiovascular centre in the medulla sends impulses along sympathetic nerves (faster) and the vagus nerve (slower) to the SAN. Adrenaline from the adrenal glands also binds to receptors on the SAN (7.13).
- **Hypothalamus and pituitary**: neurones in the hypothalamus make **ADH**, which passes down their axons to the posterior pituitary and is released into the blood (7.21). A nerve cell sends a hormonal signal.
- Plants rely on chemical coordination only (8.11): slower, and mostly through growth.

EXAM TIP

Write comparisons as paired sentences: "Nerve impulses travel along neurones, **whereas** hormones are carried in the blood." Signal, route, speed and duration cover most marks.

MARK-SCHEME LANGUAGE

- Heart rate: credit **both** routes: impulses "via the sympathetic / vagus nerve to the SAN" and "adrenaline released from the adrenal glands binds to (receptors on) the SAN" (WBI15 Oct 2023).
- Hormones: "travel in the blood", "bind to (specific) receptors on target cells", and for gene-switching signals "activate transcription factors".

Often asked as: not examined as a direct comparison (2020–2026); it comes up inside heart-rate, adrenaline and article questions.

COMMON MISTAKE

"Hormones only travel to their target cells." They go everywhere in the blood; only the target cells have the complementary receptors, so only they respond.

BEYOND THE SPEC, BUT EXAMINED

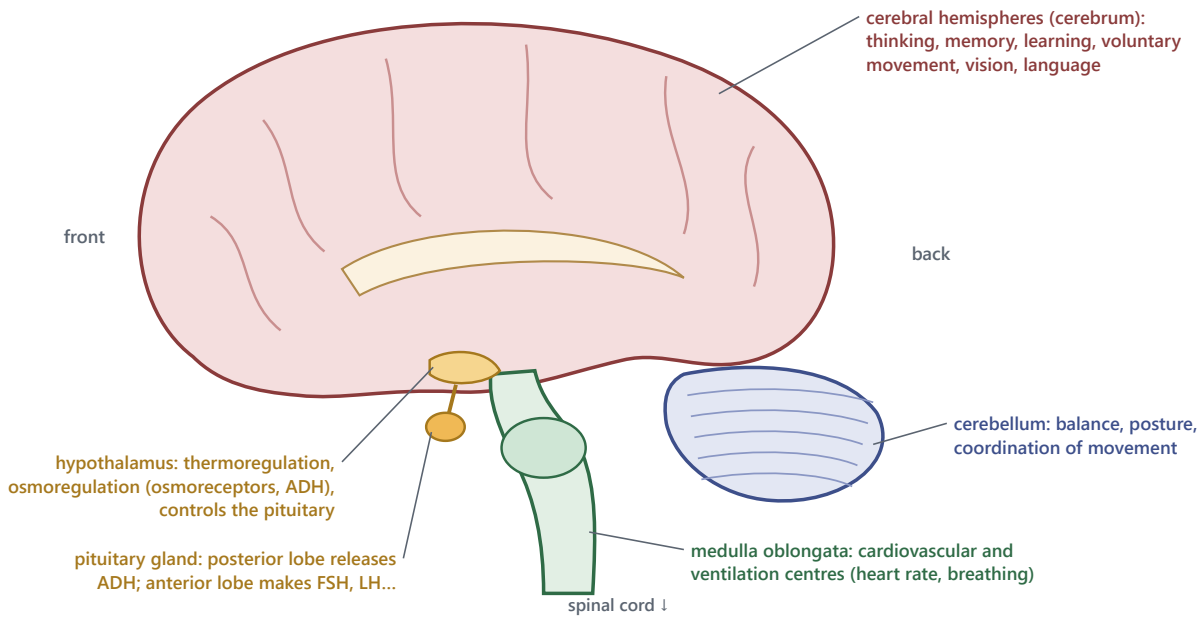
- **Myokines**: signalling proteins released by contracting muscle. They act on the muscle itself (autocrine) and on other organs via the blood (endocrine), binding to cell surface receptors and activating transcription factors (WBI15 Jun 2021 article).

SPEC 8.14

The human brain

WHAT THIS SPEC POINT ASKS

Know where the cerebral hemispheres, hypothalamus, pituitary gland, cerebellum and medulla oblongata are in the human brain, and their main functions.



A simplified section through the middle of the brain, front on the left. The rounded bulge just above the medulla is the pons.

Region	Where	Main functions
Cerebral hemispheres (cerebrum)	the large top part; two halves joined by the corpus callosum; folded outer layer (cortex) of grey matter	conscious thought, learning, memory, language, emotions; **voluntary movement** (motor cortex); interpreting sensory information, e.g. **vision** (occipital lobe at the back) and hearing. Each hemisphere controls the opposite side of the body
Hypothalamus	base of the brain, below the cerebrum, just above the pituitary	**homeostasis**: **thermoregulation** (monitors blood temperature) and **osmoregulation** (osmoreceptors; makes ADH); **controls the pituitary gland**; hunger and thirst
Pituitary gland	hangs on a stalk below the hypothalamus	secretes hormones: the **posterior lobe** stores and releases **ADH** (made in the hypothalamus); the **anterior lobe** makes and secretes FSH, LH and growth hormone
Cerebellum	back of the brain, below the cerebrum	**coordinates movement** so it's smooth and precise; **balance** and **posture**; uses information from muscles, joints and the inner ear
Medulla oblongata	the brain stem, where the brain joins the spinal cord	non-conscious (autonomic) control: the **cardiovascular centre** (heart rate, blood pressure) and the **ventilation centre** (breathing rate); reflexes such as swallowing, coughing and vomiting

- The cortex is **folded**, which gives it a large surface area, so more neurones fit in the skull.
- Memory aid: the **cerebrum** thinks; the **cerebellum** (Latin for "little brain", at the back) keeps you balanced; the **medulla** keeps you alive (heart and breathing).
- Damage questions: damage to the cerebellum → jerky, unbalanced movement; damage to the motor cortex → loss of voluntary movement on the opposite side of the body.

EXAM TIP

Examiners mark the **first** function you write. Give one precise function per region ("controls heart rate", not "reflexes"; "releases ADH", not "makes hormones"). Don't list several to hedge your bets.

MARK-SCHEME LANGUAGE

- Cerebellum: "coordination of (voluntary) movement", "posture", "balance".
- Medulla oblongata: "control of heartbeat and breathing" (or blood pressure, vomiting); "non-conscious regulation".
- Cerebral hemispheres: "voluntary behaviour", "memory", "learning", "thinking", "emotions", "interpretation of information from the retina".
- Hypothalamus: "thermoregulation", "osmoregulation", "production of hormones (ADH)", "controls the pituitary gland".
- Pituitary: "secretion / release (or storage) of ADH", or another correctly named hormone.
- Table marking: 3 correct = 2 marks, 2 correct = 1 mark. Initials are accepted if correct. Only one label line per label.

Often asked as: label a brain diagram (pituitary, cerebellum, hypothalamus, medulla, cerebrum); complete a name-and-function table (set five times); MCQs on which region holds the cardiovascular centre or controls balance.

COMMON MISTAKE

Mixing up **cerebellum** (balance, coordination) and **cerebrum** (thinking, voluntary actions).
Labelling the pons (the bulge just above) as the medulla; putting the hypothalamus in the wrong place (it sits just above the pituitary stalk).
Vague functions: "reflex actions", or "hormones" with no hormone named. The medulla was the weakest one in recent papers.

BEYOND THE SPEC, BUT EXAMINED

- The **pons** is accepted alongside the medulla for control of breathing (WBI15 Jun 2022).
- The vomiting centre is in the medulla (WBI15 Jan 2025).

SPEC 8.15

Brain imaging: CT, MRI, fMRI and PET

WHAT THIS SPEC POINT ASKS

Understand how CT, MRI, fMRI and PET scans are used in medical diagnosis and in investigating brain structure and function.

Sort the four by the question they answer:

- **Structure** (what's there?): CT and MRI.
- **Function** (which parts are active?): fMRI and PET.

8.15 CT (computed tomography)

- Thin **X-ray** beams pass through the head from many angles as the scanner rotates. Denser tissue (bone, fresh blood) absorbs more X-rays. Detectors measure what gets through, and a computer builds **cross-sectional images** (2D slices, which can be stacked into a 3D picture).
- Used for: bleeding in the brain (one kind of stroke), tumours, skull fractures. **Fast** and fairly cheap, so good in emergencies.
- Drawbacks: **ionising radiation** (a small cancer risk, so avoided in pregnancy and not repeated often); less soft-tissue detail than MRI; shows structure only.

8.15 MRI (magnetic resonance imaging)

- A strong **magnetic field** lines up the hydrogen nuclei (mostly in water). **Radio-wave** pulses knock them out of line; as they realign they give off signals that differ between tissues. A computer turns these into detailed 2D or 3D images.
- Used for: tumours, multiple sclerosis lesions, strokes, small structural changes. **Excellent soft-tissue detail** and **no ionising radiation**. A contrast **dye** can be injected to make some tissues stand out more.
- Drawbacks: expensive; slow (you must lie still); noisy and enclosed; can't be used with pacemakers or some metal implants.

8.15 fMRI (functional MRI)

- The same machine, but it follows **blood flow and oxygen**. Active brain regions use more O_2, so blood flow to them rises. **Oxygenated and deoxygenated haemoglobin** behave differently in the magnetic field, so the active regions show up.
- The person does a task during the scan (moving a finger, looking at faces), so you see **which areas are active** as it happens.

- Used for: mapping brain functions (e.g. before surgery), research on memory, language and emotion, studying brain disorders.
- Drawbacks: as for MRI; it measures blood flow, an indirect sign of neurone activity, with a short delay.

8.15 PET (positron emission tomography)

- A **radioactive tracer** is injected, usually glucose labelled with a positron-emitting isotope.
- Active, respiring cells take up more glucose, so the tracer collects there.
- The isotope emits **positrons**; each positron meets an electron and produces **gamma rays**, which the scanner detects. A computer makes a 2D or 3D image showing **metabolic activity**.
- Used for: Alzheimer's disease (areas of low activity), tumours (very active), the effects of drugs, Parkinson's disease (with tracers that bind to dopamine pathways).
- Drawbacks: a dose of radiation (tracers have a short half-life); expensive; lower resolution than MRI, so often combined with CT or MRI to show exactly where the activity is.

	CT	MRI	fMRI	PET
Uses	X-rays	magnetic field + radio waves	magnetic field + radio waves	radioactive tracer (gamma rays detected)
Shows	structure	structure	**function** (blood oxygenation)	**function** (metabolic activity)
Detail	moderate	high, especially soft tissue	high	lower
Ionising radiation?	yes	no	no	yes
Speed / cost	fast, cheaper	slow, expensive	slow, expensive	slow, expensive

WORKED EXAMPLE: LESION VOLUME FROM A SCAN

A lesion on an MRI scan is roughly spherical, 1.97 cm across. Volume of a sphere = $\frac{4}{3}\pi r^3$, with r = 1.97 ÷ 2 = 0.985 cm.

V = $\frac{4}{3} \times \pi \times 0.985^3 = 4.189 \times 0.9557 =$ **4.00 cm³** (2 d.p.) (WBI15 Oct 2023).

Halve the diameter first, and cube the radius before multiplying.

EXAM TIP

"Which scan would you use?" Match the scan to the question: **structure** → CT (quick; emergencies; bleeding) or MRI (soft-tissue detail; no radiation). **Activity** → fMRI (no radiation; fine for repeated scans) or PET (metabolism; specific chemicals). Then give one benefit and one drawback for the patient.

MARK-SCHEME LANGUAGE

- PET: "radioactive tracer / isotope (in glucose)" "injected"; "taken up by active / respiring cells"; "positrons emitted, producing gamma rays" "detected by the scanner"; "2D / 3D image formed"; more active areas take up more tracer (WBI15 Jan 2021, Jan 2022).
- CT: "uses X-rays to produce a cross-sectional (2D / 3D) image".
- MRI: "uses radio waves / a magnetic field"; "no ionising radiation". A dye is given "to enhance / increase the contrast of the image". "High-resolution image" on its own isn't credited.
- fMRI: detects "deoxyhaemoglobin / oxygen uptake", so shows "brain activity"; it takes longer (about 45 min v 5–10 min for CT) and costs more; no ionising radiation.
- Compare and contrast: similarities **and** differences, each in one sentence: "both produce images of the brain, but CT uses X-rays whereas PET uses a radioactive tracer".

Often asked as: describe how PET could be used (3–4 marks); compare two techniques; advantages and disadvantages of MRI and PET; MCQ on which scan shows function as it happens; size or volume of a lesion from a scan.

COMMON MISTAKE

Mixing up PET and fMRI. PET uses a **radioactive tracer** and detects **gamma rays**; fMRI uses **magnetic fields** and detects **blood oxygenation**. And the isotope isn't "the glucose": it is attached to the glucose.
Describing what the images show (e.g. before and after a drug) when the question asks **how the technique works**.
"Picture" or "image" with no detail. Say what kind: cross-sectional, 3D, showing activity.
Treating "advantages and disadvantages" as a comparison and leaving out the disadvantages.

BEYOND THE SPEC, BUT EXAMINED

- Named PET isotopes (carbon-11, oxygen-15) and the positron-to-gamma-ray step (WBI15 Jan 2021).
- PET to show how a drug such as cocaine changes brain activity (WBI15 Jan 2022).
- Combined PET–MRI gives activity and detailed structure in one image (WBI15 Oct 2024 article).

SPEC 8.16 Brain chemical imbalances: Parkinson's disease and depression

WHAT THIS SPEC POINT ASKS

Understand how imbalances in the brain's own chemicals can cause ill health (dopamine in Parkinson's disease, serotonin in depression), and how this knowledge helps develop new drugs.

8.16 Parkinson's disease: too little dopamine

- Dopamine-releasing neurones in the **substantia nigra** (in the midbrain) die. Their axons normally release dopamine in the basal ganglia, which help control movement.
- Less dopamine released → fewer dopamine molecules bind to postsynaptic receptors → fewer action potentials in the neurones that control movement → **tremor, stiff muscles, slow movement, poor balance**, and difficulty starting a movement.
- Symptoms usually appear only once most (about 80%) of these neurones have gone. Risk rises with age.
- Treatments:
  - **L-DOPA**, with carbidopa (8.7);
  - **dopamine agonists**: drugs that bind to dopamine receptors and activate them;

- **MAO-B inhibitors**: block the enzyme (monoamine oxidase B) that breaks dopamine down, so dopamine lasts longer;
- experimental: **stem cells** to replace the lost neurones; **gene therapy** (e.g. genes for enzymes that make dopamine).

8.16 Depression: too little serotonin

- Serotonin is released by neurones whose cell bodies are in the brain stem and whose axons reach many brain regions, including those controlling **mood**, sleep and appetite.
- Less serotonin released → fewer impulses in mood-regulating pathways. Low serotonin is **linked** with depression (low mood, loss of interest, poor sleep). The link is a **correlation**: genes, stress and environment also play a part, and low serotonin isn't proven to be the only cause.
- Treatments:
  - **SSRIs** (selective serotonin reuptake inhibitors): block the **reuptake transporter** in the presynaptic membrane, so serotonin stays in the cleft longer and keeps binding to postsynaptic receptors → more depolarisation;
  - older drugs (MAO inhibitors, tricyclics) also raise serotonin and noradrenaline levels;
  - talking therapies.

8.16 Developing new drugs

- Once you know which chemical is out of balance, and the shapes of its receptors, transporters and enzymes, you can **design drugs** that fit them: agonists, antagonists, reuptake inhibitors, enzyme inhibitors.
- Genome data and bioinformatics (8.21) help find new targets and predict who will respond to which drug (personalised medicine).
- New drugs are tested on cells and animals, then on small groups of healthy volunteers, then on patients in **double-blind, placebo-controlled** trials.

Drug type	Example	Target	Effect at the synapse
precursor	L-DOPA	dopamine-making enzyme in the brain	more dopamine made and released
agonist	dopamine agonists	dopamine receptors	activates receptors without dopamine
enzyme inhibitor	MAO-B inhibitors	enzyme that breaks dopamine down	dopamine lasts longer
reuptake inhibitor	SSRIs	serotonin transporter	serotonin stays in the cleft

MARK-SCHEME LANGUAGE

- Parkinson's with age (4 of 5): "number of dopamine-producing cells falls with age"; the age at which it falls below the stated number (read from the graph); "less dopamine released"; "fewer impulses / action potentials"; "reduced muscle contraction / loss of motor control" (WBI15 Oct 2024). Don't stop at "symptoms get worse".
- A toxin in food causing Parkinson's: "absorbed into the blood (from the gut)", "crosses the blood–brain barrier", "kills dopamine-producing cells / stops dopamine production" (WBI15 Jan 2022).
- SSRIs: "depression linked to a lack of serotonin"; "SSRI binds to the reuptake (transporter) protein"; "reuptake of serotonin inhibited"; "serotonin stays in the synapse and keeps binding to postsynaptic receptors"; "more depolarisation" (WBI15 Jun 2024).
- Error bars: overlapping means "not significantly different"; name the groups or concentrations you're comparing.

Often asked as: data on dopamine cells and age; how a toxin or mutation leads to Parkinson's symptoms; how SSRIs work; data with error bars (dopamine and eating, dopamine and heart rate); correlation or causation.

COMMON MISTAKE

Writing about dopamine in an SSRI (serotonin) question.
Stopping at "symptoms increase". Link it to movement: fewer impulses → less muscle control.
"The error bars overlap" without saying which ones, or calling overlapping bars "unreliable". Overlap means the difference is **not significant**.
Placing the dopamine loss in the "motor cortex". It happens in the substantia nigra.

BEYOND THE SPEC, BUT EXAMINED

- Mutations such as LRRK2: Parkinson's neurones grown in the lab had fewer and shorter dendrites (WBI15 Jan 2021).
- Gut bacteria release chemicals that travel in the blood, cross the blood–brain barrier and change neurotransmitter release (WBI15 Jun 2023 article).
- Standard deviation of a small data set: 45, 63, 74, 58 → mean 60, s = 12 (2 s.f.) (WBI15 Jan 2023).

SPEC 8.17

Making drugs with genetically modified organisms

WHAT THIS SPEC POINT ASKS

Know how drugs can be produced using genetically modified microorganisms, plants and animals.

- **Genetic modification**: changing an organism's DNA, usually by adding a gene from another species. The result is a **genetically modified organism (GMO)**, also called **transgenic**.
- It works because the **genetic code is universal**: a human gene is transcribed and translated into the same protein in a bacterium.
- The overall plan: isolate the gene → put it into a vector → get the vector into the host cells → select the modified cells → grow them → extract and purify the product.

8.17 Microorganisms (bacteria and yeast)

Example: **human insulin** made by *E. coli* or yeast. Others: human growth hormone, clotting factors, vaccine antigens.

1. Get the gene. For insulin, extract **mRNA** from pancreatic β-cells (which make lots of it) and use **reverse transcriptase** to make **cDNA** (8.18).
2. Cut a plasmid with a restriction endonuclease and insert the gene using DNA ligase. The plasmid also carries a promoter and a marker gene.
3. Get the plasmids into the bacteria (e.g. heat shock, 8.19) and select the ones that took up the recombinant plasmid.
4. Grow them on a large scale in a **fermenter**, with sterile conditions and the best temperature, pH, oxygen and nutrients.
5. **Extract and purify** the insulin (downstream processing).

- Insulin is two polypeptide chains, so **two genes** are needed (in one method each chain is made by different bacteria and the chains are joined afterwards).
- Why cDNA: bacteria can't remove introns, and cDNA made from mature mRNA has none.
- Limitation: bacteria have no ER or Golgi apparatus, so they can't add sugar groups (glycosylation) or fold some human proteins properly.

8.17 Plants

Examples: vaccine antigens in edible plants, antibodies in rice, tomatoes that make L-DOPA.

- The gene is put into the **Ti plasmid** of the bacterium *Agrobacterium tumefaciens*, which infects plant cells and inserts the gene into their DNA. Or DNA-coated gold particles are fired into plant cells with a **gene gun**.
- Modified cells are grown into whole plants by tissue culture. The drug is extracted from the leaves or fruit, or (for edible vaccines) the plant part is eaten.

8.17 Animals ("pharming")

Examples: **antithrombin** (an anti-clotting protein) in goats' milk; human α-1-antitrypsin in sheep's milk.

- The human gene is joined to the **promoter of a milk-protein gene**, so it is only switched on in the **mammary gland**.
- The DNA is **microinjected** into the nucleus of a **fertilised egg (zygote)**, and the embryo is implanted into a surrogate mother.
- Offspring are tested for the gene. Females that have it make the protein in their **milk**, from which it is extracted and purified. They are then bred to make more transgenic animals.

Host	How the gene gets in	Advantages	Disadvantages
Bacteria	plasmid + heat shock or electroporation	grow fast and cheaply in fermenters; large amounts; easy to purify	no introns removed (so cDNA needed); no glycosylation or correct folding of some proteins
Yeast	plasmid	eukaryotic, so some protein modification; grows in fermenters	still can't make some human proteins correctly
Plants	Ti plasmid of *Agrobacterium*, gene gun	cheap to grow on a large scale; no animal pathogens; edible vaccines need no fridge	yield varies; risk of gene spreading in pollen
Animals	microinjection into a zygote	proteins **modified correctly** (glycosylated, folded) like human ones; collected from milk without harming the animal	expensive; slow; low success rate; animal welfare concerns

MARK-SCHEME LANGUAGE

- The frame (4 of 5–6), always **in context**, naming the gene, host and product: "identify / isolate the (human) gene for X" (or "extract mRNA and use reverse transcriptase to make cDNA"); "cut with a restriction endonuclease"; "insert into a vector (plasmid / virus)"; "vector put into the host cell by a named method"; "select / grow the modified cells (in a fermenter) and extract / purify X".
- Animals: the gene goes into "udder / mammary cells" or a "fertilised egg / zygote / embryo"; the product comes from the milk (WBI15 Oct 2022).
- Eukaryotic host advantage: "has ER and Golgi", so the protein is "(properly) glycosylated / modified" (WBI15 Jan 2021).
- GM insulin compared with animal insulin: "not limited by animal slaughter", "large quantities", "no ethical / religious objection", "less immune response / rejection" (identical to human insulin), "no risk of infection from animals" (WBI15 Jan 2025).

Often asked as: "Describe how [organism] could be genetically modified to make [product]" (about four marks, in nearly every paper); advantages of a host; yield data from GM strains.

COMMON MISTAKE

A generic GM answer that never names the gene, host or product. Examiners repeatedly cap these, and re-using an old mark scheme out of context doesn't work.
Missing the first mark: say which gene ("the human growth hormone gene") and that it's isolated.
Saying the gene comes from the sheep or the tomato. It comes from the **donor** (usually human). And it goes into a **zygote** or mammary cells, not "into the sheep".
"The insulin gene" when the question involves both chains: two polypeptides need **two genes**.

BEYOND THE SPEC, BUT EXAMINED

- GloFish-style zebrafish: a coral fluorescence gene put into zebrafish embryos; glowing fish selected under UV light and then bred (WBI15 Jun 2024 article).
- Mouse cells compared with *E. coli* for making human growth hormone: more like human cells, but slower, costlier and lower yield (WBI15 Jun 2025).
- A recombinant protein drug can itself trigger an immune response: antigen presented to T helper cells, which activate B cells to make antibodies (WBI15 Oct 2022).

SPEC 8.18

Making recombinant DNA

WHAT THIS SPEC POINT ASKS

Understand how recombinant DNA is made, including the roles of restriction endonucleases and DNA ligase.

Recombinant DNA is DNA made by joining DNA from two different sources, e.g. a human gene inside a bacterial plasmid.

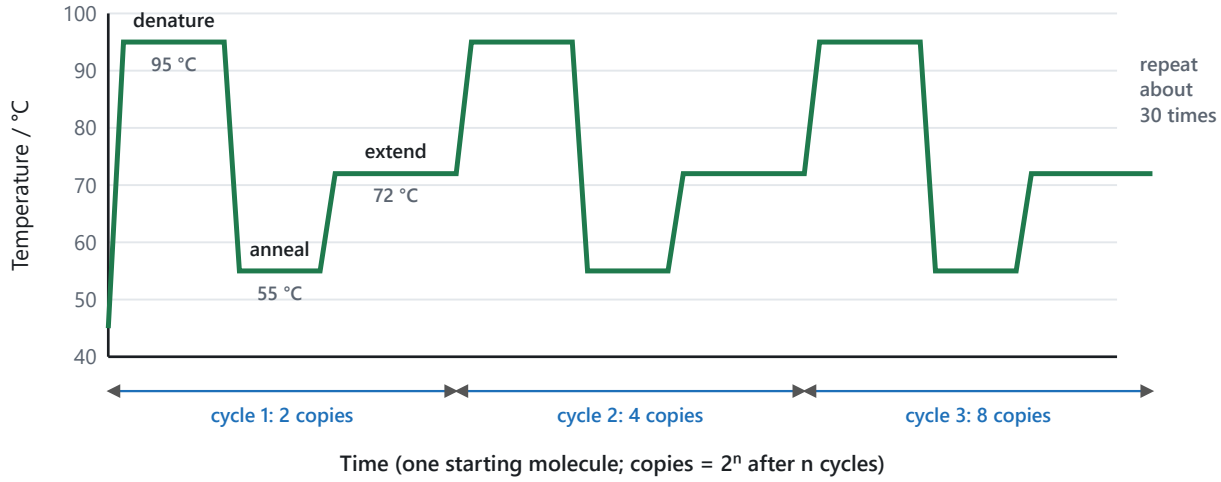
8.18 1. Getting the gene

- **Cut it out** of the donor's DNA with restriction endonucleases. You need to know where it is, and a human gene cut out this way still contains introns.
- **Make it from mRNA** (the usual route for human proteins made by bacteria): take mRNA from cells that express the gene (e.g. pancreatic β-cells for insulin, where there is lots of it), use **reverse transcriptase** to

make a single complementary DNA strand (**cDNA**), then **DNA polymerase** to make the second strand. The cDNA has **no introns**.

- **Make it artificially**, nucleotide by nucleotide, if the base sequence (or the protein's amino acid sequence) is known.

8.18 2. Copying it: PCR



Each PCR cycle runs through three temperatures and doubles the number of copies of the target DNA.

The reaction tube holds the DNA, two **primers** (short single-stranded DNA pieces complementary to the ends of the target section, one for each strand), free **DNA nucleotides**, **Taq polymerase** and a buffer. A **thermal cycler** then repeats:

1. **Denaturation, about 95 °C**: the hydrogen bonds between bases break and the two strands separate.
2. **Annealing, about 55 °C** (50–65 °C): the primers bind to the ends of the target on each single strand by complementary base pairing.
3. **Extension, about 72 °C**: Taq polymerase adds nucleotides to the primer, building a complementary strand (joining them with phosphodiester bonds).

- **Taq polymerase** comes from *Thermus aquaticus*, a bacterium from hot springs. It isn't denatured at 95 °C, so it doesn't need replacing each cycle, and 72 °C is its optimum.
- Each cycle takes a few minutes and **doubles** the amount of target DNA.

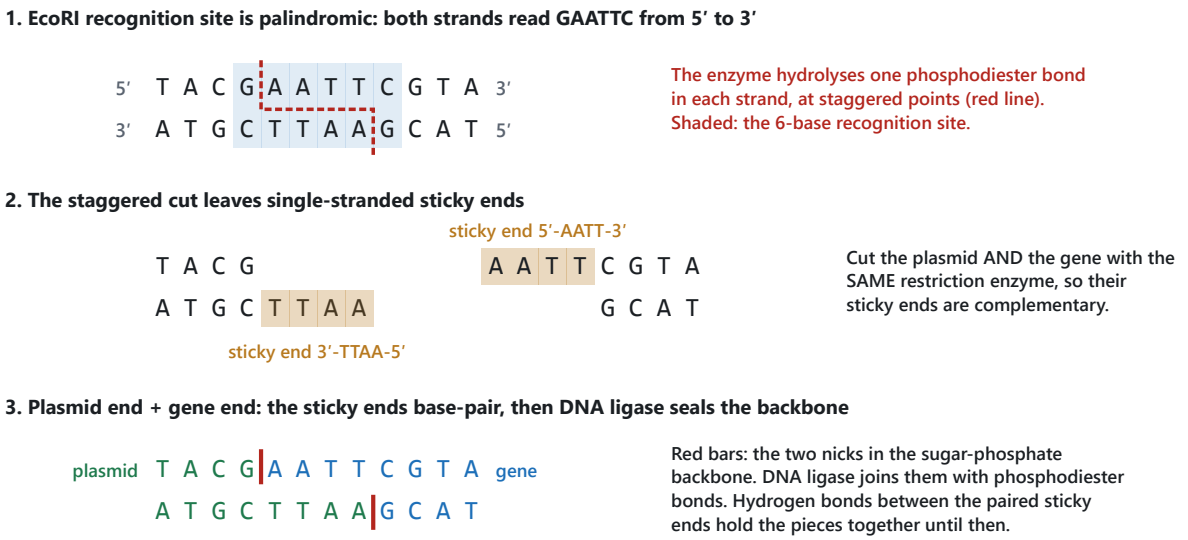
WORKED EXAMPLE: PCR COPY NUMBER

Each cycle doubles the number of DNA molecules: copies = starting number × 2^n, where n = number of cycles.
One molecule after 30 cycles: 2^{30} = 1 073 741 824 ≈ **1.07×10^9** copies.
200 molecules after 20 cycles: 200 × 2^{20} = 200 × 1 048 576 = **2.10×10^8** copies.
Cycles needed for at least 10^6 copies from one molecule: 2^{19} = 524 288 (too few), 2^{20} = 1 048 576, so **20 cycles**. At 3 minutes per cycle that's 60 minutes.

8.18 3. Cutting: restriction endonucleases

- These enzymes come from bacteria, which use them to chop up the DNA of invading viruses.
- Each one recognises a specific short base sequence, its **recognition site** (usually 4–8 base pairs), and **hydrolyses the phosphodiester bonds** in the sugar-phosphate backbone of both strands there. Its active site is complementary only to that sequence.

- Recognition sites are **palindromic**: the sequence reads the same on both strands when each is read 5′ → 3′. For EcoRI, one strand reads 5′-GAATTC-3′ and so does the other (read in its own 5′ → 3′ direction). Because the site is symmetrical, the enzyme cuts both strands at the same point within it.
- Many restriction enzymes cut the two strands at **staggered** points, leaving short single-stranded overhangs: **sticky ends** (EcoRI cuts G^AATTC, leaving AATT overhangs). Others cut straight across, leaving **blunt ends** (e.g. CCC^GGG).
- Any two pieces of DNA cut with the **same** enzyme have **complementary sticky ends**, which can base-pair with each other.



Cutting with EcoRI and joining the sticky ends of a plasmid and a gene.

8.18 4. Joining: DNA ligase

1. Cut the plasmid (the vector) with the **same restriction enzyme**, so it has sticky ends complementary to the gene's.
2. Mix the gene with the opened plasmids. The sticky ends **anneal**: hydrogen bonds form between complementary bases.
3. **DNA ligase** joins the sugar-phosphate backbones by forming **phosphodiester bonds** (condensation reactions). The result is a **recombinant plasmid**.

- The plasmid also needs a **promoter** (so the gene is transcribed in the host) and usually a **marker gene** (8.19).
- Some plasmids close up again without the gene, so the cells that took up a recombinant plasmid must be picked out (8.19).

Enzyme	Job
Restriction endonuclease	cuts DNA at a specific recognition site (hydrolyses phosphodiester bonds)
DNA ligase	joins DNA fragments (forms phosphodiester bonds)
Reverse transcriptase	makes cDNA using mRNA as the template
DNA polymerase (Taq in PCR)	builds a new DNA strand on a template

WORKED EXAMPLE: CUTTING A SEQUENCE

An enzyme recognises 5′-AAGCTT-3′ and cuts between the two A's (A^AGCTT).
Is the site palindromic? The other strand reads 3′-TTCGAA-5′, which is AAGCTT when read 5′ → 3′. **Yes.**
The sticky end left is a four-base overhang, 5′-AGCT.
A **linear** DNA molecule with 3 of these sites is cut into **4 fragments**; a **circular** plasmid with 3 sites gives **3 fragments**.

MARK-SCHEME LANGUAGE

- "cut the gene / DNA using a restriction endonuclease" ("restriction enzyme" is fine; "restricting enzyme" is ignored).
- "cut the plasmid with the **same** restriction enzyme" → "complementary sticky ends".
- "sticky ends joined using **DNA ligase**" / "forms phosphodiester bonds". Ligase was required in some sessions and not in others, so always include it.
- The mRNA route: "extract mRNA" → "reverse transcriptase" → "cDNA".
- PCR, using any temperatures given: "about 95 °C breaks hydrogen bonds / separates the strands"; "cooled (about 55 °C) so primers bind"; "about 72 °C, Taq polymerase joins nucleotides"; "repeated cycles".
- MCQs: the enzyme that cuts DNA is a restriction endonuclease; the bond joining nucleotides is a phosphodiester bond (WBI15 Oct 2023).

Often asked as: inside the "describe how an organism could be modified" questions (8.17); MCQs on enzymes and bonds; CRISPR and viral vectors in article questions.

COMMON MISTAKE

"Restriction enzymes join the fragments." They **cut**; **ligase joins**.
Not saying that the **same** enzyme cuts both the gene and the plasmid.
"The recombinant DNA forms inside the host." It's made in a test tube; the host only takes it up.
PCR: leaving out the temperatures; confusing primers with probes; saying Taq polymerase makes RNA.

BEYOND THE SPEC, BUT EXAMINED

- **CRISPR-Cas9**: a **guide RNA** complementary to a target gene leads the **Cas9** endonuclease to that site, where it cuts the DNA, so the gene can be deleted or edited (WBI15 Jun 2025).
- An adenovirus as a vector for a working gene (AAT deficiency): the same restriction enzyme and ligase put the gene into the viral DNA; the virus enters the host cell and a functional protein is made (WBI15 Jan 2022).

SPEC 8.19

Getting recombinant DNA into cells

WHAT THIS SPEC POINT ASKS

Understand how recombinant DNA can be inserted into other cells.

The DNA has to get across the cell surface membrane (and, in eukaryotes, into the nucleus), then be kept and expressed. It is carried by a **vector** or pushed in physically.

Method	How it works	Used for
Plasmid + heat shock	bacteria are chilled in calcium chloride solution (Ca^{2+} ions reduce the repulsion between negatively charged DNA and the phospholipid heads), then warmed briefly to about 42 °C; the membrane becomes more permeable and plasmids enter	bacteria
Electroporation	a brief high-voltage pulse opens temporary pores in the membrane	bacteria, plant and animal cells
Viral vector	the gene is put into a harmless (disabled) virus, which infects the host cell and injects its DNA; some viruses insert it into the host's chromosomes	animal and human cells; gene therapy (e.g. adenovirus)
Liposome	the DNA is wrapped in a phospholipid sphere, which fuses with the cell surface membrane (or is taken in by endocytosis)	gene therapy (e.g. lung cells in cystic fibrosis)
Agrobacterium tumefaciens	this soil bacterium naturally inserts part of its Ti plasmid into plant DNA; the gene is put into the Ti plasmid first	plants
Gene gun	tiny gold or tungsten pellets coated with DNA are fired into the cells	plants, especially cereals
Microinjection	DNA is injected through a very fine glass pipette into the nucleus of a fertilised egg	animals (pharming, transgenic animals)

8.19 Finding the cells that took up the gene

Only a small fraction of cells take up a plasmid, and some plasmids don't contain the gene. **Marker genes** let you find the right ones:

- **Antibiotic resistance**: the plasmid carries a gene for resistance to antibiotic A. Grow the bacteria on agar containing A: only those with a plasmid survive.
- **Insertional inactivation and replica plating**: the plasmid carries resistance genes for two antibiotics, A and B, and the new gene is inserted **inside** gene B, which stops it working.
  1. Grow the bacteria on agar with antibiotic A (the master plate): only bacteria with a plasmid grow.
  2. Press sterile velvet onto the master plate and then onto a plate with antibiotic B (the replica), so each colony is copied in the same position.
  3. Bacteria with the **recombinant** plasmid **don't grow** on B, because gene B has been disrupted.
  4. Colonies present on the master plate but **missing from the replica** are the ones you want. Pick them from the master plate.
- **Fluorescent markers**: a gene for a fluorescent protein (e.g. GFP) makes modified cells glow under UV light. No antibiotic resistance genes are needed, which is safer.

WORKED EXAMPLE: READING REPLICA PLATES

A plasmid has genes for resistance to ampicillin and to tetracycline. The insulin gene was inserted into the tetracycline resistance gene. Bacteria were grown on ampicillin, then replica-plated onto tetracycline. Colony P grows on both plates; colony Q grows on ampicillin but not on tetracycline.
P took up a plasmid **without** the insulin gene (its tetracycline gene still works). Q took up the **recombinant plasmid** (its tetracycline gene is disrupted by the insert). **Pick Q** from the master plate. Bacteria with no plasmid at all don't grow on either plate.

MARK-SCHEME LANGUAGE

- Named vectors: "plasmid", "virus (adenovirus)", "liposome". Named methods: "heat shock", "electroporation", "gene gun", "micro-injection", "*Agrobacterium tumefaciens*". The vector must be described as **entering the host cell**.
- Bacteria: "plasmid taken up by the *E. coli*"; "bacteria given heat shock" (WBI15 Oct 2023, Jan 2024). "Gene gun" for bacteria and "vector" on its own are ignored.
- Animals: into "udder / mammary cells" or a "fertilised egg / zygote / embryo"; fish: into embryos or zygotes.

Often asked as: a mark point inside the "describe how" GM questions (8.17); how to find the bacteria that took up the gene.

COMMON MISTAKE

"Put the gene into the bacteria" with no vector or method. Name the method.
Putting the gene into the wrong cells: the whole sheep rather than a zygote; a gene gun for bacteria.
Thinking the inserted DNA always lands in the right place in a chromosome. Where it inserts is random, which is one reason screening is needed.

BEYOND THE SPEC, BUT EXAMINED

- Gene therapy for inherited blindness (the RPE65 gene), with a dose–response comparison using error bars (WBI15 Jan 2021).
- Antibiotic-resistance marker genes could spread to other bacteria, which is why fluorescent markers are a safer choice.

SPEC 8.20 Microarrays: finding which genes are active

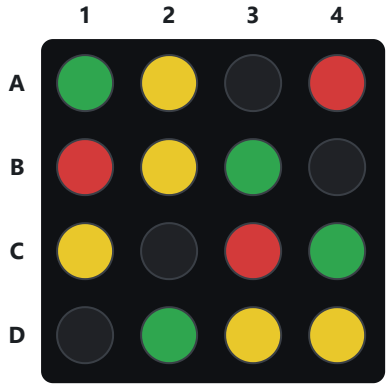
WHAT THIS SPEC POINT ASKS

Know how microarrays can be used to identify active genes.

- An **active gene** is one that is being **transcribed**, so its **mRNA** is present. Different cell types, and the same cells at different times or in disease, express different genes.
- A **microarray** (DNA chip) is a small glass slide with thousands of microscopic spots in a grid. Each spot holds many copies of one known **single-stranded DNA probe**, complementary to part of one gene.

8.20 Method: comparing two samples

1. Extract **mRNA** from both samples (e.g. healthy and tumour cells; treated and untreated cells).
2. Use **reverse transcriptase** to make **cDNA** from each mRNA sample (cDNA is more stable than mRNA).
3. Label each cDNA sample with a different **fluorescent dye** (e.g. healthy green, tumour red).
4. Mix the samples and wash them over the microarray. Each cDNA **hybridises** (complementary base pairing) with the probe for its gene.
5. **Wash off** any cDNA that hasn't bound.
6. **Scan** the slide with a laser. A computer records the colour and brightness of every spot.



Key (this question's colours)

- green: cDNA from healthy cells only (gene active only there)
- red: cDNA from tumour cells only (gene active only there)
- yellow: both bound (gene active in both)
- black: neither bound (gene not transcribed)

Each spot holds many copies of one known single-stranded DNA probe. Brightness shows how much cDNA bound, so how strongly that gene was being expressed.

A small microarray. The colour shows which sample a gene was active in; the brightness shows how strongly.

WORKED EXAMPLE: READING A MICROARRAY

Using the grid above (green = healthy cells only, red = tumour cells only, yellow = both, black = neither):

- Active **only in the tumour**: A4, B1, C3. These may be driving the cancer, so they're possible drug targets.
- **Switched off in the tumour** (active only in healthy cells): A1, B3, C4, D2.
- Active in **both**: A2, B2, C1, D3, D4. Active in **neither**: A3, B4, C2, D1.

Always use the key in the question: the colour for each sample changes from paper to paper.

8.20 Uses and limitations

- Uses: comparing healthy and diseased tissue (e.g. telling cancer types apart and choosing treatment); finding the genes a drug, hormone or toxin switches on; finding genes linked to a disease by testing many people; finding out whether a pathogen's genes are active in host tissue.
- Limitations:
  - it measures mRNA, not the amount of **protein** made;
  - it gives **relative**, not absolute, amounts;
  - it only detects genes that have a probe on the chip;
  - it produces huge amounts of data, which need **bioinformatics** (8.21).

MARK-SCHEME LANGUAGE

- "extract mRNA (from the cells / named tissue)"; "use a microarray / DNA probes"; "fluorescent dyes / tags / labels" (not just "fluorescence"); "compare stimulated and unstimulated / healthy and diseased"; "use bioinformatics / computer algorithms to analyse the data" (WBI15 Oct 2022 article).
- Short form: "use a microarray" "to detect mRNA", with cDNA or fluorescent labels (WBI15 Jan 2024).
- Active gene = "mRNA present / gene being transcribed".
- A separate route (don't mix it with the microarray one): mRNA → reverse transcriptase → cDNA → PCR → gel electrophoresis → compare bands.

Often asked as: "Suggest how scientists could find out which genes are active in…" (about ten papers, 2–4 marks, often in the article question).

COMMON MISTAKE

Knowing the words "microarray" and "bioinformatics" but giving no detail. Mention **mRNA**, **reverse transcriptase / cDNA**, **fluorescent tags**, **hybridisation** and the **comparison**.
Not saying which tissue the mRNA comes from (e.g. the infected brain tissue).
Mixing steps from two different routes. Each route is marked separately.

BEYOND THE SPEC, BUT EXAMINED

- Sequencing the genomes of individuals with and without a trait and comparing them is credited as another way to find the genes involved (WBI15 Oct 2024, Oct 2025 articles).

SPEC 8.21

Bioinformatics

WHAT THIS SPEC POINT ASKS

Understand what is meant by bioinformatics.

- **Bioinformatics** is the development and use of **computer software, databases and statistical methods (algorithms)** to **store, organise and analyse** large amounts of biological data, such as DNA and protein sequences and microarray results.
- It's needed because sequencing and microarrays produce far more data than anyone could analyse by hand: a human genome is about 3×10^9 base pairs.
- Sequence databases are shared online, so scientists anywhere can compare their data with everyone else's.
- Uses:
  - finding genes in a newly sequenced genome and predicting the protein (and its shape) each codes for;
  - comparing sequences between species to work out how closely related they are;
  - analysing microarray data to find the genes switched on in a disease, and possible drug targets;
  - comparing the genomes of people with and without a disease (genetic risk, personalised medicine);
  - tracking new variants of pathogens.
- Limits: the results are only as good as the data put in; huge storage needs; privacy of personal genetic information.

WORKED EXAMPLE: COMPARING SEQUENCES

The same 20-base stretch of a gene is compared in three species:
A: `ATGGCTTACGATCCGTAAGC`
B: `ATGGCTTACGTTCCGTAAGC` (1 base different from A)
C: `ATGCCTAACGTTCAGTATGC` (5 bases different from A)
Similarity A–B = 19 ÷ 20 × 100 = **95%**; A–C = 15 ÷ 20 × 100 = **75%**.
A and B are more closely related: they share a more recent common ancestor. Bioinformatics software does the same thing for millions of bases at once.

MARK-SCHEME LANGUAGE

- "use of bioinformatics / computers / databases / algorithms to analyse the data"; "reference to the use of a computer database" is accepted.
- Showing a gene came from a plant (whitefly): "determine the DNA base sequence", "use bioinformatics to compare it with plant genomes", "the sequence is found only in plants and not in other whitefly" (WBI15 Jun 2024). Finish with that conclusion.
- Homology between species: "sequence the genes", "compare the sequences for similarity", "the products have similar amino acid sequences / functions".

Often asked as: one mark point inside microarray and genome-comparison questions; never asked as a stand-alone definition in 2020–2026.

COMMON MISTAKE

"Bioinformatics finds the gene." Say what is compared with what (sequences, gene activity) and what result would answer the question.
Gel band patterns that "are similar". They must be the **same** (match).

BEYOND THE SPEC, BUT EXAMINED

- Comparing human and zebrafish genes to justify using zebrafish as a model for human toxicity (WBI15 Jan 2022 article).

SPEC 8.22

Risks and benefits of GMOs

WHAT THIS SPEC POINT ASKS

Understand the risks and benefits of using genetically modified organisms.

Area	Benefits	Risks
Crops	higher yields; **pest resistance** (Bt crops make a protein toxic to insect larvae, so less insecticide is sprayed); **herbicide tolerance** (the crop survives a herbicide such as glyphosate, so weeds can be killed without harming it); drought, salt and disease resistance; better nutrition (**Golden Rice** makes β-carotene, which the body turns into vitamin A); longer shelf life; more food from the same land	**gene flow**: pollen carries the gene to wild relatives, producing **superweeds**; pests evolve resistance to the Bt toxin (**superpests**); **monocultures** reduce **biodiversity**; harm to insects that aren't pests; unknown long-term effects of eating GM food (e.g. allergy to a new protein)
Medicine	human proteins (insulin, growth hormone) identical to our own, so fewer immune reactions than animal versions; cheap, plentiful and pure; no infection risk from animal or human sources; plant vaccines need no refrigeration; gene therapy	viral vectors can trigger immune responses or insert in the wrong place in the genome; antibiotic-resistance marker genes could pass to harmful bacteria (**horizontal gene transfer**)
Animals	drugs collected in milk; faster-growing or disease-resistant animals; animal models for research	animal welfare; escaped GM animals could outcompete or breed with wild populations
Society and economics	cheaper food; less pesticide spraying and cost; food security	seed patented by large companies; farmers may have to buy new seed every year, so poorer farmers lose out; moral or religious objections; consumer choice and labelling

8.22 Reducing the risks

- Grow GM crops in secure greenhouses, or with **buffer zones** of non-GM plants round them; keep GM and non-GM fields apart.
- Make GM plants sterile so their pollen can't spread the gene.
- Use fluorescent markers rather than antibiotic-resistance genes.
- Test thoroughly before release; regulate and label GM food.

EXAM TIP

For a six-mark "discuss" or "assess" question: use the data you're given (quote figures), give **both** benefits and risks with a reason for each, and finish with a judgement if the command word is "assess". The top level needs **both sides** in detail. "It's cheaper" or "it's unethical" needs a reason.

MARK-SCHEME LANGUAGE

- Risks (3 of 4): "accumulation of engineered material in the environment / other species"; "cross-breeding with wild species / pollen transfer"; "unknown long-term effects of eating GMOs"; "superweeds / superpests"; a named health effect such as allergy. Reducing a risk: "grow in secure greenhouses", "separate GM and non-GM crops". Reduce the risk you named; don't bring in a new one (WBI15 Oct 2021).
- GM crops: benefits are yield, less pest damage, less herbicide and pesticide use, better nutrition and food supply; risks are monoculture, reduced biodiversity, horizontal gene transfer (antibiotic resistance), herbicide-resistant weeds and unknown long-term effects (WBI15 Oct 2022).
- Resistance evolving: "only the individuals with the gene / allele survive", "reproduce", "pass on the allele" (WBI15 Jun 2024).

Often asked as: six-mark "discuss / assess the risks and benefits" with a table and a graph (GM soybean and corn; a GM fungus in a warming world); risks and benefits of GM zebrafish; advantages and disadvantages of a GM drug source.

COMMON MISTAKE

Vague risks such as "it could be dangerous". Give the mechanism: pollen spreads the herbicide-resistance gene to a wild relative, which becomes a superweed.
Discussing only the risks. Benefits are needed for the top level.
Describing the data without linking it to a benefit or a risk.

BEYOND THE SPEC, BUT EXAMINED

- **Horizontal gene transfer**: a gene that passed from a plant into whitefly, letting the whitefly break down plant poisons (WBI15 Jun 2024 article).
- GM uptake data: GM soybean rose from 7% to 95%, and GM corn from 6% to 88%, of the US planted area between 1996 and 2017 (WBI15 Oct 2022).